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# UNITED STATES SECURITIES AND EXCHANGE COMMISSION

**Washington, D.C. 20549**

# Form 10-K

(Mark one)

☑ **ANNUAL REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934**

**For the fiscal year ended October 28, 2018**

**or**

☐ **TRANSITION REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934**

**For the transition period from to**

**Commission file number 000-06920**

# Applied Materials, Inc.

*(Exact name of registrant as specified in its charter)*

| **Delaware** | **94-1655526** |
|---|---|
| *(State or other jurisdiction of incorporation or organization)* | *(I.R.S. Employer Identification No.)* |
| **3050 Bowers Avenue, P.O. Box 58039 Santa Clara, California** | **95052-8039** |
| *(Address of principal executive offices)* | *(Zip Code)* |

**Registrant's telephone number, including area code:**
**(408) 727-5555**

**Securities registered pursuant to Section 12(b) of the Act:**

| **Title of Each Class** | **Name of Each Exchange on Which Registered** |
|---|---|
| Common Stock, par value $.01 per share | The NASDAQ Stock Market LLC |

**Securities registered pursuant to Section 12(g) of the Act: None**

Indicate by check mark if the registrant is a well-known seasoned issuer, as defined in Rule 405 of the Securities Act. Yes ☑ No ☐

Indicate by check mark if the registrant is not required to file reports pursuant to Section 13 or Section 15(d) of the Act. Yes ☐ No ☑

Indicate by check mark whether the registrant (1) has filed all reports required to be filed by Section 13 or 15(d) of the Securities Exchange Act of 1934 during the preceding 12 months (or for such shorter period that the registrant was required to file such reports), and (2) has been subject to such filing requirements for the past 90 days. Yes ☑ No ☐

Indicate by check mark whether the registrant has submitted electronically every Interactive Data File required to be submitted pursuant to Rule 405 of Regulation S-T (§232.405 of this chapter) during the preceding 12 months (or for such shorter period that the registrant was required to submit). Yes ☑ No ☐

Indicate by check mark if disclosure of delinquent filers pursuant to Item 405 of Regulation S-K (§229.405 of this chapter) is not contained herein, and will not be contained, to the best of registrant's knowledge, in definitive proxy or information statements incorporated by reference in Part III of this Form 10-K or any amendment to this Form 10-K. ☑

Indicate by check mark whether the registrant is a large accelerated filer, an accelerated filer, a non-accelerated filer, a smaller reporting company or an emerging growth company. See definitions of "large accelerated filer," "accelerated filer," "smaller reporting company" and "emerging growth company" in Rule 12b-2 of the Exchange Act.

| | |
|---|---|
| Large accelerated filer ☑ | Accelerated filer ☐ |
| Non-accelerated filer ☐ | Smaller reporting company ☐ |
| | Emerging growth company ☐ |

If an emerging growth company, indicate by check mark if the registrant has elected not to use the extended transition period for complying with any new or revised financial accounting standards provided pursuant to Section 13(a) of the Exchange Act. ☐

Indicate by check mark whether the registrant is a shell company (as defined in Rule 12b-2 of the Act). Yes ☐ No ☑

Aggregate market value of the voting stock held by non-affiliates of the registrant as of April 29, 2018, based upon the closing sale price reported by the NASDAQ Global Select Market on that date: $49,674,881,854

Number of shares outstanding of the registrant's Common Stock, $.01 par value, as of December 7, 2018: 958,606,093

**DOCUMENTS INCORPORATED BY REFERENCE:**

Portions of Part III will be provided in accordance with Instruction G(3) to Form 10-K no later than February 25, 2019.

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**Caution Regarding Forward-Looking Statements**

This Annual Report on Form 10-K of Applied Materials, Inc. and its subsidiaries, including "Management's Discussion and Analysis of Financial Condition and Results of Operations" in Item 7, contains forward-looking statements that involve a number of risks and uncertainties.

Examples of forward-looking statements include those regarding Applied's future financial or operating results, customer demand and spending, end-use demand, market and industry trends and outlooks, cash flows and cash deployment strategies, declaration of dividends, share repurchases, business strategies and priorities, costs and cost controls, products, competitive positions, management's plans and objectives for future operations, research and development, strategic acquisitions and investments, growth opportunities, restructuring activities, backlog, working capital, liquidity, investment portfolio and policies, taxes, supply chain, manufacturing, properties, legal proceedings and claims, and other statements that are not historical facts, as well as their underlying assumptions. Forward-looking statements may contain words such as "may," "will," "should," "could," "would," "expect," "plan," "anticipate," "believe," "estimate," "potential" and "continue," the negative of these terms, or other comparable terminology. All forward-looking statements are subject to risks and uncertainties and other important factors, including those discussed in Part I, Item 1A, "Risk Factors," below and elsewhere in this report. These and many other factors could affect Applied's future financial condition and operating results and could cause actual results to differ materially from expectations based on forward-looking statements made in this document or elsewhere by Applied or on its behalf. Forward-looking statements are based on management's estimates, projections and expectations as of the date hereof, and Applied undertakes no obligation to revise or update any such statements.

The following information should be read in conjunction with the Consolidated Financial Statements and the accompanying Notes to Consolidated Financial Statements included in this report.

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**APPLIED MATERIALS, INC.**

**FORM 10-K FOR THE FISCAL YEAR ENDED OCTOBER 28, 2018**

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## PART I

**Item 1:** ***Business***

Incorporated in 1967, Applied Materials, Inc. (Applied) is a Delaware corporation. A global company with a broad set of capabilities in materials engineering, Applied provides manufacturing equipment, services and software to the semiconductor, display and related industries. With its diverse technology capabilities, Applied delivers products and services that improve device performance, yield and cost. Applied's customers include manufacturers of semiconductor chips, liquid crystal and organic light-emitting diode (OLED) displays, and other electronic devices. These customers may use what they manufacture in their own end products or sell the items to other companies for use in advanced electronic components. Applied's fiscal year ends on the last Sunday in October.

Applied operates in three reportable segments: Semiconductor Systems, Applied Global Services, and Display and Adjacent Markets. A summary of financial information for each reportable segment is found in Note 15 of Notes to Consolidated Financial Statements. A discussion of factors that could affect operations is set forth under "Risk Factors" in Item 1A, which is incorporated herein by reference.

### *Semiconductor Systems*

Applied's Semiconductor Systems segment develops, manufactures and sells a wide range of manufacturing equipment used to fabricate semiconductor chips, also referred to as integrated circuits (ICs). The Semiconductor Systems segment includes semiconductor capital equipment used for many steps of the chip making process including the transfer of patterns into device structures, transistor and interconnect fabrication, metrology, inspection and review, and packaging technologies for connecting finished IC die. Applied's patterning systems and technologies address challenges resulting from shrinking pattern dimensions and the growing complexity in vertical stacking found in today's most advanced semiconductor devices. Applied's transistor and interconnect products and technologies enable continued device scaling of 3D transistors. Applied's metrology, inspection and review systems' imaging capabilities and algorithms employ optical and e-beam technologies to meet the most advanced technical demands, such as self-aligned double and quad patterning, extreme ultraviolet layers, measurement-intensive optimal proximity correction mask qualification, and new 3D architectures. Applied's packaging technologies address challenges resulting from the increasing integration of multiple IC dies in a single package. Applied delivers leading-edge capabilities that enable chipmakers to establish accurate statistical process control, ramp up production runs rapidly, and achieve consistently high production yields. The majority of Applied's new equipment sales are to leading integrated device manufacturers and foundries worldwide.

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| Technologies | Product(s) |
|---|---|
| **Epitaxy**<br>Epitaxy (or epi) is a technique for growing silicon (e.g. silicon with another element) as a uniform crystalline structure on a wafer to form high quality material for the device circuity. Epi technology is used in device transistors to enhance chip speed. | Centura RP Epi |
| **Ion Implant**<br>Ion implantation is a key technology for forming transistors and is used many times during chip fabrication. During ion implantation, wafers are bombarded by a beam of electrically-charged ions, called dopants, which can change the electrical properties of the exposed semiconductor material. | VIISta Systems |
| **Oxidation/Nitridation**<br>Applied's systems provide critical oxidation steps - like memory gate oxide, shallow trench isolation and liner oxide - for advanced device scaling. | Vantage, Radiance and Centura Systems |
| **Rapid Thermal Processing (RTP)**<br>RTP is used primarily for annealing, which modifies the properties of deposited films. Applied's single-wafer RTP systems are also used for growing high quality oxide and oxynitride films. | Vantage Systems |
| **Physical Vapor Deposition (PVD)**<br>PVD is used to deposit high quality metal films. Applications include metal gate, silicides, contact liner/barrier, interconnect copper barrier seed and metal hard mask. | Endura Systems |
| **Chemical Vapor Deposition (CVD)**<br>CVD is used to deposit dielectric and metal films on a wafer. During the CVD process, gases that contain atoms of the material to be deposited react on the wafer surface, forming a thin film of solid material. | Endura, Centura and Producer Systems |
| **Chemical Mechanical Planarization (CMP)**<br>CMP is used to planarize a wafer surface, a process that allows subsequent photolithography patterning and material deposition steps to occur with greater accuracy, resulting in more uniform film layers with minimal thickness variations. | Reflexion Systems |
| **Electrochemical Deposition (ECD)**<br>ECD is a process by which metal atoms from a chemical fluid (an electrolyte) are deposited on the surface of an immersed object. | Raider and Nokota Platforms |
| **Atomic Layer Deposition (ALD)**<br>ALD technology enables ultra thin film growth of either a conducting or insulating material with uniform coverage in nanometer-sized structures. | Olympia System |
| **Etch**<br>Etching is used many times throughout the IC manufacturing process to selectively remove material from the surface of a wafer. Applied offers systems for etching dielectric, metal, and silicon films to meet the requirements of advanced processing. | Centris and Producer Systems |
| **Selective Removal**<br>Selective removal is a new etch technology intended to remove a material of a particular composition without damaging materials of different composition that coexist on the wafer. | Producer Systems |
| **Metrology and Inspection**<br>Metrology and inspection tools are used to locate, measure, and analyze defects and features on the wafer during various stages of the fabrication processes. Applied enables customers to characterize and control critical dimension (CD) and defect issues, especially at advanced generation technology nodes. | SEMVision G7 Defect Analysis<br>PROVision eBeam Inspection<br>UVision 8 Inspection<br>VeritySEM 5i Metrology<br>Aera4 Mask Inspection |

***Applied Global Services***

The Applied Global Services (AGS) segment provides integrated solutions to optimize equipment and fab performance and productivity, including spares, upgrades, services, remanufactured earlier generation equipment and factory automation software for semiconductor, display and other products. Customer demand for products and services is fulfilled through a global distribution system with 93 locations and trained service engineers located in close proximity to customer sites in more than a dozen countries to support over 40,000 installed Applied semiconductor, display and other manufacturing systems worldwide. Applied offers the following general types of services and products under the Applied Global Services segment.

**AGS Solutions and Technology**

**Technology-enabled Services**

A comprehensive service product portfolio that combines service technology and tool specific performance commitments in order to optimize customer factory productivity.

**Fab Consulting**

Experts using advanced analytical tools to solve production problems that have the greatest impact on customer fab productivity.

**Supply Chain Assurance Programs**

Spare parts product portfolio offers options to balance inventory, cost and risk to efficiently meet fab requirements.

**Subfab Equipment**

Applied SubFab solutions lower costs, save energy, reduce environmental impact, and meet Environmental Protection Agency reporting regulations for greenhouse gas emissions.

**Legacy Equipment**

Comprehensive 200mm equipment and upgrades portfolio to address a full spectrum of production needs and extend tool lifetime. Applied 200mm equipment supports market inflections and new technology for a broad variety of devices including analog, power, and MEMS.

**Automation Software**

Applied SmartFactory automation software portfolio coordinates and streamlines every aspect of a factory-the processes, equipment and people-to provide competitive advantage to customers.

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***Display and Adjacent Markets***

The Display and Adjacent Markets segment is comprised of products for manufacturing liquid crystal displays (LCDs), organic light-emitting diodes (OLEDs), and other display technologies for TVs, monitors, laptops, personal computers (PCs), electronic tablets, smart phones, and other consumer-oriented devices as well as equipment for processing flexible substrates. While similarities exist between the technologies utilized in semiconductor and display fabrication, the most significant differences are in the size and composition of the substrate. Substrates used to manufacture display panels and other devices are typically glass, although newer flexible materials are entering the market. Display and Adjacent Markets industry growth depends primarily on consumer demand for increasingly larger and more advanced TVs and high resolution displays for mobile devices as well as new form factors, including thin, light and curved displays, and new applications such as virtual reality. The Display and Adjacent Markets segment offers a variety of technologies and products, including:

| Display and Adjacent Markets Technologies | Product(s) |
|---|---|
| **Array Test**<br>LCD display substrates are inspected at many stages of production to maximize yield, minimize scrap, optimize equipment utilization, and monitor manufacturing processes. At the completion of the array stage, the performance of the millions of individual pixels on each display is tested. | Electron Beam Array Tester |
| **Defect Review**<br>Defects are identified during inspection steps and reviewed by a scanning electron microscope and other analyses to determine defect root cause and composition. | Electron Beam Review (EBR) |
| **Chemical Vapor Deposition (CVD)**<br>During CVD processing, gases containing atoms or molecules are introduced into the process chamber. The gases form reactive radicals or ions, which undergo chemical reactions to form thin films on the heated substrate. | AKT PECVD Systems |
| **Physical Vapor Deposition (PVD)**<br>PVD is used to deposit high quality films of metals, alloys, transparent conductors and semiconductors. In Display, these films are used for contact, interconnect, transparent electrodes and transistor materials in TFT-LCD and OLED display backplanes, as well as for transparent electrodes in color filters and touch panels. | AKT Aristo and PiVot Systems |
| **Flexible Technologies**<br>Flexible coating systems utilize physical vapor deposition, thermal evaporation, chemical vapor deposition, and e-beam technology to deposit thin layers of metal onto flexible substrates. | TopBeam, TopMet and SmartWeb Systems |

**Backlog**

Applied manufactures systems to meet demand represented by order backlog and customer commitments. Backlog as of October 29, 2017 consisted of: (1) orders for which written authorizations have been accepted and assigned shipment dates are within the next 12 months, or shipment has occurred but revenue has not been recognized; and (2) contractual service revenue and maintenance fees to be earned within the next 12 months.

Backlog by reportable segment as of October 29, 2017 was as follows:

| | 2017 | |
|---|---|---|
| | (In millions, except percentages) | |
| Semiconductor Systems | $ 2,991 | 49% |
| Applied Global Services | 1,130 | 19% |
| Display and Adjacent Markets | 1,847 | 31% |
| Corporate and Other | 63 | 1% |
| Total | $ 6,031 | 100% |

In anticipation of the adoption of the new revenue recognition standard in the first quarter of fiscal 2019, Applied has removed the 12 month threshold in relation to backlog in order to more closely align backlog to performance obligations as defined under the new revenue recognition standard. As such, backlog as of October 28, 2018 consists of: (1) orders for which written authorizations have been accepted, or shipment has occurred but revenue has not been recognized; and (2) contractual service revenue and maintenance fees.

Backlog by reportable segment as of October 28, 2018 was as follows:

| | 2018 | |
|---|---|---|
| | (In millions, except percentages) | |
| Semiconductor Systems | $ 2,479 | 41% |
| Applied Global Services | 1,751 | 29% |
| Display and Adjacent Markets | 1,836 | 30% |
| Corporate and Other | 26 | —% |
| Total | $ 6,092 | 100% |

Of the total backlog as of October 28, 2018, approximately 15% is not reasonably expected to be filled within the next 12 months.

Applied's backlog on any particular date is not necessarily indicative of actual sales for any future periods, due to the potential for customer changes in delivery schedules or order cancellations. Customers may delay delivery of products or cancel orders prior to shipment, subject to possible cancellation penalties. Delays in delivery schedules or a reduction of backlog during any particular period could have a material adverse effect on Applied's business and results of operations.

**Manufacturing, Raw Materials and Supplies**

Applied's manufacturing activities consist primarily of assembly, test and integration of various proprietary and commercial parts, components and subassemblies that are used to manufacture systems. Applied has implemented a distributed manufacturing model under which manufacturing and supply chain activities are conducted in various countries, including Germany, Israel, Italy, Singapore, Taiwan, the United States and other countries in Asia. Applied uses numerous vendors, including contract manufacturers, to supply parts and assembly services for the manufacture and support of its products, including some systems being completed at customer sites.

Although Applied makes reasonable efforts to assure that parts are available from multiple qualified suppliers, this is not always possible. Accordingly, some key parts may be obtained from only a single supplier or a limited group of suppliers. Applied seeks to reduce costs and to lower the risks of manufacturing and service interruptions by selecting and qualifying alternate suppliers for key parts; monitoring the financial condition of key suppliers; maintaining appropriate inventories of key parts; qualifying new parts on a timely basis; and ensuring quality and performance of parts.

**Research, Development and Engineering**

Applied's long-term growth strategy requires continued development of new materials engineering capabilities, including products and platforms that enable expansion into new and adjacent markets. Applied's significant investments in research, development and engineering (RD&E) must generally enable it to deliver new products and technologies before the emergence of strong demand, thus allowing customers to incorporate these products into their manufacturing plans during early-stage technology selection. Applied works closely with its global customers and ecosystem partners to design systems and processes that meet planned technical and production requirements.

Applied's product development and engineering organizations are located primarily in the United States, as well as in Canada, China, Europe, India, Israel, Singapore and Taiwan. In addition, certain outsourced RD&E activities, process support and customer demonstrations are performed in the United States, India, China, Singapore and Taiwan.

**Marketing and Sales**

Because of the highly technical nature of its products, Applied markets and sells products worldwide almost entirely through a direct sales force.

Applied has operations in many countries, with some of its business activities concentrated in certain geographic areas, and global and regional economic conditions can impact the company's business and financial results. Applied's business is based on capital equipment investments by major semiconductor, display and other manufacturers, and is subject to significant variability in customer demand for Applied's products. Customers' expenditures depend on many factors, including: general economic conditions; anticipated market demand and pricing for semiconductors, display technologies and other electronic devices; the development of new technologies; customers' factory utilization; capital resources and financing; and government policies and incentives. In addition, a significant driver in the semiconductor and display industries is end-demand for mobile consumer products, which is characterized by seasonality that impacts the timing of customer investments in manufacturing equipment and, in turn, Applied's business.

Information on net sales to unaffiliated customers and long-lived assets attributable to Applied's geographic regions is included in Note 15 of Notes to Consolidated Financial Statements. The following companies accounted for at least 10 percent of Applied's net sales in each fiscal year, which were for products and services in multiple reportable segments.

| | 2018 | 2017 | 2016 |
|---|---|---|---|
| Samsung Electronics Co., Ltd. | 13% | 23% | 13% |
| Taiwan Semiconductor Manufacturing Company Limited | 11% | 15% | 16% |
| Micron Technology, Inc. | * | * | 11% |
| Intel Corporation | 11% | * | 11% |

* Less than 10%

**Competition**

The industries in which Applied operates are highly competitive and characterized by rapid technological change. Applied's ability to compete generally depends on its ability to commercialize its technology in a timely manner, continually improve its products, and develop new products that meet constantly evolving customer requirements. Significant competitive factors include technical capability and differentiation, productivity, cost-effectiveness and the ability to support a global customer base. The importance of these factors varies according to customers' needs, including product mix and respective product requirements, applications, and the timing and circumstances of purchasing decisions. Substantial competition exists in all areas of Applied's business. Competitors range from small companies that compete in a single region, which may benefit from policies and regulations that favor domestic companies, to global, diversified companies. Applied's ability to compete requires a high level of investment in RD&E, marketing and sales, and global customer support activities. Management believes that many of Applied's products have strong competitive positions.

The competitive environment for each segment is described below.

The semiconductor industry is driven by demand for advanced electronic products, including smartphones and other mobile devices, servers, personal computers, automotive devices, storage, and other products. The growth of data and emerging end-market drivers such as artificial intelligence, augmented and virtual reality, the Internet of Things and smart vehicles are also creating new opportunities for the industry. As a result, products within the Semiconductor Systems segment are subject to significant changes in customer requirements, including transitions to smaller dimensions, increasingly complex chip architectures, new materials and an increasing number of applications. While certain existing technologies may be adapted to new requirements, some applications create the need for an entirely different technological approach. The rapid pace of technological change can quickly diminish the value of current technologies and products and create opportunities for existing and new competitors. Applied offers a variety of technologically-differentiated products that must continuously evolve to satisfy customers' requirements to compete effectively in the marketplace. Applied allocates resources among its numerous product offerings and therefore may decide not to invest in an individual product to the same degree as competitors who specialize in fewer products. There are a number of competitors serving the semiconductor manufacturing equipment industry, which has experienced increasing consolidation. Some of these competitors offer a single product line and others offer multiple product lines, and range from suppliers serving a single region to global, diversified companies.

Products and services within the Applied Global Services segment complement Semiconductor Systems and Display and Adjacent Markets segments' products in markets that are characterized by demanding worldwide service requirements and a diverse group of numerous competitors. To compete effectively, Applied offers products and services to improve tool performance, lower overall cost of ownership, and increase yields and productivity of customers' fab operations. Significant competitive factors include productivity, cost-effectiveness, and the level of technical service and support. The importance of these factors varies according to customers' needs and the type of products or services offered.

Products in the Display and Adjacent Markets segment are generally subject to strong competition from a number of major competitors primarily in Asia. Applied holds established market positions with its technically-differentiated LCD and OLED manufacturing solutions for PECVD, color filter PVD, PVD array, PVD touch panel, and TFT array testing, although its market position could change quickly due to customers' evolving requirements. Important factors affecting the competitive position of Applied's Display and Adjacent Markets products include: industry trends, Applied's ability to innovate and develop new products, and the extent to which Applied's products are technically-differentiated, as well as which customers within a highly concentrated customer base are making capital equipment investments and Applied's existing position at these customers.

**Patents and Licenses**

Applied's competitive position significantly depends upon its research, development, engineering, manufacturing and marketing capabilities, as well as its patent position. Protection of Applied's technology assets through enforcement of its intellectual property rights, including patents, is important. Applied's practice is to file patent applications in the United States and other countries for inventions that it considers significant. Applied has approximately 12,500 patents in the United States and other countries, and additional applications are pending for new inventions. Although Applied does not consider its business materially dependent upon any one patent, the rights of Applied and the products made and sold under its patents, taken as a whole, are a significant element of its business. In addition to its patents, Applied possesses other intellectual property, including trademarks, know-how, trade secrets, and copyrights.

Applied enters into patent and technology licensing agreements with other companies when it is determined to be in its best interest. Applied pays royalties under existing patent license agreements for the use, in several of its products, of certain patented technologies. Applied also receives royalties from licenses granted to third parties. Royalties received from or paid to third parties have not been material to Applied's consolidated results of operations.

In the normal course of business, Applied periodically receives and makes inquiries regarding possible patent infringement. In responding to such inquiries, it may become necessary or useful for Applied to obtain or grant licenses or other rights. However, there can be no assurance that such licenses or rights will be available to Applied on commercially reasonable terms, or at all. If Applied is not able to resolve or settle claims, obtain necessary licenses on commercially reasonable terms, or successfully prosecute or defend its position, Applied's business, financial condition and results of operations could be materially and adversely affected.

**Environmental Matters**

Applied maintains a number of environmental, health, and safety programs that are primarily preventative in nature. As part of these programs, Applied regularly monitors ongoing compliance with applicable laws and regulations. In addition, Applied has trained personnel to conduct investigations of any environmental, health, or safety incidents, including, but not limited to, spills, releases, or possible contamination.

Compliance with federal, state and local environmental, health and safety laws and regulations, including those regulating the discharge of materials into the environment, remedial agreements, and other actions relating to the environment have not had, and are not expected to have, a material effect on Applied's capital expenditures, competitive position, financial condition, or results of operations.

The most recent report on Applied's environmental, health and safety activities can be found in Applied's latest Citizenship Report on its website at http://www.appliedmaterials.com/news/citizenship_report.html. The Citizenship Report is updated periodically. This website address is intended to be an inactive textual reference only. None of the information on, or accessible through, Applied's website is part of this Form 10-K or is incorporated by reference herein.

**Employees**

At October 28, 2018, Applied employed approximately 21,000 regular employees. In the high-technology industry, competition for highly-skilled employees is intense. Applied believes that its future success is highly dependent upon its continued ability to attract, retain and motivate qualified employees.

**Executive Officers of the Registrant**

The following table and notes set forth information about Applied's executive officers:

| Name of Individual | Position |
|---|---|
| Gary E. Dickerson(1) | President, Chief Executive Officer |
| Ginetto Addiego(2) | Senior Vice President, Engineering, Operations and Quality |
| Daniel J. Durn(3) | Senior Vice President, Chief Financial Officer |
| Steve Ghanayem(4) | Senior Vice President, New Markets and Alliances Group |
| Thomas F. Larkins(5) | Senior Vice President, General Counsel |
| Omkaram Nalamasu(6) | Senior Vice President, Chief Technology Officer |
| Prabu Raja(7) | Senior Vice President, Semiconductor Products Group |
| Ali Salehpour(8) | Senior Vice President, Services, Display and Flexible Technology |
| Charles Read(9) | Corporate Vice President, Corporate Controller and Chief Accounting Officer |

(1) Mr. Dickerson, age 61, was named President of Applied in June 2012 and appointed Chief Executive Officer and a member of the Board of Directors in September 2013. Before joining Applied, he served as Chief Executive Officer and a director of Varian Semiconductor Equipment Associates, Inc. (Varian) from 2004 until its acquisition by Applied in November 2011. Prior to Varian, Mr. Dickerson served 18 years with KLA-Tencor Corporation (KLA-Tencor), a supplier of process control and yield management solutions for the semiconductor and related industries, where he held a variety of operations and product development roles, including President and Chief Operating Officer. Mr. Dickerson started his semiconductor career in manufacturing and engineering management at General Motors' Delco Electronics Division and then AT&T Technologies.

(2) Dr. Addiego, age 59, has been Senior Vice President, Engineering, Operations and Quality since June 2015. He served as Senior Vice President, Engineering from March 2014 to June 2015. He previously was with Applied from 1996 to 2005, leading various product groups as well as global organizations, including Global Operations, Facilities and Real Estate, Foundation Engineering, and Information Technology. From March 2011 to March 2014, Dr. Addiego was President and Chief Operating Officer of Ultra Clean Technology Corp., a public company listed on NASDAQ and a supplier of critical subsystems for the semiconductor capital equipment, medical device, energy, research, and flat panel industries. From February 2005 to March 2011, Dr. Addiego worked at Novellus Systems, Inc., a provider of advanced process equipment for the semiconductor industry, where he served as Executive Vice President of Corporate Global Operations responsible for Central Engineering, Facilities, Real Estate, Human Resources and Information Technology, and Chief Administrative Officer.

(3) Mr. Durn, age 52, has been Senior Vice President and Chief Financial Officer of Applied since August 2017. Previously, Mr. Durn was Executive Vice President and Chief Financial Officer of NXP Semiconductors N.V., a semiconductor manufacturer (NXP), from December 2015 to August 2017. Mr. Durn served as Senior Vice President of Finance and Chief Financial Officer of Freescale Semiconductor, Inc., from June 2014 until its merger with NXP in December 2015. Prior to Freescale, Mr. Durn was Chief Financial Officer and Executive Vice President of Finance and Administration at GlobalFoundries, a semiconductor foundry, which he joined in December 2011.

(4) Mr. Ghanayem, age 53, has been Senior Vice President, New Markets and Alliances Group of Applied since November 2017. He has served in various senior management, product development and operational roles since joining Applied in 1989, including Group Vice President and General Manager of the Transistor and Interconnect Group.

(5) Mr. Larkins, age 57, has been Senior Vice President, General Counsel of Applied since November 2012 and was Corporate Secretary from November 2012 to March 2018. Previously, Mr. Larkins was employed by Honeywell International Inc., a diversified global technology and manufacturing company, where he was Vice President, Corporate Secretary and Deputy General Counsel from 2002 until joining Applied. Mr. Larkins served in various other positions at Honeywell (formerly AlliedSignal) after joining the company in 1997.

(6) Dr. Nalamasu, age 60, has been Senior Vice President, Chief Technology Officer since June 2013, and President of Applied Ventures, LLC, Applied's venture capital arm, since November 2013. He had served as Group Vice President, Chief Technology Officer from January 2012 to June 2013, and as Corporate Vice President, Chief Technology Officer from January 2011 to January 2012. Upon joining Applied in June 2006 until January 2011, Dr. Nalamasu was an Appointed Vice President of Research and served as Deputy Chief Technology Officer and General Manager for the Advanced Technologies Group. From 2002 to 2006, Dr. Nalamasu was a NYSTAR distinguished professor of Materials Science and Engineering at Rensselaer Polytechnic Institute, where he also served as Vice President of Research from 2005 to 2006. Prior to Rensselaer, Dr. Nalamasu served in several leadership roles at Bell Laboratories.

(7) Dr. Raja, age 56, has been Senior Vice President, Semiconductor Products Group of Applied since November 2017. He previously served in various senior management, product development and operational roles since joining Applied in 1995, including Group Vice President and General Manager of the Patterning and Packaging Group.

(8) Mr. Salehpour, age 57, has been Senior Vice President, Services, Display and Flexible Technology since September 2013. He previously served as Group Vice President, General Manager Energy and Environmental Solutions and Display Business Groups, since joining Applied in November 2012. Prior to Applied, Mr. Salehpour worked at KLA-Tencor for 16 years, where he served as a Senior Vice President and General Manager and worked for 10 years in senior management positions at Schlumberger Test Systems.

(9) Mr. Read, age 52, has been Corporate Vice President, Corporate Controller and Chief Accounting Officer of Applied since joining the Company in September 2013. Prior to Applied, Mr. Read worked at Brocade Communications Systems, Inc., a provider of semiconductor and software-based network solutions, since October 2002, where he most recently served as Vice President, Corporate Controller. Prior to Brocade, Mr. Read worked at KPMG LLP, an audit, tax and advisory firm, from 1996 to 2002.

**Available Information**

Applied's website is http://www.appliedmaterials.com. Applied makes available free of charge, on or through its website, its annual, quarterly and current reports, and any amendments to those reports, as soon as reasonably practicable after electronically filing such reports with, or furnishing them to, the SEC. The SEC's website, www.sec.gov, contains reports, proxy and information statements, and other information regarding issuers that file electronically with the SEC. These website addresses are intended to be an inactive textual references only. None of the information on, or accessible through, these websites is part of this Form 10-K or is incorporated by reference herein.

**Item 1A:** ***Risk Factors***

The following risk factors could materially and adversely affect Applied's business, financial condition or results of operations and cause reputational harm, and should be carefully considered in evaluating the Company and its business, in addition to other information presented elsewhere in this report.

***The industries that Applied serves can be volatile and difficult to predict.***

As a supplier to the global semiconductor and display and related industries, Applied is subject to variable industry conditions, as demand for manufacturing equipment and services can change depending on several factors, including the nature and timing of technology inflections and advances in fabrication processes, the timing and requirements of new and emerging technologies and market drivers, production capacity relative to demand for chips and display technologies, end-user demand, customers' capacity utilization, production volumes, access to affordable capital, consumer buying patterns and general economic conditions. Applied's industries historically have been cyclical, and are subject to volatility and sudden changes in customer requirements for new manufacturing capacity and advanced technology. These changes can affect the timing and amounts of customer investments in technology and manufacturing equipment, and can have a significant impact on Applied's net sales, operating expenses, gross margins and net income. The amount and mix of capital equipment spending between different products and technologies can have a significant impact on the Company's results of operations.

To meet rapidly changing demand in the industries it serves, Applied must accurately forecast demand and effectively manage its resources and production capacity across its businesses, and may incur unexpected or additional costs to align its business operations. During periods of increasing demand for its products, Applied must have sufficient manufacturing capacity and inventory to meet customer demand; effectively manage its supply chain; attract, retain and motivate a sufficient number of qualified employees; and continue to control costs. During periods of decreasing demand, Applied must reduce costs and align its cost structure with prevailing market conditions; effectively manage its supply chain; and motivate and retain key employees. If Applied does not effectively manage these challenges during periods of changing demand, its business performance and results of operations may be adversely impacted. Even with effective allocation of resources and management of costs, during periods of decreasing demand, Applied's gross margins and earnings may be adversely impacted.

***Applied is exposed to risks associated with an uncertain global economy.***

Uncertain global economic and business conditions, along with uncertainties and volatility in the financial markets, national debt and fiscal concerns in various regions, pose challenges to the industries in which Applied operates. Markets for semiconductors and displays depend largely on business and consumer spending and demand for electronic products. Economic uncertainty and related factors exacerbate negative trends in business and consumer spending and may cause certain Applied customers to push out, cancel, or refrain from purchasing for equipment or services, which may have an adverse impact on Applied's revenues, results of operations and financial condition. Uncertain market conditions, difficulties in obtaining capital, or reduced profitability may also cause some customers to scale back operations, exit businesses, merge with other manufacturers, or file for bankruptcy protection and potentially cease operations, which can also result in lower sales, additional inventory or bad debt expense for Applied. Economic and industry uncertainty may similarly affect suppliers, which could impair their ability to deliver parts and negatively affect Applied's ability to manage operations and deliver its products. These conditions may also lead to consolidation or strategic alliances among other equipment manufacturers, which could adversely affect Applied's ability to compete effectively.

Uncertain economic and industry conditions also make it more challenging for Applied to forecast its operating results, make business decisions, and identify and prioritize the risks that may affect its businesses, sources and uses of cash, financial condition and results of operations. If Applied does not appropriately manage its business operations in response to changing economic and industry conditions, it could have a significant negative impact on its business performance and financial condition. Applied may be required to implement additional cost reduction efforts, including restructuring activities, which may adversely affect Applied's ability to capitalize on opportunities. Even during periods of economic uncertainty or lower revenues, Applied must continue to invest in research and development and maintain a global business infrastructure to compete effectively and support its customers, which can have a negative impact on its operating margins and earnings.

Applied maintains an investment portfolio that is subject to general credit, liquidity, market and interest rate risks. The risks to Applied's investment portfolio may be exacerbated if financial market conditions deteriorate and, as a result, the value and liquidity of the investment portfolio, as well as returns on pension assets, could be negatively impacted and lead to impairment charges. Applied also maintains cash balances in various bank accounts globally in order to fund normal operations. If any of these financial institutions becomes insolvent, it could limit Applied's ability to access cash in the affected accounts, which could affect its ability to manage its operations.

***Applied is exposed to the risks of operating a global business.***

Applied has product development, engineering, manufacturing, sales and other operations distributed throughout many countries, and some of its business activities are concentrated in certain geographic areas. Moreover, in fiscal 2018, approximately 91 percent of Applied's net sales were to customers in regions outside the United States. As a result of the global nature of its operations, Applied's business performance and results of operations may be adversely affected by a number of factors, including:

- uncertain global economic and political business conditions and demands;
- political and social attitudes, laws, rules, regulations and policies within countries that favor domestic companies over non-domestic companies, including customer- or government-supported efforts to promote the development and growth of local competitors;
- global trade issues and changes in and uncertainties with respect to trade policies, including the ability to obtain required import and export licenses, trade sanctions, tariffs, and international trade disputes;
- customer- or government-supported efforts to influence Applied to conduct more of its operations and sourcing in a particular country, such as Korea and China;
- variations among, and changes in, local, regional, national or international laws and regulations, including contract, intellectual property, cybersecurity, data privacy, labor, tax, and import/export laws, and the interpretation and application of such laws and regulations;
- ineffective or inadequate legal protection of intellectual property rights in certain countries;
- positions taken by governmental agencies regarding possible national commercial and/or security issues posed by international business operations;
- fluctuating raw material, commodity, energy and shipping costs;
- delays or restrictions in shipping materials or finished products between countries;
- geographically diverse operations and projects, and our ability to maintain appropriate business processes, procedures and internal controls, and comply with environmental, health and safety, anti-corruption and other regulatory requirements;
- supply chain interruptions, and service interruptions from utilities, transportation, data hosting or telecommunications providers, or other events beyond our control;
- a diverse workforce with different experience levels, languages, cultures, customs, business practices and worker expectations, and differing employment practices and labor issues;
- variations in the ability to develop relationships with local customers, suppliers and governments;
- fluctuations in interest rates and currency exchange rates, including the relative strength or weakness of the U.S. dollar against the Japanese yen, euro, Taiwanese dollar, Israeli shekel, Chinese yuan or Singapore dollar;
- the need to provide sufficient levels of technical support in different locations around the world;
- performance of third party providers of outsourced functions, including certain engineering, software development, manufacturing, information technology and other activities;
- political instability, natural disasters, pandemics, social unrest, terrorism or acts of war in locations where Applied has operations, suppliers or sales, or that may influence the value chain of the industries that Applied serves;
- challenges in hiring and integration of an increasing number of workers in new countries;
- the increasing need for a mobile workforce to work in or travel to different regions; and
- uncertainties with respect to economic growth rates in various countries, including for the manufacture and sale of semiconductors and displays in the developing economies of certain countries.

***International trade disputes could result in increases in tariffs and other trade restrictions and protectionist measures that could have an adverse impact on our operations.***

We sell a significant majority of our products into countries outside of the United States and we purchase a significant portion of equipment and supplies from suppliers outside of the United States. The United States and other countries have levied tariffs and taxes on certain goods. Increases in tariffs, additional taxes or other trade restrictions and retaliatory measures may impact end-user demand and customer investment in manufacturing equipment, increase our manufacturing costs, decrease margins, reduce the competitiveness of our products, or inhibit our ability to sell products or purchase necessary equipment and supplies, which could have a material adverse effect on our business, results of operations, or financial condition.

Certain international sales depend on our ability to obtain export licenses, and our inability to obtain such licenses could potentially limit our markets and impact our business. In addition, government authorities may impose conditions that require the use of local suppliers or partnerships with local companies, require the license or other transfer of intellectual property, or engage in other efforts to promote local businesses and local competitors, which could have a significant adverse impact on Applied's business. Many of these challenges are present in China and Korea, markets that represent significant long-term growth opportunities for Applied businesses.

***Applied is exposed to risks as a result of ongoing changes in the various industries in which it operates.***

The global semiconductor, display and related industries in which Applied operates are characterized by ongoing changes affecting some or all of these industries that impact demand for and the profitability of Applied's products and its consolidated results of operations, including:

- the nature, timing and degree of visibility of changes in end demand for electronic products, including those related to fluctuations in consumer buying patterns tied to seasonality or the introduction of new products, and the effects of these changes on customers' businesses and on demand for Applied's products;
- increasing capital requirements for building and operating new fabrication plants and customers' ability to raise the necessary capital;
- trade, regulatory or tax policies impacting the timing of customers' investment in new or expanded fabrication plants;
- differences in growth rates among the semiconductor, display and other industries in which Applied operates;
- the increasing importance of establishing, improving and maintaining strong relationships with customers;
- the increasing cost and complexity for customers to move from product design to volume manufacturing, which may slow the adoption rate of new manufacturing technology;
- the need for customers to continually reduce the total cost of manufacturing system ownership;
- the heightened importance to customers of system reliability and productivity and the effect on demand for fabrication systems as a result of their increasing productivity, device yield and reliability;
- manufacturers' ability to reconfigure and re-use fabrication systems which can reduce demand for new equipment;
- the increasing importance of, and difficulties in, developing products with sufficient differentiation to influence customers' purchasing decisions;
- requirements for shorter cycle times for the development, manufacture and installation of manufacturing equipment;
- price and performance trends for semiconductor devices and displays, and the corresponding effect on demand for such products;
- the increasing importance of the availability of spare parts to maximize the time that customers' systems are available for production;
- the increasing role for and complexity of software in Applied products; and
- the increasing focus on reducing energy usage and improving the environmental impact and sustainability associated with manufacturing operations.

***Applied is exposed to risks as a result of ongoing changes specific to the semiconductor industry.***

The largest proportion of Applied's consolidated net sales and profitability is derived from sales of manufacturing equipment in the Semiconductor Systems segment to the global semiconductor industry. In addition, a majority of the revenues of Applied Global Services is from sales to semiconductor manufacturers. The semiconductor industry is characterized by ongoing changes particular to this industry that impact demand for and the profitability of Applied's semiconductor equipment and service products, including:

- the increasing frequency and complexity of technology transitions and inflections, and Applied's ability to timely and effectively anticipate and adapt to these changes;
- the increasing cost of research and development due to many factors, including shrinking geometries, the use of new materials, new and more complex device structures, more applications and process steps, increasing chip design costs, and the increasing cost and complexity of integrated manufacturing processes;
- the need to reduce product development time, despite the increasing difficulty of technical challenges;
- the growing number of types and varieties of semiconductors and number of applications across multiple substrate sizes;

- the increasing cost and complexity for semiconductor manufacturers to move more technically advanced capability and smaller geometries to volume manufacturing, and the resulting impact on the rates of technology transition and investment in capital equipment;
- challenges in generating organic growth given semiconductor manufacturers' levels of capital expenditures and the allocation of capital investment to market segments that Applied does not serve, such as lithography, or segments where Applied's products have lower relative market presence;
- the importance of increasing market positions in segments with growing demand;
- semiconductor manufacturer's ability to reconfigure and re-use equipment, and the resulting effect on their need to purchase new equipment and services;
- shorter cycle times between order placements by customers and product shipment require greater reliance on forecasting of customer investment, which may lead to inventory write-offs and manufacturing inefficiencies that decrease gross margin;
- competitive factors that make it difficult to enhance position, including challenges in securing development-tool-of-record (DTOR) and production-tool-of-record (PTOR) positions with customers;
- consolidation in the semiconductor industry, including among semiconductor manufacturers and among manufacturing equipment suppliers;
- shifts in sourcing strategies by computer and electronics companies that impact the equipment requirements of Applied's foundry customers;
- the concentration of new wafer starts in Korea and Taiwan, where Applied's service penetration and service-revenue-per-wafer-start have been lower than in other regions;
- investment in semiconductor manufacturing capabilities in China, which may be affected by changes in economic conditions and governmental policies in China; and
- the increasing fragmentation of semiconductor markets, leading certain markets to become too small to support the cost of a new fabrication plant, while others require less technologically advanced products.

If Applied does not accurately forecast, and allocate appropriate resources and investment towards addressing, key technology changes and inflections, successfully develop and commercialize products to meet demand for new technologies, and effectively address industry trends, its business and results of operations may be adversely impacted.

***Applied is exposed to risks as a result of ongoing changes specific to the display industry.***

The global display industry historically has experienced considerable volatility in capital equipment investment levels, due in part to the limited number of display manufacturers, the concentrated nature of end-use applications, production capacity relative to end-use demand, and panel manufacturer profitability. Industry growth depends primarily on consumer demand for increasingly larger and more advanced TVs, and on demand for advanced smartphones and mobile device displays, which demand is highly sensitive to cost and improvements in technologies and features. The display industry is characterized by ongoing changes particular to this industry that impact demand for and the profitability of Applied's display products and services, including:

- the importance of new types of display technologies, such as organic light-emitting diode (OLED), low temperature polysilicon (LTPS), flexible displays and metal oxide, and new touch panel films;
- the increasing cost of research and development, and complexity of technology transitions and inflections, and Applied's ability to timely and effectively anticipate and adapt to these changes;
- the timing and extent of an expansion of manufacturing facilities in China, which may be affected by changes in economic conditions and governmental policies in China;
- the importance of increasing market positions in products and technologies with growing demand;
- the rate of transition to larger substrate sizes for TVs and to new display technologies for TVs and mobile applications, and the resulting effect on capital intensity in the industry and on Applied's product differentiation, gross margin and return on investment; and
- the variability in demand for display manufacturing equipment, concentration of display manufacturer customers and their ability to successfully commercialize new products and technologies, and uncertainty with respect to future display technology end-use applications and growth drivers.

If Applied does not successfully develop and commercialize products to meet demand for new and emerging display technologies, or if industry demand for display manufacturing equipment and technologies slows, Applied's business and its results of operations may be adversely impacted.

***The industries in which Applied operates are highly competitive and subject to rapid technological and market changes.***

Applied operates in a highly competitive environment in which innovation is critical, and its future success depends on many factors, including the development of new technologies and effective commercialization and customer acceptance of its equipment, services and related products, and its ability to increase its position in its current markets, expand into adjacent and new markets, and optimize operational performance. The development, introduction and support of a broadening set of products in a geographically diverse and competitive environment, and that may require greater collaboration with customers and other industry participants, have grown more complex and expensive over time. Furthermore, new or improved products may entail higher costs and lower profits. To compete successfully, Applied must:

- identify and address technology inflections, market changes, new applications, customer requirements and end-use demand;
- develop new products and disruptive technologies, improve and develop new applications for existing products, and adapt products for use by customers in different applications and markets with varying technical requirements;
- differentiate its products from those of competitors, meet customers' performance specifications, appropriately price products, and achieve market acceptance;
- maintain operating flexibility to enable responses to changing markets, applications, customers and customer requirements;
- enhance its worldwide operations across its businesses to reduce cycle time, enable continuous quality improvement, reduce costs, and enhance design for manufacturability and serviceability;
- focus on product development and sales and marketing strategies that address customers' high value problems and strengthen customer relationships;
- effectively allocate resources between its existing products and markets, the development of new products, and expanding into new and adjacent markets;
- improve the productivity of capital invested in R&D activities;
- accurately forecast demand, work with suppliers and meet production schedules for its products;
- improve its manufacturing processes and achieve cost efficiencies across product offerings;
- adapt to changes in value offered by companies in different parts of the supply chain;
- qualify products for evaluation and volume manufacturing with its customers; and
- implement changes in its design engineering methodology to reduce material costs and cycle time, increase commonality of platforms and types of parts used in different systems, and improve product life cycle management.

If Applied does not successfully anticipate technology inflections, develop and commercialize new products and technologies, and respond to changes in customer requirements and market trends, its business performance and results of operations may be adversely impacted.

***Applied is exposed to risks associated with a highly concentrated customer base.***

Applied's customer base is highly concentrated, and has become increasingly concentrated as a result of continued consolidation. Applied's customer base is also geographically concentrated. A relatively limited number of manufacturers account for a substantial portion of Applied's business. As a result, the actions of even a single customer can expose Applied's business and results of operations to greater volatility. The mix and type of customers, and sales to any single customer, including as a result of changes in government policy, may vary significantly from quarter to quarter and from year to year, and have a significant impact on Applied's net sales, gross margins and net income. Applied's products are configured to customer specifications, and changing, rescheduling or canceling orders may result in significant, non-recoverable costs. If customers do not place orders, or they substantially reduce, delay or cancel orders, Applied may not be able to replace the business, which may have a significant adverse impact on its results of operations and financial condition. The concentration of Applied's customer base increases its risks related to the financial condition of its customers, and the deterioration in financial condition of a single customer or the failure of a single customer to perform its obligations could have a material adverse effect on Applied's results of operations and cash flow. To the extent its customers experience liquidity constraints, Applied may incur additional bad debt expense, which may have a significant impact on its results of operations. Major customers may also seek pricing, payment, intellectual property-related, or other commercial terms that are less favorable to Applied, which may have a negative impact on Applied's business, cash flow, revenue and gross margins.

***Applied is exposed to risks associated with business combinations, acquisitions and strategic investments.***

Applied engages in acquisitions of or investments in companies, technologies or products in existing, related or new markets for Applied. Business combinations, acquisitions and investments involve numerous risks to Applied's business, financial condition and operating results, including but not limited to:

- diversion of management's attention and disruption of ongoing businesses;
- contractual restrictions on the conduct of Applied's business during the pendency of a proposed transaction;
- inability to complete proposed transactions due to the failure to obtain regulatory or other approvals, litigation or other disputes, and any ensuing obligation to pay a termination fee;
- the failure to realize expected returns from acquired businesses;
- requirements imposed by government regulators in connection with their review of a transaction, which may include, among other things, divestitures and restrictions on the conduct of Applied's existing business or the acquired business;
- ineffective integration of operations, systems, technologies, products or employees, which can impact the ability to realize anticipated synergies or other benefits;
- failure to commercialize technologies from acquired businesses or developed through strategic investments;
- dependence on unfamiliar supply chains or relatively small supply partners;
- inability to capitalize on characteristics of new markets that may be significantly different from Applied's existing markets and where competitors may have stronger market positions and customer relationships;
- failure to retain and motivate key employees of acquired businesses;
- the potential impact of the announcement or consummation of a proposed transaction on relationships with third parties;
- potential changes in Applied's credit rating, which could adversely impact the Company's access to and cost of capital;
- reductions in cash balances or increases in debt obligations to finance activities associated with a transaction, which reduce the availability of cash flow for general corporate or other purposes, including share repurchases and dividends;
- exposure to new operational risks, rules, regulations, worker expectations, customs and practices to the extent acquired businesses are located in regions where Applied has not historically conducted business;
- challenges associated with managing new, more diverse and more widespread operations, projects and people;
- inability to obtain and protect intellectual property rights in key technologies;
- inadequacy or ineffectiveness of an acquired company's internal financial controls, disclosure controls and procedures, cybersecurity, privacy policies and procedures, or environmental, health and safety, anti-corruption, human resource, or other policies or practices;
- impairment of acquired intangible assets and goodwill as a result of changing business conditions, technological advancements or worse-than-expected performance of the segment;
- the risk of litigation or claims associated with a proposed or completed transaction;
- unknown, underestimated or undisclosed commitments or liabilities; and
- the inappropriate scale of acquired entities' critical resources or facilities for business needs.

Applied also makes strategic investments in other companies, including companies formed as joint ventures, which may decline in value or not meet desired objectives. The success of these investments depends on various factors over which Applied may have limited or no control and, particularly with respect to joint ventures, requires ongoing and effective cooperation with strategic partners. In addition, new legislation or additional regulations may affect or impair our ability to invest in certain countries or require us to obtain regulatory approvals to do so. The risks to Applied's strategic investment portfolio may be exacerbated by unfavorable financial market and macroeconomic conditions and, as a result, the value of the investment portfolio could be negatively impacted and lead to impairment charges.

***Applied's indebtedness and debt covenants could adversely affect its financial condition and business.***

Applied has $5.4 billion in aggregate principal amount of senior unsecured notes outstanding. Under the indenture governing the senior unsecured notes, it may be required to offer to repurchase the notes at a price equal to 101% of the principal amount, plus accrued and unpaid interest, upon a change of control of Applied and a contemporaneous downgrade of the notes below investment grade. Applied also has in place a $1.5 billion committed revolving credit agreement. While no amounts were outstanding under this credit agreement at October 28, 2018, Applied may borrow amounts in the future under the agreement. Applied may also enter into new financing arrangements. Applied's ability to satisfy its debt obligations is dependent upon the results of its business operations and subject to other risks discussed in this section. Significant changes in Applied's credit rating or changes in the interest rate environment could have a material adverse consequence on Applied's access to and cost of capital for future financings, and financial condition. If Applied fails to satisfy its debt obligations, or comply with financial and other debt covenants, it may be in default and any borrowings may become immediately due and payable, and such default may also constitute a default under other of Applied's obligations. There can be no assurance that Applied would have sufficient financial resources or be able to arrange financing to repay any borrowings at such time.

***Applied is exposed to risks associated with expanding into new and related markets and industries.***

As part of its growth strategy, Applied must successfully expand into related or new markets and industries, either with its existing products or with new products developed internally, or those developed in collaboration with third parties, or obtained through acquisitions. Applied's ability to successfully expand its business into new and related markets and industries may be adversely affected by a number of factors, including:

- the need to devote additional resources to develop new products for, and operate in, new markets;
- the need to develop new sales and technical marketing strategies, cultivate relationships with new customers and meet different customer service requirements;
- differing rates of profitability and growth among multiple businesses;
- Applied's ability to anticipate demand, capitalize on opportunities, and avoid or minimize risks;
- the complexity of managing multiple businesses with variations in production planning, execution, supply chain management and logistics;
- the adoption of new business models, business processes and systems;
- the complexity of entering into and effectively managing strategic alliances or partnering opportunities;
- new materials, processes and technologies;
- the need to attract, motivate and retain employees with skills and expertise in these new areas;
- new and more diverse customers and suppliers, including some with limited operating histories, uncertain or limited funding, evolving business models or locations in regions where Applied does not have, or has limited, operations;
- new or different competitors with potentially more financial or other resources, industry experience and established customer relationships;
- entry into new industries and countries, with differing levels of government involvement, laws and regulations, and business, employment and safety practices;
- third parties' intellectual property rights; and
- the need to comply with, or work to establish, industry standards and practices.

In addition, Applied from time to time receives funding from United States and other government agencies for certain strategic development programs to increase its research and development resources and address new market opportunities. As a condition to this government funding, Applied may be subject to certain record-keeping, audit, intellectual property rights-sharing and/or other obligations.

***Manufacturing interruptions or delays, or the failure to accurately forecast customer demand, could affect Applied's ability to meet customer demand, lead to higher costs, or result in excess or obsolete inventory.***

Applied's business depends on its timely supply of equipment, services and related products that meet the rapidly changing technical and volume requirements of its customers, which depends in part on the timely delivery of parts, including components and subassemblies, from suppliers, including contract manufacturers. The inability to timely obtain sufficient quantities of parts can have an adverse impact on Applied's manufacturing operations and ability to meet customer demand for equipment, spares and services. Some key parts are subject to long lead-times or obtainable only from a single supplier or limited group of suppliers, and some sourcing or subassembly is provided by suppliers located in countries other than the countries where Applied conducts its manufacturing. Variable industry conditions and the volatility of demand for manufacturing equipment increase capital, technical, operational and other risks for Applied and for companies throughout its supply chain. These conditions may cause some suppliers to scale back operations, exit businesses, merge with other companies, or file for bankruptcy protection and possibly cease operations. Applied may also experience significant interruptions of its manufacturing operations, delays in its ability to deliver products or services, increased costs or customer order cancellations as a result of:

- the failure or inability to accurately forecast demand and obtain sufficient quantities of quality parts on a cost-effective basis;
- volatility in the availability and cost of materials;
- difficulties or delays in obtaining required import or export approvals;
- shipment delays due to transportation interruptions or capacity constraints;
- information technology or infrastructure failures, including those of a third party supplier or service provider; and
- natural disasters or other events beyond Applied's control (such as earthquakes, floods or storms, regional economic downturns, pandemics, social unrest, political instability, terrorism, or acts of war), particularly where it conducts manufacturing.

If a supplier fails to meet Applied's requirements concerning quality, cost, protection of intellectual property, socially-responsible business practices, or other performance factors, Applied may transfer its business to alternative sources, which could entail manufacturing delays, additional costs, or other difficulties. If Applied is unable to meet its customers' demand for a prolonged period due to its inability to obtain certain parts or components, it could affect its ability to manage its operations, and have an adverse impact on Applied's business, results of operations and customer relationships. In addition, if Applied needs to rapidly increase its business and manufacturing capacity to meet increases in demand or expedited shipment schedules, this may exacerbate any interruptions in Applied's manufacturing operations and supply chain and the associated effect on Applied's working capital. Moreover, if actual demand for Applied's products is different than expected, Applied may purchase more/fewer parts than necessary or incur costs for canceling, postponing or expediting delivery of parts. If Applied purchases inventory in anticipation of customer demand that does not materialize, or if customers reduce or delay orders, Applied may incur excess inventory charges.

***The ability to attract, retain and motivate key employees is vital to Applied's success.***

Applied's success, competitiveness and ability to execute on its global strategies and maintain a culture of innovation depend in large part on its ability to attract, retain and motivate employees with key skills and experience. Achieving this objective may be difficult due to many factors, including fluctuations in global economic and industry conditions, management changes, Applied's organizational structure, increasing global competition for talent, the availability of qualified employees, cost reduction activities (including workforce reductions and unpaid shutdowns), availability of career development opportunities, the ability to obtain necessary authorizations for workers to provide services outside their home countries, and the attractiveness of Applied's compensation and benefit programs, including its share-based programs. The loss or retirement of employees present particular challenges to the extent they involve the departure of knowledgeable and experienced employees and the resulting need to identify and train existing or new workers to perform necessary functions, which may result in unexpected costs, reduced productivity, and/or difficulties with respect to internal processes and controls.

***Applied is exposed to various risks related to protection and enforcement of intellectual property rights.***

Applied's success depends in significant part on the protection of its patents, trade secrets, copyrights and other intellectual property rights. Infringement of Applied's rights by a third party, such as the unauthorized manufacture or sale of equipment or spare parts, could result in uncompensated lost market and revenue opportunities for Applied. Policing any unauthorized use of intellectual property is difficult and costly and Applied cannot be certain that the measures it has implemented will prevent misuse. Applied's ability to enforce its intellectual property rights is subject to litigation risks, as well as uncertainty as to the protection and enforceability of those rights in some countries. If Applied seeks to enforce its intellectual property rights, it may be subject to claims that those rights are invalid or unenforceable, and others may seek counterclaims against Applied, which could have a negative impact on its business. If Applied is unable to enforce and protect intellectual property rights, or if they are circumvented, invalidated, rendered obsolete by the rapid pace of technological change, it could have an adverse impact on its competitive position and business. In addition, changes in intellectual property laws or their interpretation, such as recent changes in U.S. patent laws, may impact Applied's ability to protect and assert its intellectual property rights, increase costs and uncertainties in the prosecution of patent applications and enforcement or defense of issued patents, and diminish the value of Applied's intellectual property.

Third parties may also assert claims against Applied and its products. Claims that Applied's products infringe the rights of others, whether or not meritorious, can be expensive and time-consuming to defend and resolve, and may divert the efforts and attention of management and personnel. The inability to obtain rights to use third party intellectual property on commercially reasonable terms could have an adverse impact on Applied's business. In addition, Applied may face claims based on the theft or unauthorized use or disclosure of third-party trade secrets and other confidential business information. Any such incidents and claims could severely harm Applied's business and reputation, result in significant expenses, harm its competitive position, and prevent Applied from selling certain products, all of which could have a significant adverse impact on Applied's business and results of operations.

***Applied is exposed to risks related to cybersecurity threats and incidents.***

In the conduct of its business, Applied collects, uses, transmits and stores data on information technology systems, including systems owned and maintained by Applied or third party providers. These data include confidential information and intellectual property belonging to Applied or its customers or other business partners, as well as personally-identifiable information of individuals. All information systems are subject to disruption, breach or failure. Applied has experienced, and expects to continue to be subject to, cybersecurity threats and incidents ranging from employee error or misuse to individual attempts to gain unauthorized access to these information systems, to sophisticated and targeted measures known as advanced persistent threats, none of which have been material to the Company to date. Applied devotes significant resources to network security, data encryption and other measures to protect its systems and data from unauthorized access or misuse. However, depending on their nature and scope, cybersecurity incidents could result in business disruption; the misappropriation of intellectual property, and corruption or loss of confidential information and critical data (Applied's and that of third parties); reputational damage; litigation with third parties; diminution in the value of Applied's investment in research, development and engineering; data privacy issues; and increased cybersecurity protection and remediation costs. Compliance with, and changes to, laws and regulations concerning privacy and information security could result in significant expense, and any failure to comply could result in proceedings against Applied by regulatory authorities or other third parties.

***Applied is exposed to various risks related to legal proceedings.***

Applied from time to time is, and in the future may be involved in legal proceedings or claims regarding patent infringement, intellectual property rights, antitrust, environmental regulations, securities, contracts, product performance, product liability, unfair competition, misappropriation of trade secrets, employment, workplace safety, and other matters. Applied also on occasion receives notification from customers who believe that Applied owes them indemnification or other obligations related to claims made against such customers by third parties.

Legal proceedings and claims, whether with or without merit, and associated internal investigations, may be time-consuming and expensive to prosecute, defend or conduct; divert management's attention and other Applied resources; inhibit Applied's ability to sell its products; result in adverse judgments for damages, injunctive relief, penalties and fines; and negatively affect Applied's business. There can be no assurance regarding the outcome of current or future legal proceedings, claims or investigations.

***The failure to successfully implement enterprise resource planning and other information systems changes could adversely impact Applied's business and results of operations.***

Applied periodically implements new or enhanced enterprise resource planning and related information systems in order to better manage its business operations, align its global organizations and enable future growth. Implementation of new business processes and information systems requires the commitment of significant personnel, training and financial resources, and entails risks to Applied's business operations. If Applied does not successfully implement enterprise resource planning and related information systems improvements, or if there are delays or difficulties in implementing these systems, Applied may not realize anticipated productivity improvements or cost efficiencies, and may experience interruptions in service and operational difficulties, such as its ability to track orders, timely manufacture and ship products, project inventory requirements, effectively manage its supply chain and allocate human resources, aggregate financial data and report operating results, and otherwise effectively manage its business, all of which could result in quality issues, reputational harm, lost market and revenue opportunities, and otherwise adversely affect Applied's business, financial condition and results of operations.

***Applied may incur impairment charges to goodwill or long-lived assets.***

Applied has a significant amount of goodwill and other acquired intangible assets related to acquisitions. Goodwill and purchased intangible assets with indefinite useful lives are not amortized, but are reviewed for impairment annually during the fourth quarter of each fiscal year, and more frequently when events or changes in circumstances indicate that the carrying value of an asset may not be recoverable. The review compares the fair value for each of Applied's reporting units to its associated carrying value, including goodwill. Factors that could lead to impairment of goodwill and intangible assets include adverse industry or economic trends, reduced estimates of future cash flows, declines in the market price of Applied common stock, changes in Applied's strategies or product portfolio, and restructuring activities. Applied's valuation methodology for assessing impairment requires management to make judgments and assumptions based on historical experience and projections of future operating performance. Applied may be required to record future charges to earnings during the period in which an impairment of goodwill or intangible assets is determined to exist.

***Applied is exposed to risks associated with operating in jurisdictions with complex and changing tax laws.***

Applied is subject to income taxes in the United States and foreign jurisdictions. Significant judgment is required to determine and estimate worldwide tax liabilities. Applied's provision for income taxes and effective tax rates could be affected by numerous factors, including changes in applicable tax laws, interpretations of applicable tax laws, amount and composition of pre-tax income in jurisdictions with differing tax rates, and valuation of deferred tax assets. An increase in Applied's provision for income taxes and effective tax rate could have a material adverse impact on Applied's results of operations and financial condition.

On December 22, 2017, the U.S. government enacted tax legislation referred to as the Tax Cuts and Jobs Act (the "Tax Act"). The U.S. government may issue regulations that significantly affect how the Tax Act is interpreted. The U.S. Securities and Exchange Commission issued Staff Accounting Bulletin No. 118 (SAB 118), which provides guidance on accounting for income tax effects of the Tax Act. SAB 118 provides a measurement period for companies to complete this accounting, which will not extend beyond one year from the Tax Act enactment date. Pursuant to SAB 118, provisional adjustments were recorded when reasonable estimates could be determined. No adjustments were recorded when reasonable estimates could not be determined. These provisional estimates will be revised as information becomes available and guidance is issued by the Internal Revenue Service, and changes in Applied's provision for income taxes and effective tax rates could have a material impact on Applied's results of operations and financial condition.

Consistent with the international nature of its business, Applied conducts certain manufacturing, supply chain, and other operations in Asia, bringing these activities closer to customers and reducing operating costs. In certain foreign jurisdictions, conditional reduced income tax rates have been granted to Applied. To obtain the benefit of these tax incentives, Applied must meet requirements relating to various activities. Applied's ability to realize benefits from these incentives could be materially affected if, among other things, applicable requirements are not met or Applied incurs net losses in these jurisdictions.

In addition, Applied is subject to examination by the Internal Revenue Service and other tax authorities, and from time to time amends previously filed tax returns. Applied regularly assesses the likelihood of favorable or unfavorable outcomes resulting from these examinations and amendments to determine the adequacy of its provision for income taxes, which requires estimates and judgments. Although Applied believes its tax estimates are reasonable, there can be no assurance that the tax authorities will agree with such estimates. Applied may have to engage in litigation to achieve the results reflected in the estimates, which may be time-consuming and expensive. There can be no assurance that Applied will be successful or that any final determination will not be materially different from the treatment reflected in Applied's historical income tax provisions and effective tax rates.

***Applied is subject to risks associated with environmental and safety regulations.***

Applied is subject to environmental and safety regulations in connection with its global business operations, including but not limited to: regulations related to the development, manufacture and use of its products; handling, discharge, recycling and disposal of hazardous materials used in its products or in producing its products; the operation of its facilities; and the use of its real property. The failure or inability to comply with existing or future environmental and safety regulations could result in: significant remediation or other legal liabilities; the imposition of penalties and fines; restrictions on the development, manufacture, sale or use of certain of its products; limitations on the operation of its facilities or ability to use its real property; and a decrease in the value of its real property. Applied could be required to alter its manufacturing and operations and incur substantial expense in order to comply with environmental and safety regulations. Any failure to comply with environmental and safety regulations could subject Applied to significant costs and liabilities that could adversely affect Applied's business, financial condition and results of operations.

***Applied is exposed to various risks related to the global regulatory environment.***

As a public company with global operations, Applied is subject to the laws of the United States and multiple foreign jurisdictions and the rules and regulations of various governing bodies, which may differ among jurisdictions, including those related to financial and other disclosures, accounting standards, corporate governance, intellectual property, tax, trade, antitrust, employment, immigration and travel regulations, privacy, and anti-corruption. Changing, inconsistent or conflicting laws, rules and regulations, and ambiguities in their interpretation and application create uncertainty and challenges, and compliance with laws, rules and regulations may be onerous and expensive, divert management time and attention from revenue-generating activities, and otherwise adversely impact Applied's business operations. Violations of law, rules and regulations could result in fines, criminal sanctions, restrictions on Applied's business, and damage to its reputation, and could have an adverse impact on its business operations, financial condition and results of operations.

**Item 1B:** ***Unresolved Staff Comments***

None.

**Item 2:** ***Properties***

Information concerning Applied's properties is set forth below:

| (Square feet in thousands) | United States | Other Countries | Total |
|---|---|---|---|
| Owned | 4,530 | 2,417 | 6,947 |
| Leased | 1,037 | 1,341 | 2,378 |
| Total | 5,567 | 3,758 | 9,325 |

Because of the interrelation of Applied's operations, properties within a country may be shared by the segments operating within that country. The Company's headquarters offices are in Santa Clara, California. Products in Semiconductor Systems are manufactured in Santa Clara, California; Austin, Texas; Gloucester, Massachusetts; Kalispell, Montana; Rehovot, Israel; and Singapore. Remanufactured equipment products in the Applied Global Services segment are produced primarily in Austin, Texas. Products in the Display and Adjacent Markets segment are manufactured in Alzenau, Germany and Tainan, Taiwan. Other products are manufactured in Treviso, Italy.

Applied also owns and leases offices, plants and warehouse locations in many locations throughout the world, including in Europe, Japan, North America (principally the United States), Israel, China, India, Korea, Southeast Asia and Taiwan. These facilities are principally used for manufacturing; research, development and engineering; and marketing, sales and customer support.

Applied also owns a total of approximately 269 acres of buildable land in Montana, Texas, California, Israel and Italy that could accommodate additional building space.

Applied considers the properties that it owns or leases as adequate to meet its current and future requirements. Applied regularly assesses the size, capability and location of its global infrastructure and periodically makes adjustments based on these assessments.

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**Item 3:** ***Legal Proceedings***

The information set forth under "Legal Matters" in Note 14 of Notes to Consolidated Financial Statements is incorporated herein by reference.

**Item 4:** ***Mine Safety Disclosures***

None.

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## PART II

**Item 5:** ***Market for Registrant's Common Equity, Related Stockholder Matters and Issuer Purchases of Equity Securities***

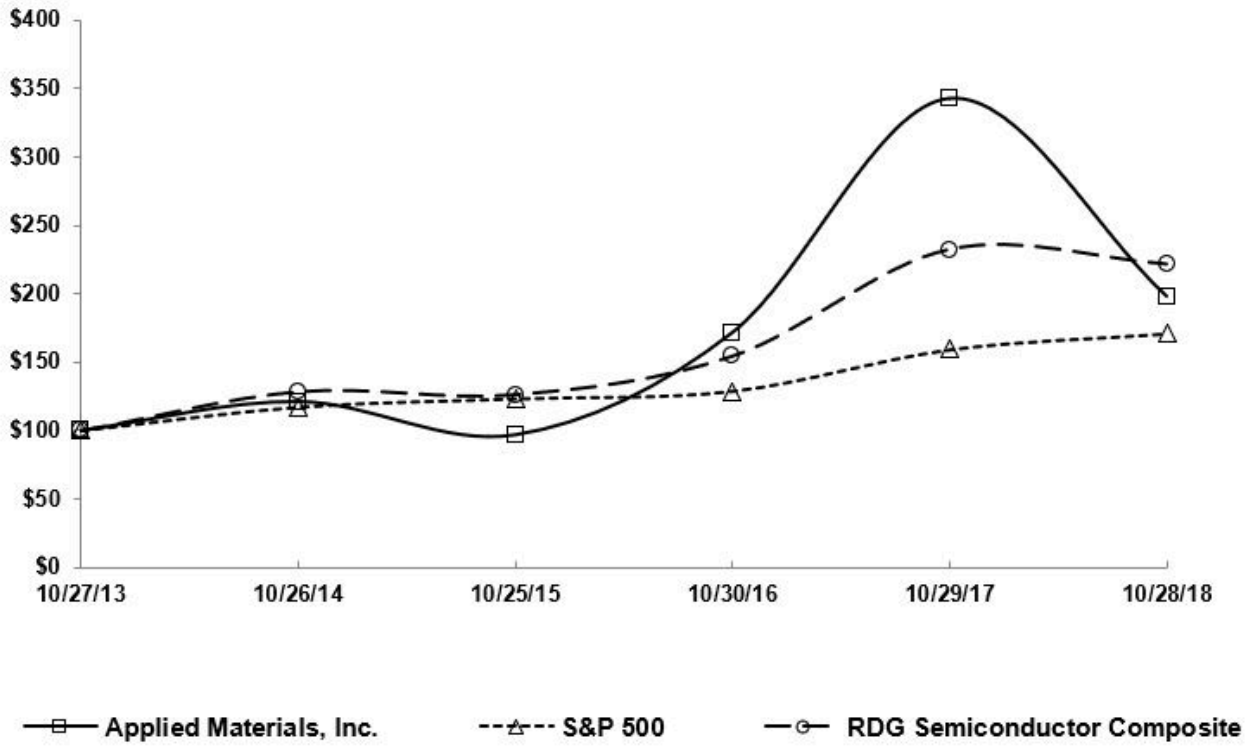
**Market Information**

Applied's common stock is traded on the NASDAQ Global Select Market under the symbol AMAT. As of December 7, 2018, there were 2,854 registered holders of Applied common stock.

**Performance Graph**

The performance graph below shows the five-year cumulative total stockholder return on Applied common stock during the period from October 27, 2013 through October 28, 2018. This is compared with the cumulative total return of the Standard & Poor's 500 Stock Index and the RDG Semiconductor Composite Index over the same period. The comparison assumes $100 was invested on October 27, 2013 in Applied common stock and in each of the foregoing indices and assumes reinvestment of dividends, if any. Dollar amounts in the graph are rounded to the nearest whole dollar. The performance shown in the graph represents past performance and should not be considered an indication of future performance.

**COMPARISON OF 5 YEAR CUMULATIVE TOTAL RETURN***
Among Applied Materials, Inc., the S&P 500 Index
and the RDG Semiconductor Composite Index




*Assumes $100 invested on 10/27/13 in stock or 10/31/13 in index, including reinvestment of dividends.
Indexes calculated on month-end basis.
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| | 10/27/2013 | 10/26/2014 | 10/25/2015 | 10/30/2016 | 10/29/2017 | 10/28/2018 |
|---|---|---|---|---|---|---|
| Applied Materials | 100.00 | 121.04 | 96.67 | 171.69 | 343.16 | 198.27 |
| S&P 500 Index | 100.00 | 117.27 | 123.37 | 128.93 | 159.40 | 171.11 |
| RDG Semiconductor Composite Index | 100.00 | 128.42 | 126.26 | 154.41 | 232.29 | 221.61 |

**Issuer Purchases of Equity Securities**

The following table provides information as of October 28, 2018 with respect to the shares of common stock repurchased by Applied during the fourth quarter of fiscal 2018 pursuant to a publicly announced stock repurchase program approved by the Board of Directors in February 2018, which authorized up to an aggregate of $6.0 billion in repurchases.

| Period | Total Number of Shares Purchased | Average Price Paid per Share | Aggregate Price Paid | Total Number of Shares Purchased as Part of Publicly Announced Programs * | Maximum Dollar Value of Shares That May Yet be Purchased Under the Programs |
|---|---|---|---|---|---|
| | | | (In millions, except per share amounts) | | |
| Month #1 | | | | | |
| (July 30, 2018 to August 26, 2018) | 7.9 | $ 46.69 | $ 369 | 7.9 | $ 4,709 |
| Month #2 | | | | | |
| (August 27, 2018 to September 23, 2018) | 3.7 | $ 40.13 | 148 | 3.7 | $ 4,561 |
| Month #3 | | | | | |
| (September 24, 2018 to October 28, 2018) | 6.5 | $ 35.78 | 234 | 6.5 | $ 4,327 |
| Total | 18.1 | $ 41.43 | $ 751 | 18.1 | |

**Item 6:** ***Selected Financial Data***

The following selected financial information has been derived from Applied's historical audited consolidated financial statements and should be read in conjunction with the consolidated financial statements and the accompanying notes for the corresponding fiscal years:

| Fiscal Year[1] | 2018 | 2017 | 2016 | 2015 | 2014 |
|---|---|---|---|---|---|
| | (In millions, except percentages and per share amounts) | | | | |
| Net sales | $ 17,253 | $ 14,537 | $ 10,825 | $ 9,659 | $ 9,072 |
| Gross profit | $ 7,817 | $ 6,532 | $ 4,511 | $ 3,952 | $ 3,843 |
| Gross margin | 45.3% | 44.9% | 41.7% | 40.9% | 42.4% |
| Research, development and engineering | $ 2,019 | $ 1,774 | $ 1,540 | $ 1,451 | $ 1,428 |
| Operating income | $ 4,796 | $ 3,868 | $ 2,152 | $ 1,693 | $ 1,520 |
| Operating margin | 27.8% | 26.6% | 19.9% | 17.5% | 16.8% |
| Income before income taxes | $ 4,694 | $ 3,731 | $ 2,013 | $ 1,598 | $ 1,448 |
| Net income | $ 3,313 | $ 3,434 | $ 1,721 | $ 1,377 | $ 1,072 |
| Earnings per diluted share | $ 3.23 | $ 3.17 | $ 1.54 | $ 1.12 | $ 0.87 |
| Long-term debt | $ 5,309 | $ 5,304 | $ 3,143 | $ 3,342 | $ 1,947 |
| Cash dividends declared per common share | $ 0.70 | $ 0.40 | $ 0.40 | $ 0.40 | $ 0.40 |
| Total assets | $ 17,773 | $ 19,419 | $ 14,588 | $ 15,308 | $ 13,174 |

(1) Each fiscal year ended on the last Sunday in October. Fiscal 2018, 2017, 2015, and 2014 each contained 52 weeks, and fiscal 2016 contained 53 weeks.

**Item 7:** ***Management's Discussion and Analysis of Financial Condition and Results of Operations***

**Introduction**

Management's Discussion and Analysis of Financial Condition and Results of Operations (MD&A) is intended to facilitate an understanding of Applied's business and results of operations. This MD&A should be read in conjunction with Applied's Consolidated Financial Statements and the accompanying Notes to Consolidated Financial Statements included elsewhere in this Form 10-K. The following discussion contains forward-looking statements and should also be read in conjunction with the cautionary statement set forth at the beginning of this Form 10-K. MD&A consists of the following sections:

- *Overview:* a summary of Applied's business and measurements
- *Results of Operations:* a discussion of operating results
- *Segment Information:* a discussion of segment operating results
- *Recent Accounting Pronouncements:* a discussion of new accounting pronouncements and its impact to Applied's consolidated financial statements
- *Financial Condition, Liquidity and Capital Resources:* an analysis of cash flows, sources and uses of cash
- *Off-Balance Sheet Arrangements and Contractual Obligations*
- *Critical Accounting Policies and Estimates:* a discussion of critical accounting policies that require the exercise of judgments and estimates
- *Non-GAAP Adjusted Results:* a presentation of results reconciling GAAP to non-GAAP adjusted measures

**Overview**

Applied provides manufacturing equipment, services and software to the semiconductor, display, and related industries. Applied's customers include manufacturers of semiconductor wafers and chips, liquid crystal and organic light-emitting diode (OLED) displays, and other electronic devices. These customers may use what they manufacture in their own end products or sell the items to other companies for use in advanced electronic components. Each of Applied's businesses is subject to variable industry conditions, as demand for manufacturing equipment and services can change depending on supply and demand for chips, display technologies, and other electronic devices, as well as other factors, such as global economic and market conditions, and the nature and timing of technological advances in fabrication processes.

Applied operates in three reportable segments: Semiconductor Systems, Applied Global Services, and Display and Adjacent Markets. A summary of financial information for each reportable segment is found in Note 15 of Notes to Consolidated Financial Statements. A discussion of factors that could affect Applied's operations is set forth under "Risk Factors" in Part I, Item 1A, which is incorporated herein by reference. Product development and manufacturing activities occur primarily in the United States, Europe, Israel, and Asia. Applied's broad range of equipment and service products are highly technical and are sold primarily through a direct sales force.

Applied's results are driven primarily by customer spending on capital equipment and services to support key technology transitions or to increase production volume in response to worldwide demand for semiconductors and displays. Spending by semiconductor customers, which include companies that operate in the foundry, memory and logic markets, is driven by demand for advanced electronic products, including smartphones and other mobile devices, servers, personal computers, automotive devices, storage, and other products. The growth of data and emerging end-market drivers such as artificial intelligence, augmented and virtual reality, the Internet of Things and smart vehicles are also creating new opportunities for the industry. As a result, products within the Semiconductor Systems segment are subject to significant changes in customer requirements, including transitions to smaller dimensions, increasingly complex chip architectures, new materials and an increasing number of applications. Demand for display manufacturing equipment spending depends primarily on consumer demand for increasingly larger and more advanced TVs as well as larger and higher resolution displays for next-generation mobile devices, and investments in new types of display technologies. While certain existing technologies may be adapted to new requirements, some applications create the need for an entirely different technological approach. The timing of customer investment in manufacturing equipment is also affected by the timing of next-generation process development and the timing of capacity expansion to meet end-market demand. In light of these conditions, Applied's results can vary significantly year-over-year, as well as quarter-over-quarter.

Applied's strategic priorities include developing products that help solve customers' challenges at technology inflections; expanding its served market opportunities in the semiconductor and display industries; and growing its services business. Applied's long-term growth strategy requires continued development of new materials engineering capabilities, including products and platforms that enable expansion into new and adjacent markets. Applied's significant investments in research, development and engineering must generally enable it to deliver new products and technologies before the emergence of strong demand, thus allowing customers to incorporate these products into their manufacturing plans during early-stage technology selection. Applied works closely with its global customers to design systems and processes that meet their planned technical and production requirements.

The following table presents certain significant measurements for the past three fiscal years:

| | | | | Change | |
|---|---|---|---|---|---|
| | 2018 | 2017 | 2016 | 2018 over 2017 | 2017 over 2016 |
| | (In millions, except per share amounts and percentages) | | | | |
| Net sales | $ 17,253 | $ 14,537 | $ 10,825 | $ 2,716 | $ 3,712 |
| Gross margin | 45.3% | 44.9% | 41.7% | 0.4 points | 3.2 points |
| Operating income | $ 4,796 | $ 3,868 | $ 2,152 | $ 928 | $ 1,716 |
| Operating margin | 27.8% | 26.6% | 19.9% | 1.2 points | 6.7 points |
| Net income | $ 3,313 | $ 3,434 | $ 1,721 | $ (121) | $ 1,713 |
| Earnings per diluted share | $ 3.23 | $ 3.17 | $ 1.54 | $ 0.06 | $ 1.63 |

Fiscal 2018 and 2017 contained 52 weeks, and fiscal 2016 contained 53 weeks.

Fiscal 2018 included a one-time expense related to the enactment of recent U.S. tax legislation that reduced diluted earnings per share by $1.08.

Investment in semiconductor and display manufacturing equipment and services continued to be a strong driver of revenue during fiscal 2018. Semiconductor equipment customers made investments in new capacity and technology transitions, and Applied's overall semiconductor systems revenue increased as compared to the prior year. Demand for dynamic random-access memory (DRAM) and NAND increased in fiscal 2018 although overall spending by memory customers was lower towards the second half of the year. Logic customers' spending remained strong while foundry customers' spending decreased compared to the prior year and Applied continues to see these customers optimize existing capacity and re-prioritize their capital spending plans on longer lead-time equipment not in Applied's product portfolio. Display equipment spending during fiscal 2018 reflected continued investment in new technology and manufacturing equipment for producing larger LCD TVs and in new equipment for mobile devices. Applied also continued to see strong growth in demand for spares and services from customers as compared to the prior year.

While Applied anticipates major technology trends to continue driving long-term growth in the semiconductor industry, the trends characterizing the second half of 2018 are likely to continue into early fiscal 2019, with lower spending by memory customers, and foundry customers prioritizing spending on longer lead-time equipment not in Applied's product portfolio. Applied also expects lower spending for display manufacturing equipment in fiscal 2019, although long-term demand drivers remain in place. Applied anticipates continued growth in semiconductor spares and services spending in fiscal 2019.

**Results of Operations**

***Net Sales***

Net sales for the periods indicated were as follows:

| | 2018 | | 2017 | | 2016 | | Change 2018 over 2017 | Change 2017 over 2016 |
|---|---|---|---|---|---|---|---|---|
| | (In millions, except percentages) | | | | | | | |
| Semiconductor Systems | $ 10,903 | 63% | $ 9,517 | 65% | $ 6,873 | 64% | 15% | 38% |
| Applied Global Services | 3,754 | 22% | 3,017 | 21% | 2,589 | 24% | 24% | 17% |
| Display and Adjacent Markets | 2,498 | 14% | 1,900 | 13% | 1,206 | 11% | 31% | 58% |
| Corporate and Other | 98 | 1% | 103 | 1% | 157 | 1% | (5)% | (34)% |
| Total | $ 17,253 | 100% | $ 14,537 | 100% | $ 10,825 | 100% | 19% | 34% |

Net sales in fiscal 2018 compared to fiscal 2017 increased due to increased customer investments across all segments. The Semiconductor Systems segment continued to represent the largest contributor of net sales and to the increase in net sales.

Net sales in fiscal 2017 compared to fiscal 2016 increased due to increased customer investments in all segments, with the majority of the increases resulting from investments in semiconductor equipment.

Net sales by geographic region, determined by the location of customers' facilities to which products were shipped, were as follows:

| | 2018 | | 2017 | | 2016 | | Change 2018 over 2017 | Change 2017 over 2016 |
|---|---|---|---|---|---|---|---|---|
| | (In millions, except percentages) | | | | | | | |
| China | $ 5,113 | 30% | $ 2,746 | 19% | $ 2,259 | 21% | 86% | 22% |
| Korea | 3,603 | 21% | 4,052 | 28% | 1,883 | 17% | (11)% | 115% |
| Taiwan | 2,732 | 16% | 3,291 | 23% | 2,843 | 26% | (17)% | 16% |
| Japan | 2,405 | 14% | 1,518 | 10% | 1,279 | 12% | 58% | 19% |
| Southeast Asia | 802 | 4% | 640 | 4% | 803 | 7% | 25% | (20)% |
| Asia Pacific | 14,655 | 85% | 12,247 | 84% | 9,067 | 83% | 20% | 35% |
| United States | 1,532 | 9% | 1,474 | 10% | 1,143 | 11% | 4% | 29% |
| Europe | 1,066 | 6% | 816 | 6% | 615 | 6% | 31% | 33% |
| Total | $ 17,253 | 100% | $ 14,537 | 100% | $ 10,825 | 100% | 19% | 34% |

The changes in net sales in all regions other than Korea for fiscal 2018 compared to fiscal 2017 primarily reflected changes in semiconductor equipment spending, including product and customer mix, and increased spending in semiconductor spares and services. The increase in net sales to customers in China also reflected increased investments in display manufacturing equipment. The decrease in net sales to customers in Korea primarily reflected decreased investments in display manufacturing equipment.

The changes in net sales from customers in all regions for fiscal 2017 compared to fiscal 2016 primarily reflected increased investments in semiconductor equipment and changes in semiconductor equipment customer mix. In addition, the increase in net sales from customers in China also reflected increased investments from display manufacturing equipment customers.

***Gross Margin***

Gross margins for the periods indicated were as follows:

| | | | | Change | |
|---|---|---|---|---|---|
| | 2018 | 2017 | 2016 | 2018 over 2017 | 2017 over 2016 |
| Gross margin | 45.3% | 44.9% | 41.7% | 0.4 points | 3.2 points |

Gross margin in fiscal 2018 increased slightly compared to fiscal 2017, primarily due to higher net sales and materials cost savings. Gross margin in fiscal 2017 increased compared to fiscal 2016, primarily due to higher net sales, favorable product mix and materials cost savings.

Gross margin during fiscal 2018, 2017 and 2016 included $87 million, $69 million and $62 million, respectively, of share-based compensation expense.

***Research, Development and Engineering***

Research, Development and Engineering (RD&E) expenses for the periods indicated were as follows:

| | | | | Change | |
|---|---|---|---|---|---|
| | 2018 | 2017 | 2016 | 2018 over 2017 | 2017 over 2016 |
| | | | (In millions) | | |
| Research, development and engineering | $ 2,019 | $ 1,774 | $ 1,540 | $ 245 | $ 234 |

Applied's future operating results depend to a considerable extent on its ability to maintain a competitive advantage in the equipment and service products it provides. Development cycles range from 12 to 36 months depending on whether the product is an enhancement of an existing product, which typically has a shorter development cycle, or a new product, which typically has a longer development cycle. Most of Applied's existing products resulted from internal development activities and innovations involving new technologies, materials and processes. In certain instances, Applied acquires technologies, either in existing or new product areas, to complement its existing technology capabilities and to reduce time to market.

Management believes that it is critical to continue to make substantial investments in RD&E to assure the availability of innovative technology that meets the current and projected requirements of its customers' most advanced designs. Applied has maintained and intends to continue its commitment to investing in RD&E in order to continue to offer new products and technologies.

In fiscal 2018, Applied increased its RD&E investments across Semiconductor Systems and Display and Adjacent Markets, including etch, e-beam inspection and other materials engineering solutions to improve chip performance and enable advanced displays. Applied's investments in etch were focused on supporting the adoption of precision etch technology for enabling continued scaling of 3D logic and memory chips. The investment in e-beam inspection was in support of strengthening our e-beam portfolio, which helps achieve higher yields at advanced nodes by detecting the most challenging defects and monitoring and controlling process marginality. Applied also invested in materials engineering solutions to support patterning applications as well as to improve transistor performance and device power efficiency. In Display and Adjacent Markets, RD&E investments were focused on expanding the company's market opportunity with new display technologies.

RD&E expenses increased in fiscal 2018 compared to the prior year and also in fiscal 2017 compared to fiscal 2016, primarily due to additional headcount and increased research and development spending in Semiconductor Systems and Display and Adjacent Market segments. These increases reflect Applied's ongoing investments in product development initiatives, consistent with the Company's growth strategy. Applied continued to prioritize existing RD&E investments in technical capabilities and critical research and development programs in current and new markets, with a focus on semiconductor technologies.

RD&E expense during fiscal 2018, 2017 and 2016 included $96 million, $83 million and $76 million, respectively, of share-based compensation expense.

***Marketing and Selling***

Marketing and selling expenses for the periods indicated were as follows:

| | 2018 | 2017 | 2016 | Change 2018 over 2017 | Change 2017 over 2016 |
|---|---|---|---|---|---|
| | | | (In millions) | | |
| Marketing and selling | $ 521 | $ 456 | $ 429 | $ 65 | $ 27 |

Marketing and selling expenses increased in fiscal 2018 compared to fiscal 2017 primarily due to additional headcount. Marketing and selling expenses increased in fiscal 2017 compared to fiscal 2016, primarily due to additional headcount, partially offset by the reduction in bad debt provision recorded during fiscal 2017.

Marketing and selling expenses for fiscal years 2018, 2017 and 2016 included $31 million, $28 million and $26 million, respectively, of share-based compensation expense.

***General and Administrative***

General and administrative (G&A) expenses for the periods indicated were as follows:

| | 2018 | 2017 | 2016 | Change 2018 over 2017 | Change 2017 over 2016 |
|---|---|---|---|---|---|
| | | | (In millions) | | |
| General and administrative | $ 481 | $ 434 | $ 390 | $ 47 | $ 44 |

General and administrative expenses in fiscal 2018 increased compared to fiscal 2017 primarily due to additional headcount and unfavorable impact from foreign exchange fluctuation, partially offset by lower variable compensation expense. General and administrative expenses in fiscal 2017 increased compared to fiscal 2016, primarily due to higher variable compensation and additional headcount.

G&A expenses during fiscal 2018, 2017 and 2016 included $44 million, $40 million and $37 million, respectively, of share-based compensation expense.

***Interest Expense and Interest and Other Income (loss), net***

Interest expense and interest and other income (loss), net for the periods indicated were as follows:

| | 2018 | 2017 | 2016 | Change 2018 over 2017 | Change 2017 over 2016 |
|---|---|---|---|---|---|
| | | | (In millions) | | |
| Interest expense | $ 234 | $ 198 | $ 155 | $ 36 | $ 43 |
| Interest and other income, net | $ 132 | $ 61 | $ 16 | $ 71 | $ 45 |

Interest expenses incurred were primarily associated with the senior unsecured notes issued in June 2011, September 2015, and March 2017. Interest expense increased in fiscal 2018 compared to the prior year and also in fiscal 2017 compared to fiscal 2016 due to the issuance of senior unsecured notes in March 2017.

Interest and other income, net primarily includes interest earned on cash and investments, realized gains or losses on sales of securities and impairment of strategic investments. Interest and other income, net in fiscal 2018 increased compared to fiscal 2017 primarily driven by realized gains on sales of securities and higher interest income from investments. Interest and other income, net in fiscal 2017 increased compared to the prior year primarily due to higher interest income from investments, partially offset by higher impairment of strategic investments in fiscal 2017.

***Income Taxes***

Provision for income taxes and effective tax rates for the periods indicated were as follows:

| | | | | Change | |
|---|---|---|---|---|---|
| | 2018 | 2017 | 2016 | 2018 over 2017 | 2017 over 2016 |
| | (In millions, except percentages) | | | | |
| Provision for income taxes | $ 1,381 | $ 297 | $ 292 | $ 1,084 | $ 5 |
| Effective income tax rate | 29.4% | 8.0% | 14.5% | 21.4 points | (6.5) points |

Applied's provision for income taxes and effective tax rate are affected by the geographical composition of pre-tax income which includes jurisdictions with differing tax rates, conditional reduced tax rates and other income tax incentives. It is also affected by events that are not consistent from period to period, such as changes in income tax laws and the resolution of prior years' income tax filings.

On December 22, 2017, the U.S. government enacted the Tax Cuts and Jobs Act (the "Tax Act"). The Tax Act includes a reduction to the U.S. federal corporate tax rate from 35.0 percent to 21.0 percent and requires a one-time transition tax on certain unrepatriated earnings of foreign subsidiaries payable over eight years. U.S. deferred tax assets and liabilities were subject to remeasurement due to the reduction of the U.S. federal corporate tax rate. Applied has a blended U.S. federal corporate tax rate of 23.4 percent for fiscal 2018 based on the number of days before and after the effective date of the Tax Act.

The U.S. Securities and Exchange Commission issued Staff Accounting Bulletin No. 118 (SAB 118), which provides guidance on accounting for the income tax effects of the Tax Act. SAB 118 provides a measurement period for companies to complete this accounting. Pursuant to SAB 118, provisional adjustments were recorded when reasonable estimates could be determined. These provisional estimates will be revised as information becomes available and as guidance is issued by the Internal Revenue Service. The accounting for the income tax effects of the Tax Act will be completed during the measurement period, which will not extend beyond one year from the Tax Act enactment date. Applied continues to evaluate certain unrepatriated earnings of foreign subsidiaries used to calculate the transition tax. The remeasurement of U.S. deferred tax assets and liabilities is complete.

The effective tax rate for fiscal 2018 was higher than fiscal 2017 primarily due to tax expense of $1.1 billion for the transition tax and remeasurement of deferred tax assets as a result of the Tax Act, partially offset by changes in the geographical composition of income, tax benefits from the lower U.S. federal corporate tax rate, adoption of authoritative guidance for share-based compensation, and the resolution of tax liabilities for uncertain tax positions. In addition, fiscal 2017 included tax benefits from the recognition of previously unrecognized foreign tax credits.

The effective tax rate for fiscal 2017 was lower than fiscal 2016 primarily due to the recognition of previously unrecognized foreign tax credits and changes in the geographical composition of income. In addition, fiscal 2016 included unfavorable resolutions and changes related to income tax liabilities for uncertain tax positions as well as the reinstatement of the U.S. federal R&D tax credit retroactive to its expiration in December of 2015 which did not reoccur in fiscal 2017.

**Segment Information**

Applied reports financial results in three segments: Semiconductor Systems, Applied Global Services, and Display and Adjacent Markets. A description of the products and services, as well as financial data, for each reportable segment can be found in Note 15 of Notes to Consolidated Financial Statements.

The Corporate and Other category includes revenues from products, as well as costs of products sold, for fabricating solar photovoltaic cells and modules and certain operating expenses that are not allocated to its reportable segments and are managed separately at the corporate level. These operating expenses include costs for share-based compensation; certain management, finance, legal, human resource, and RD&E functions provided at the corporate level; and unabsorbed information technology and occupancy. In addition, Applied does not allocate to its reportable segments restructuring and asset impairment charges and any associated adjustments related to restructuring actions, unless these actions pertain to a specific reportable segment.

The results for each reportable segment are discussed below.

***Semiconductor Systems Segment***

The Semiconductor Systems segment is comprised primarily of capital equipment used to fabricate semiconductor chips. Semiconductor industry spending on capital equipment is driven by demand for advanced electronic products, including smartphones and other mobile devices, servers, personal computers, automotive devices, storage, and other products, and the nature and timing of technological advances in fabrication processes, and as a result is subject to variable industry conditions. Development efforts are focused on solving customers' key technical challenges in transistor, interconnect, patterning and packaging performance as devices scale to advanced technology nodes.

Investment in semiconductor manufacturing equipment continued to be a strong driver of revenue during fiscal 2018. Semiconductor equipment customers made investments in new capacity and technology transitions, and overall semiconductor systems revenue increased. Demand for logic, DRAM, and NAND increased during fiscal 2018. Foundry customers' spending decreased compared to the prior year and Applied continues to see these customers optimize existing capacity and re-prioritize their capital spending plans on longer lead-time equipment not in Applied's product portfolio.

Certain significant measures for the periods indicated were as follows:

| | | | | Change | | | |
|---|---|---|---|---|---|---|---|
| | 2018 | 2017 | 2016 | 2018 over 2017 | | 2017 over 2016 | |
| | (In millions, except percentages and ratios) | | | | | | |
| Net sales | $ 10,903 | $ 9,517 | $ 6,873 | $ 1,386 | 15% | $ 2,644 | 38% |
| Operating income | $ 3,634 | $ 3,173 | $ 1,807 | $ 461 | 15% | $ 1,366 | 76% |
| Operating margin | 33.3% | 33.3% | 26.3% | | —% | | 7.0 points |

Net sales for Semiconductor Systems by end use application for the periods indicated were as follows:

| | 2018 | 2017 | 2016 |
|---|---|---|---|
| Foundry | 24% | 41% | 40% |
| Dynamic random-access memory (DRAM) | 27% | 16% | 16% |
| Flash memory | 36% | 34% | 31% |
| Logic and other | 13% | 9% | 13% |
| | 100% | 100% | 100% |

Net sales for fiscal 2018 increased compared to fiscal 2017 primarily due to higher spending from memory and logic customers, offset by lower spending from foundry customers. Although operating margin remained flat, operating income for fiscal 2018 increased compared to fiscal 2017 primarily due to favorable changes in product mix and higher net sales, partially offset by higher RD&E expenses. Six customers represented at least 10 percent of this segment's net sales, and together they accounted for approximately 76 percent of this segment's net sales for fiscal 2018.

Net sales for fiscal 2017 increased compared to fiscal 2016 primarily due to higher spending from foundry and memory customers. Operating income and operating margin for fiscal 2017 increased compared to prior year primarily due to favorable changes in product mix and higher net sales, partially offset by higher RD&E expenses.

The following regions accounted for at least 30 percent of total net sales for the Semiconductor Systems segment for one or more of past three fiscal years:

| | 2018 | | 2017 | | 2016 | | Change 2018 over 2017 | | Change 2017 over 2016 | |
|---|---|---|---|---|---|---|---|---|---|---|
| | (In millions, except percentages) | | | | | | | | | |
| Korea | $ 2,886 | 26% | $ 2,962 | 31% | $ 1,177 | 17% | $ (76) | (3)% | $ 1,785 | 152% |
| Taiwan | $ 1,895 | 17% | $ 2,638 | 28% | $ 2,165 | 32% | $ (743) | (28)% | $ 473 | 22% |

***Applied Global Services Segment***

The Applied Global Services segment provides integrated solutions to optimize equipment and fab performance and productivity, including spares, upgrades, services, certain remanufactured earlier generation equipment and factory automation software for semiconductor, display and solar products. Customer demand for products and services is fulfilled through a global distribution system with trained service engineers located in close proximity to customer sites.

Demand for Applied Global Services' service solutions are driven by Applied's large and growing installed base of manufacturing systems, and customers' needs to shorten ramp times, improve device performance and yield, and optimize factory output and operating costs. Industry conditions that affect Applied Global Services' sales of spares and services are primarily characterized by increases in semiconductor manufacturers' wafer starts and continued strong utilization rates, growth of the installed base of equipment, growing service intensity of newer tools, and the company's ability to sell more comprehensive service agreements.

Certain significant measures for the periods indicated were as follows:

| | 2018 | 2017 | 2016 | Change 2018 over 2017 | | Change 2017 over 2016 | |
|---|---|---|---|---|---|---|---|
| | (In millions, except percentages and ratios) | | | | | | |
| Net sales | $ 3,754 | $ 3,017 | $ 2,589 | $ 737 | 24% | $ 428 | 17% |
| Operating income | $ 1,102 | $ 817 | $ 682 | $ 285 | 35% | $ 135 | 20% |
| Operating margin | 29.4% | 27.1% | 26.3% | 2.3 points | | 0.8 points | |

There was no single region that accounted for at least 30 percent of total net sales for the Applied Global Services segment for any of the past three fiscal years.

Net sales increased in fiscal 2018 compared to the prior year and also in fiscal 2017 compared to fiscal 2016 primarily due to higher customer spending for spares and services. Two customers represented at least 10 percent of this segment's net sales, and together they accounted for approximately 23 percent of this segment's net sales for fiscal 2018.

Operating income and operating margin for fiscal 2018 increased compared to the prior year and also in fiscal 2017 compared to fiscal 2016, reflecting higher net sales partially offset by increased headcount to support business growth.

***Display and Adjacent Markets Segment***

The Display and Adjacent Markets segment encompasses products for manufacturing liquid crystal and OLED displays, and other display technologies for TVs, monitors, laptops, personal computers, electronic tablets, smart phones, and other consumer-oriented devices, equipment upgrades and flexible coating systems. The segment is focused on expanding its presence through technologically-differentiated equipment for manufacturing large-scale LCD TVs, OLEDs, low temperature polysilicon (LTPS), metal oxide, and touch panel sectors; and development of products that provide customers with improved performance and yields. Display industry growth depends primarily on consumer demand for increasingly larger and more advanced TVs as well as larger and higher resolution displays for next-generation mobile devices.

The market environment for Applied's Display and Adjacent Markets segment in fiscal 2018 was characterized by increased demand for manufacturing equipment for TV and mobile devices. Uneven demand patterns in the Display and Adjacent Markets segment can cause significant fluctuations quarter-over-quarter, as well as year-over-year.

Certain significant measures for the periods presented were as follows:

| | | | | Change | | | |
|---|---|---|---|---|---|---|---|
| | 2018 | 2017 | 2016 | 2018 over 2017 | | 2017 over 2016 | |
| | (In millions, except percentages and ratios) | | | | | | |
| Net sales | $ 2,498 | $ 1,900 | $ 1,206 | $ 598 | 31% | $ 694 | 58% |
| Operating income | $ 679 | $ 502 | $ 245 | $ 177 | 35% | $ 257 | 105% |
| Operating margin | 27.2% | 26.4% | 20.3% | 0.8 points | | 6.1 points | |

Net sales for fiscal 2018 increased compared to fiscal 2017 primarily due to higher customer investments in TV display manufacturing equipment. Operating income and operating margin for fiscal 2018 increased compared fiscal 2017, reflecting higher net sales and favorable product mix, partially offset by higher RD&E spending. Three customers, that represented at least 10 percent of this segment’s net sales, accounted for approximately 58 percent of net sales for this segment in fiscal 2018, with one customer accounting for approximately 33 percent of net sales.

Net sales for fiscal 2017 increased compared to prior year primarily due to higher customer investments in mobile and TV display manufacturing equipment. Operating income and operating margin for fiscal 2017 increased compared to the prior year, reflecting higher net sales and favorable product mix, partially offset by increased RD&E spending.

The following regions accounted for at least 30 percent of total net sales for the Display and Adjacent Markets segment for one or more of the periods presented:

| | | | | | | | Change | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | 2018 | | 2017 | | 2016 | | 2018 over 2017 | | 2017 over 2016 | |
| | (In millions, except percentages) | | | | | | | | | |
| China | $ 2,004 | 80% | $ 961 | 51% | $ 449 | 37% | $ 1,043 | 109% | $ 512 | 114% |
| Korea | $ 211 | 8% | $ 750 | 39% | $ 492 | 41% | $ (539) | (72)% | $ 258 | 52% |

**Recent Accounting Pronouncements**

For a description of recent accounting pronouncements, including the expected dates of adoption and estimated effects, if any, on Applied’s consolidated financial statements, see Note 1, “Summary of Significant Accounting Policies,” of the Notes to Consolidated Financial Statements.

**Financial Condition, Liquidity and Capital Resources**

Applied's cash, cash equivalents and investments consist of the following:

| | October 28, 2018 | October 29, 2017 |
|---|---|---|
| | (In millions) | |
| Cash and cash equivalents | $ 3,440 | $ 5,010 |
| Short-term investments | 590 | 2,266 |
| Long-term investments | 1,568 | 1,143 |
| Total cash, cash-equivalents and investments | $ 5,598 | $ 8,419 |

*Sources and Uses of Cash*

A summary of cash provided by (used in) operating, investing, and financing activities is as follows:

| | 2018 | 2017 | 2016 |
|---|---|---|---|
| | | (In millions) | |
| Cash provided by operating activities | $ 3,787 | $ 3,789 | $ 2,566 |
| Cash provided by (used in) investing activities | $ 571 | $ (2,526) | $ (425) |
| Cash provided by (used in) financing activities | $ (5,928) | $ 341 | $ (3,532) |

In March 2016, the Financial Accounting Standards Board issued authoritative guidance that simplifies several aspects of the accounting for share-based payment transactions, including forfeitures, income tax, and classification on the statement of cash flows. Applied adopted this guidance in the first quarter of fiscal 2018. Upon adoption, Applied elected to apply the presentation requirements for cash flows related to excess tax benefits and employee taxes paid for withheld shares retrospectively. Adopting this guidance increased cash provided by operating activities by $180 million and $100 million with corresponding net decreases in cash provided by financing activities for fiscal 2017 and 2016, respectively. See Note 1 of Notes to Consolidated Financial Statements for additional discussion of this adoption.

*Operating Activities*

Cash from operating activities for fiscal 2018 was $3.8 billion, which reflects net income adjusted for the effect of non-cash charges and changes in working capital components. Non-cash charges included depreciation, amortization, share-based compensation and deferred income taxes. Cash provided from operating activities remained flat from fiscal 2017 to fiscal 2018 due to the increase in income taxes payable, offset by lower deferred revenue and higher increase in accounts receivable. Cash provided by operating activities increased from fiscal 2016 to fiscal 2017 primarily due to higher net income, offset by a smaller increase in customer deposits and a greater increase in inventory in fiscal 2017.

Applied has agreements with various financial institutions to sell accounts receivable and discount promissory notes from selected customers. Applied sells its accounts receivable without recourse. Applied, from time to time, also discounts letters of credit issued by customers through various financial institutions. The discounting of letters of credit depends on many factors, including the willingness of financial institutions to discount the letters of credit and the cost of such arrangements. Applied sold $1.6 billion, $746 million and $75 million of accounts receivable during fiscal 2018, 2017 and 2016, respectively. Applied discounted letters of credit issued by customers of $37 million in fiscal 2018. There was no discounting of promissory notes in fiscal 2018. Applied did not discount letters of credit issued by customers or discount promissory notes during fiscal 2017 or 2016.

Applied's working capital was $6.7 billion at October 28, 2018 and $8.8 billion at October 29, 2017.

Days sales outstanding at the end of fiscal 2018, 2017 and 2016 was 58 days, 54 days, and 63 days, respectively. Days sales outstanding varies due to the timing of shipments and payment terms, and the increase from fiscal 2017 to fiscal 2018 was primarily due to better revenue linearity in fiscal 2017. Days sales outstanding decreased from fiscal 2016 to fiscal 2017 primarily due to better collections performance and increase in factoring of accounts receivable.

*Investing Activities*

Applied generated $571 million in cash from investing activities in fiscal 2018. Applied used $2.5 billion of cash in investing activities in fiscal 2017 and $425 million in fiscal 2016. Capital expenditures in fiscal 2018, 2017 and 2016 were $622 million, $345 million and $253 million, respectively. Capital expenditures in fiscal 2018 were primarily for real property acquisitions and improvements in North America and Taiwan, as well as investments in demonstration, testing and laboratory tools. Capital expenditures in fiscal 2017 and fiscal 2016 were primarily for demonstration and test equipment and laboratory tools in North America. Proceeds from sales and maturities of investments, net of purchase of investments was $1.2 billion for fiscal 2018. Purchases of investments, net of proceeds from sales and maturities of investments were $2.1 billion and $156 million for fiscal 2017 and 2016, respectively. Investing activities also included investments in technology to allow Applied to access new market opportunities or emerging technologies.

Applied's investment portfolio consists principally of investment grade money market mutual funds, U.S. Treasury and agency securities, municipal bonds, corporate bonds and mortgage-backed and asset-backed securities, as well as equity securities. Applied regularly monitors the credit risk in its investment portfolio and takes appropriate measures, which may include the sale of certain securities, to manage such risks prudently in accordance with its investment policies.

*Financing Activities*

Applied used $5.9 billion of cash in financing activities in fiscal 2018, consisting primarily of repurchases of common stock of $5.3 billion, cash dividends to stockholders of $605 million and tax withholding payments for vested equity awards of $164 million, offset by proceeds from common stock issuances of $124 million.

Applied generated $341 million of cash from financing activities in fiscal 2017, consisting primarily of net proceeds received from the issuance of senior unsecured notes of $2.2 billion, proceeds from common stock issuances of $97 million, partially offset by cash used for repurchases of common stock of $1.2 billion, cash dividends to stockholders of $430 million and debt repayments of $205 million.

Applied used $3.5 billion of cash in financing activities in fiscal 2016, consisting primarily $1.2 billion in debt repayments, $1.9 billion in repurchases of its common stock, and $444 million in cash dividends to stockholders, offset by proceeds from common stock issuances of $88 million.

In September 2017, Applied's Board of Directors approved a common stock repurchase program authorizing up to $3.0 billion in repurchases. In February 2018, the Board of Directors approved a new common stock repurchase program authorizing up to an additional $6.0 billion in repurchases. At October 28, 2018, $4.3 billion remained available for future stock repurchases under this repurchase program.

During fiscal 2018, Applied's Board of Directors declared three quarterly cash dividends of $0.20 per share and one quarterly cash dividend of $0.10 per share. During each of fiscal 2017 and 2016, Applied's Board of Directors declared four quarterly cash dividends in the amount of $0.10 per share. Applied currently anticipates that cash dividends will continue to be paid on a quarterly basis, although the declaration of any future cash dividend is at the discretion of the Board of Directors and will depend on Applied's financial condition, results of operations, capital requirements, business conditions and other factors, as well as a determination by the Board of Directors that cash dividends are in the best interests of Applied's stockholders.

Applied has credit facilities for unsecured borrowings in various currencies of up to $1.6 billion, of which $1.5 billion is comprised of a committed revolving credit agreement with a group of banks that is scheduled to expire in September 2021. This agreement provides for borrowings in United States dollars at interest rates keyed to one of two rates selected by Applied for each advance and includes financial and other covenants with which Applied was in compliance at October 28, 2018. Remaining credit facilities in the amount of approximately $71 million are with Japanese banks. Applied's ability to borrow under these facilities is subject to bank approval at the time of the borrowing request, and any advances will be at rates indexed to the banks' prime reference rate denominated in Japanese yen. No amounts were outstanding under any of these facilities at both October 28, 2018 and October 29, 2017, and Applied has not utilized these credit facilities.

In fiscal 2011, Applied established a short-term commercial paper program of up to $1.5 billion. At October 28, 2018 and October 29, 2017, Applied did not have any commercial paper outstanding, but may issue commercial paper notes under this program from time to time in the future.

In March 2017, Applied issued senior unsecured notes in the aggregate principal amount of $2.2 billion. Applied had senior unsecured notes in the aggregate principal amount of $5.4 billion outstanding as of October 28, 2018. The indentures governing these notes include covenants with which Applied was in compliance at October 28, 2018. In May 2017, Applied completed the redemption of the entire outstanding $200 million in principal amount of senior notes due in October 2017. See Note 10 of Notes to Consolidated Financial Statements for additional discussion of existing debt. Applied may seek to refinance its existing debt and may incur additional indebtedness depending on Applied's capital requirements and the availability of financing.

*Others*

During 2018 and 2017, Applied reduced $1 million and $17 million, respectively, of its allowance for doubtful accounts as a result of an overall lower risk profile of Applied's customers. Applied recorded a bad debt provision of $3 million in fiscal 2016. While Applied believes that its allowance for doubtful accounts at October 28, 2018 is adequate, it will continue to closely monitor customer liquidity and economic conditions.

On December 22, 2017, the U.S. government enacted the Tax Cuts and Jobs Act. The Tax Act requires a one-time transition tax on certain unrepatriated earnings of foreign subsidiaries payable over eight years, with eight percent due in each of the first five years starting with fiscal 2018. For fiscal 2018, Applied realized tax expense of $1.1 billion associated with the Tax Act, primarily due to the transition tax. The transition tax eliminated the requirement that cash, cash equivalents and marketable securities held by certain foreign subsidiaries be subject to U.S income taxes if repatriated for U.S. operations. Accordingly, cash, cash equivalents and marketable securities held by all foreign subsidiaries may be repatriated for U.S. operation without being subject to further U.S. income taxes.

Although cash requirements will fluctuate based on the timing and extent of factors such as those discussed above, Applied's management believes that cash generated from operations, together with the liquidity provided by existing cash balances and borrowing capability, will be sufficient to satisfy Applied's liquidity requirements for the next 12 months. For further details regarding Applied's operating, investing and financing activities, see the Consolidated Statements of Cash Flows in this report.

For details on standby letters of credit and other agreements with banks, see Off-Balance Sheet Arrangements below.

**Off-Balance Sheet Arrangements**

In the ordinary course of business, Applied provides standby letters of credit or other guarantee instruments to third parties as required for certain transactions initiated by either Applied or its subsidiaries. As of October 28, 2018, the maximum potential amount of future payments that Applied could be required to make under these guarantee agreements was approximately $58 million. Applied has not recorded any liability in connection with these guarantee agreements beyond that required to appropriately account for the underlying transaction being guaranteed. Applied does not believe, based on historical experience and information currently available, that it is probable that any amounts will be required to be paid under these guarantee agreements.

Applied also has agreements with various banks to facilitate subsidiary banking operations worldwide, including overdraft arrangements, issuance of bank guarantees, and letters of credit. As of October 28, 2018, Applied has provided parent guarantees to banks for approximately $149 million to cover these arrangements.

Applied also has operating leases for various facilities. Total rent expense for fiscal 2018, 2017 and 2016 was $50 million, $34 million and $38 million, respectively.

**Contractual Obligations**

The following table summarizes Applied's contractual obligations as of October 28, 2018:

| | Payments Due by Period | | | | |
|---|---|---|---|---|---|
| Contractual Obligations | Total | Less Than 1 Year | 1-3 Years | 3-5 Years | More Than 5 Years |
| | | | (In millions) | | |
| Debt obligations | $ 5,350 | $ — | $ 1,350 | $ — | $ 4,000 |
| Interest expense associated with debt obligations | 3,137 | 219 | 422 | 342 | 2,154 |
| Operating lease obligations | 173 | 50 | 68 | 33 | 22 |
| Income tax from change in U.S. tax laws[1] | 1,001 | 80 | 160 | 160 | 601 |
| Purchase obligations[2] | 2,044 | 1,914 | 115 | 15 | — |
| Other long-term liabilities[3,4] | 20 | — | 2 | 2 | 16 |
| Total | $ 11,725 | $ 2,263 | $ 2,117 | $ 552 | $ 6,793 |

1 Represents an estimate of a provisional tax amount for the transition tax liability associated with the deemed repatriation of accumulated foreign earnings as a result from the enactment of the Tax Cuts and Jobs Act into law on December 22, 2017.

2 Represents Applied's agreements to purchase goods and services consisting of Applied's outstanding purchase orders for goods and services.

3 Other long-term liabilities in the table do not include pension, post-retirement and deferred compensation plans due to the uncertainty in the timing of future payments. Applied evaluates the need to make contributions to its pension and post-retirement benefit plans after considering the funded status of the plans, movements in the discount rate, performance of the plan assets and related tax consequences. Payments to the plans would be dependent on these factors and could vary across a wide range of amounts and time periods. Payments for deferred compensation plans are dependent on activity by participants, making the timing of payments uncertain. Information on Applied's pension, post-retirement benefit and deferred compensation plans is presented in Note 12, Employee Benefit Plans, of the consolidated financial statements.

4 Applied's other long-term liabilities in the Consolidated Balance Sheets include deferred tax liabilities, gross unrecognized tax benefits and related gross interest and penalties. As of October 28, 2018, the gross liability for unrecognized tax benefits that was not expected to result in payment of cash within one year was $374 million. Interest and penalties related to uncertain tax positions that were not expected to result in payment of cash within one year of October 28, 2018 was $26 million. At this time, Applied is unable to make a reasonably reliable estimate of the timing of payments due to uncertainties in the timing of tax audit outcomes; therefore, such amounts are not included in the above contractual obligation table.

**Critical Accounting Policies and Estimates**

The preparation of consolidated financial statements and related disclosures in conformity with accounting principles generally accepted in the United States of America requires management to make judgments, assumptions and estimates that affect the amounts reported. Note 1 of Notes to Consolidated Financial Statements describes the significant accounting policies used in the preparation of the consolidated financial statements. Certain of these significant accounting policies are considered to be critical accounting policies.

A critical accounting policy is defined as one that is both material to the presentation of Applied's consolidated financial statements and that requires management to make difficult, subjective or complex judgments that could have a material effect on Applied's financial condition or results of operations. Specifically, these policies have the following attributes: (1) Applied is required to make assumptions about matters that are highly uncertain at the time of the estimate; and (2) different estimates Applied could reasonably have used, or changes in the estimate that are reasonably likely to occur, would have a material effect on Applied's financial condition or results of operations.

Estimates and assumptions about future events and their effects cannot be determined with certainty. Applied bases its estimates on historical experience and on various other assumptions believed to be applicable and reasonable under the circumstances. These estimates may change as new events occur, as additional information is obtained and as Applied's operating environment changes. These changes have historically been minor and have been included in the consolidated financial statements as soon as they became known. In addition, management is periodically faced with uncertainties, the outcomes of which are not within its control and will not be known for prolonged periods of time. These uncertainties include those discussed in Part I, Item 1A, "Risk Factors." Based on a critical assessment of its accounting policies and the underlying judgments and uncertainties affecting the application of those policies, management believes that Applied's consolidated financial statements are fairly stated in accordance with accounting principles generally accepted in the United States of America, and provide a meaningful presentation of Applied's financial condition and results of operations.

Management believes that the following are critical accounting policies and estimates:

***Revenue Recognition***

Applied recognizes revenue when all four revenue recognition criteria have been met: persuasive evidence of an arrangement exists; delivery has occurred or services have been rendered; sales price is fixed or determinable; and collectability is probable. Each sale arrangement may contain commercial terms that differ from other arrangements. In addition, Applied frequently enters into contracts that contain multiple deliverables. Judgment is required to properly identify the accounting units of the multiple deliverable transactions and to determine the manner in which revenue should be allocated among the accounting units. Moreover, judgment is used in interpreting the commercial terms and determining when all criteria of revenue recognition have been met in order for revenue recognition to occur in the appropriate accounting period. While changes in the allocation of the estimated sales price between the units of accounting will not affect the amount of total revenue recognized for a particular sales arrangement, any material changes in these allocations could impact the timing of revenue recognition, which could have a material effect on Applied's financial condition and results of operations.

***Warranty Costs***

Applied provides for the estimated cost of warranty when revenue is recognized. Estimated warranty costs are determined by analyzing specific product, current and historical configuration statistics and regional warranty support costs. Applied's warranty obligation is affected by product and component failure rates, material usage and labor costs incurred in correcting product failures during the warranty period. As Applied's customer engineers and process support engineers are highly trained and deployed globally, labor availability is a significant factor in determining labor costs. The quantity and availability of critical replacement parts is another significant factor in estimating warranty costs. Unforeseen component failures or exceptional component performance can also result in changes to warranty costs. If actual warranty costs differ substantially from Applied's estimates, revisions to the estimated warranty liability would be required, which could have a material adverse effect on Applied's business, financial condition and results of operations.

***Allowance for Doubtful Accounts***

Applied maintains an allowance for doubtful accounts for estimated losses resulting from the inability of its customers to make required payments. This allowance is based on historical experience, credit evaluations, specific customer collection history and any customer-specific issues Applied has identified. Changes in circumstances, such as an unexpected material adverse change in a major customer's ability to meet its financial obligation to Applied or its payment trends, may require Applied to further adjust its estimates of the recoverability of amounts due to Applied, which could have a material adverse effect on Applied's business, financial condition and results of operations.

***Inventory Valuation***

Inventories are generally stated at the lower of cost or net realizable value, with cost determined on a first-in, first-out basis. The carrying value of inventory is reduced for estimated obsolescence by the difference between its cost and the estimated net realizable value based upon assumptions about future demand. Applied evaluates the inventory carrying value for potential excess and obsolete inventory exposures by analyzing historical and anticipated demand. In addition, inventories are evaluated for potential obsolescence due to the effect of known and anticipated engineering change orders and new products. If actual demand were to be substantially lower than estimated, additional adjustments for excess or obsolete inventory may be required, which could have a material adverse effect on Applied's business, financial condition and results of operations.

***Goodwill and Intangible Assets***

Applied reviews goodwill and intangible assets for impairment whenever events or changes in circumstances indicate that the carrying amount of these assets may not be recoverable, and also annually reviews goodwill and intangibles with indefinite lives for impairment. Intangible assets, such as purchased technology, are generally recorded in connection with a business acquisition. The value assigned to intangible assets is usually based on estimates and judgments regarding expectations for the success and life cycle of products and technology acquired. If actual product acceptance differs significantly from the estimates, Applied may be required to record an impairment charge to reduce the carrying value of the reporting unit to its estimated fair value.

To test goodwill for impairment, Applied first performs a qualitative assessment to determine whether it is more likely than not that the fair value of a reporting unit is less than its carrying value. If it is concluded that this is the case, Applied then performs the two-step goodwill impairment test. Otherwise, the two-step goodwill impairment test is not required. Under the two-step goodwill impairment test, Applied would in the first step compare the estimated fair value of each reporting unit to its carrying value. If the carrying value of a reporting unit exceeds its estimated fair value, Applied would then perform the second step of the impairment test in order to determine the implied fair value of the reporting unit's goodwill. If Applied determines that the carrying value of a reporting unit's goodwill exceeds its implied fair value, Applied would record an impairment charge equal to the difference.

Applied determines the fair value of each reporting unit based on a weighting of an income and a market approach. Applied bases the fair value estimates on assumptions that it believes to be reasonable but that are unpredictable and inherently uncertain. Under the income approach, Applied estimates the fair value based on discounted cash flow method.

The estimates used in the impairment testing are consistent with the discrete forecasts that Applied uses to manage its business, and considers any significant developments during the period. Under the discounted cash flow method, cash flows beyond the discrete forecasts are estimated using a terminal growth rate, which considers the long-term earnings growth rate specific to the reporting units. The estimated future cash flows are discounted to present value using each reporting unit's weighted average cost of capital. The weighted average cost of capital measures a reporting unit's cost of debt and equity financing weighted by the percentage of debt and equity in a reporting unit's target capital structure. In addition, the weighted average cost of capital is derived using both known and estimated market metrics, and is adjusted to reflect both the timing and risks associated with the estimated cash flows. The tax rate used in the discounted cash flow method is the median tax rate of comparable companies and reflects Applied's current international structure, which is consistent with the market participant perspective. Under the market approach, Applied uses the guideline company method which applies market multiples to forecasted revenues and earnings before interest, taxes, depreciation and amortization. Applied uses market multiples that are consistent with comparable publicly-traded companies and considers each reporting unit's size, growth and profitability relative to its comparable companies.

Management uses significant judgment when assessing goodwill for potential impairment, especially in emerging markets. Indicators of potential impairment include, but are not limited to, challenging economic conditions, an unfavorable industry or economic environment or other severe decline in market conditions. Such conditions could have the effect of changing one of the critical assumptions or estimates used for the fair value calculation, resulting in an unexpected goodwill impairment charge, which could have a material adverse effect on Applied's business, financial condition and results of operations. See Note 9 of Notes to Consolidated Financial Statements for additional discussion of goodwill impairment.

***Income Taxes***

Applied's provision for income taxes and effective tax rate are affected by the geographical composition of pre-tax income which includes jurisdictions with differing tax rates, conditional reduced tax rates and other income tax incentives. It is also affected by events that are not consistent from period to period, such as changes to income tax laws and the resolution of prior years' income tax filings.

Applied recognizes a current tax liability for the estimated amount of income tax payable on tax returns for the current fiscal year. Deferred tax assets and liabilities are recognized for the estimated future tax effects of temporary differences between the book and tax bases of assets and liabilities. Deferred tax assets are also recognized for net operating loss and tax credit carryforwards. Deferred tax assets are offset by a valuation allowance to the extent it is more likely than not that they are not expected to be realized.

Applied recognizes tax benefits from uncertain tax positions only if it is more likely than not that the tax position will be sustained upon examination by the taxing authorities based on the technical merits of the position. The tax benefits recognized from such positions are estimated based on the largest benefit that has a greater than 50% likelihood of being realized upon ultimate settlement. Any changes in judgment related to uncertain tax positions are recognized in Applied's provision for income taxes in the quarter in which such change occurs. Interest and penalties related to uncertain tax positions are recognized in Applied's provision for income taxes.

The calculation of Applied's provision for income taxes and effective tax rate involves significant judgment in estimating the impact of uncertainties in the application of complex tax laws. Resolution of these uncertainties in a manner inconsistent with Applied's expectations could have an adverse material impact on Applied's results of operations and financial condition.

**Non-GAAP Adjusted Financial Results**

Management uses non-GAAP adjusted financial measures to evaluate the Company's operating and financial performance and for planning purposes, and as performance measures in its executive compensation program. Applied believes these measures enhance an overall understanding of its performance and investors' ability to review the Company's business from the same perspective as the Company's management and facilitate comparisons of this period's results with prior periods on a consistent basis by excluding items that management does not believe are indicative of Applied's ongoing operating performance.

The non-GAAP adjusted financial measures presented below are adjusted to exclude the impact of certain costs, expenses, gains and losses, including certain items related to mergers and acquisitions; restructuring charges and any associated adjustments; impairments of assets, or investments; gain or loss on sale of strategic investments; certain income tax items and other discrete adjustments. Additionally, fiscal 2018 non-GAAP results exclude estimated discrete income tax expense items associated with recent U.S. tax legislation. Reconciliations of these non-GAAP measures to the most directly comparable financial measures calculated and presented in accordance with GAAP are provided in the financial tables presented below. There are limitations in using non-GAAP financial measures because the non-GAAP financial measures are not prepared in accordance with generally accepted accounting principles, may be different from non-GAAP financial measures used by other companies, and may exclude certain items that may have a material impact upon our reported financial results. The presentation of this additional information is not meant to be considered in isolation or as a substitute for the directly comparable financial measures prepared in accordance with GAAP.

The following tables present a reconciliation of the GAAP and non-GAAP adjusted consolidated results for the past three fiscal years:

APPLIED MATERIALS, INC.
UNAUDITED RECONCILIATION OF GAAP TO NON-GAAP ADJUSTED RESULTS

| (In millions, except percentages) | 2018 | 2017 | 2016 |
|---|---|---|---|
| Non-GAAP Adjusted Gross Profit | | | |
| Reported gross profit - GAAP basis | $ 7,817 | $ 6,532 | $ 4,511 |
| Certain items associated with acquisitions[1] | 179 | 172 | 167 |
| Inventory reversals related to restructuring[2] | — | — | (2) |
| Non-GAAP adjusted gross profit | $ 7,996 | $ 6,704 | $ 4,676 |
| Non-GAAP adjusted gross margin | 46.3% | 46.1% | 43.2% |
| Non-GAAP Adjusted Operating Income | | | |
| Reported operating income - GAAP basis | $ 4,796 | $ 3,868 | $ 2,152 |
| Certain items associated with acquisitions[1] | 197 | 191 | 188 |
| Acquisition integration and deal costs | 5 | 3 | 2 |
| Reversals related to restructuring, net[2] | — | — | (3) |
| Other gains, losses or charges, net | — | (12) | 8 |
| Non-GAAP adjusted operating income | $ 4,998 | $ 4,050 | $ 2,347 |
| Non-GAAP adjusted operating margin | 29.0% | 27.9% | 21.7% |
| Non-GAAP Adjusted Net Income | | | |
| Reported net income - GAAP basis[4] | $ 3,313 | $ 3,434 | $ 1,721 |
| Certain items associated with acquisitions[1] | 197 | 191 | 188 |
| Acquisition integration and deal costs | 5 | 3 | 2 |
| Reversals related to restructuring, net[2] | — | — | (3) |
| Loss on early extinguishment of debt | — | 5 | 5 |
| Impairment (gain on sale) of strategic investments, net | (25) | (3) | 3 |
| Other gains, losses or charges, net | — | (12) | 8 |
| Income tax effect of changes in applicable U.S. tax laws[3] | 1,112 | — | — |
| Resolution of prior years' income tax filings, reinstatement of federal R&D tax credit and other tax items | (26) | (79) | 45 |
| Income tax effect of non-GAAP adjustments[5] | (7) | (14) | (19) |
| Non-GAAP adjusted net income | $ 4,569 | $ 3,525 | $ 1,950 |

1 These items are incremental charges attributable to completed acquisitions, consisting of amortization of purchased intangible assets.
2 Results for fiscal 2016 included adjustments associated with the cost reductions in the solar business.
3 Charges to income tax provision related to a one-time transition tax and a decrease in U.S. deferred tax assets as a result of the recent U.S. tax legislation.
4 Amounts for fiscal 2017 included the recognition of the previously unrecognized foreign tax credits.
5 Adjustment to provision for income taxes related to non-GAAP adjustments reflected in income before income taxes.

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APPLIED MATERIALS, INC.
UNAUDITED RECONCILIATION OF GAAP TO NON-GAAP ADJUSTED RESULTS

| **(In millions, except per share amounts)** | **2018** | **2017** | **2016** |
|---|---|---|---|
| Non-GAAP Adjusted Earnings Per Diluted Share | | | |
| Reported earnings per diluted share - GAAP basis | $ 3.23 | $ 3.17 | $ 1.54 |
| Certain items associated with acquisitions | 0.18 | 0.16 | 0.16 |
| Impairment (gain on sale) of strategic investments, net | (0.02) | — | — |
| Income tax effect of change in applicable U.S. tax laws | 1.08 | — | — |
| Other gains, losses or charges, net | — | (0.01) | 0.01 |
| Resolution of prior years' income tax filings and other tax items | (0.02) | (0.07) | 0.04 |
| Non-GAAP adjusted earnings per diluted share | $ 4.45 | $ 3.25 | $ 1.75 |
| Weighted average number of diluted shares | 1,026 | 1,084 | 1,116 |

The following table presents a reconciliation of the GAAP and non-GAAP adjusted segment results for the past three fiscal years:

APPLIED MATERIALS, INC.
UNAUDITED RECONCILIATION OF GAAP TO NON-GAAP ADJUSTED RESULTS

| (In millions, except percentages) | 2018 | 2017 | 2016 |
|---|---|---|---|
| Semiconductor Systems Non-GAAP Adjusted Operating Income | | | |
| Reported operating income - GAAP basis | $ 3,634 | $ 3,173 | $ 1,807 |
| Certain items associated with acquisitions[1] | 183 | 184 | 184 |
| Non-GAAP adjusted operating income | $ 3,817 | $ 3,357 | $ 1,991 |
| Non-GAAP adjusted operating margin | 35.0% | 35.3% | 29.0% |
| AGS Non-GAAP Adjusted Operating Income | | | |
| Reported operating income - GAAP basis | $ 1,102 | $ 817 | $ 682 |
| Certain items associated with acquisitions[1] | — | 1 | 1 |
| Acquisition integration costs | 2 | 3 | — |
| Non-GAAP adjusted operating income | $ 1,104 | $ 821 | $ 683 |
| Non-GAAP adjusted operating margin | 29.4% | 27.2% | 26.4% |
| Display and Adjacent Markets Non-GAAP Adjusted Operating Income | | | |
| Reported operating income - GAAP basis | $ 679 | $ 502 | $ 245 |
| Certain items associated with acquisitions[1] | 14 | 5 | — |
| Acquisition integration costs | 1 | — | — |
| Non-GAAP adjusted operating income | $ 694 | $ 507 | $ 245 |
| Non-GAAP adjusted operating margin | 27.8% | 26.7% | 20.3% |

1 These items are incremental charges attributable to completed acquisitions, consisting of amortization of purchased intangible assets.

Note: The reconciliation of GAAP and non-GAAP adjusted segment results above does not include certain revenues, costs of products sold and operating expenses that are reported within corporate and other and included in consolidated operating income.

**Item 7A:** ***Quantitative and Qualitative Disclosures About Market Risk***

Applied is exposed to interest rate risk related to its investment portfolio and debt issuances. Applied's investment portfolio includes fixed-income securities with a fair value of approximately $2.0 billion at October 28, 2018. These securities are subject to interest rate risk and will decline in value if interest rates increase. Based on Applied's investment portfolio at October 28, 2018, an immediate 100 basis point increase in interest rates would result in a decrease in the fair value of the portfolio of approximately $26 million. While an increase in interest rates reduces the fair value of the investment portfolio, Applied will not realize the losses in the consolidated statement of operations unless the individual fixed-income securities are sold prior to recovery or the loss is determined to be other-than-temporary. At October 28, 2018, the aggregate principal and estimated fair values amounts of long-term debt issued by Applied were both $5.4 billion. A hypothetical decrease in interest rates of 100 basis points would result in an increase in the fair value of Applied's long-term debt issuances of approximately $497 million at October 28, 2018.

Certain operations of Applied are conducted in foreign currencies, such as Japanese yen, euro, Israeli shekel and Taiwanese dollar. Applied enters into currency forward exchange and option contracts to hedge a portion of, but not all, existing and anticipated foreign currency denominated transactions generally expected to occur within the next 24 months. Gains and losses on these contracts are generally recognized in income at the time that the related transactions being hedged are recognized. Because the effect of movements in currency exchange rates on currency forward exchange and option contracts generally offsets the related effect on the underlying items being hedged, these financial instruments are not expected to subject Applied to risks that would otherwise result from changes in currency exchange rates. Applied does not use derivative financial instruments for trading or speculative purposes.

**Item 8:** ***Financial Statements and Supplementary Data***

The consolidated financial statements required by this Item are set forth on the pages indicated at Item 15(a).

**Item 9:** ***Changes in and Disagreements with Accountants on Accounting and Financial Disclosure***

None.

**Item 9A:** ***Controls and Procedures***

**Disclosure Controls and Procedures**

As of the end of the period covered by this report, management of Applied conducted an evaluation, under the supervision and with the participation of Applied's Chief Executive Officer and Chief Financial Officer, of the effectiveness of Applied's disclosure controls and procedures, as such term is defined in Rule 13a-15(e) of the Securities Exchange Act of 1934 (the Exchange Act). Based upon that evaluation, Applied's Chief Executive Officer and Chief Financial Officer concluded that Applied's disclosure controls and procedures were effective as of the end of the period covered by this report in ensuring that information required to be disclosed was recorded, processed, summarized and reported within the time periods specified in the SEC's rules and forms, and to provide reasonable assurance that information required to be disclosed by Applied in such reports is accumulated and communicated to the Company's management, including its Chief Executive Officer and Chief Financial Officer, as appropriate to allow timely decisions regarding required disclosure.

**Management's Report on Internal Control over Financial Reporting**

Applied's management is responsible for establishing and maintaining adequate internal control over financial reporting, as such term is defined in Rule 13a-15(f) of the Exchange Act. Under the supervision and with the participation of Applied's Chief Executive Officer and Chief Financial Officer, management of Applied conducted an evaluation of the effectiveness of Applied's internal control over financial reporting based upon the framework in "Internal Control — Integrated Framework (2013)" issued by the Committee of Sponsoring Organizations of the Treadway Commission. Based on that evaluation, Applied's management concluded that Applied's internal control over financial reporting was effective as of October 28, 2018.

KPMG LLP, an independent registered public accounting firm, has audited the consolidated financial statements included in this Form 10-K and, as part of the audit, has issued a report, included herein, on the effectiveness of Applied's internal control over financial reporting as of October 28, 2018.

**Changes in Internal Control over Financial Reporting**

During the fourth quarter of fiscal 2018, there were no changes in the internal control over financial reporting that materially affected, or are reasonably likely to materially affect, Applied's internal control over financial reporting.

**Inherent Limitations of Disclosure Controls and Procedures and Internal Control over Financial Reporting**

It should be noted that any system of controls, however well designed and operated, can provide only reasonable, and not absolute, assurance that the objectives of the system will be met. In addition, the design of any control system is based in part upon certain assumptions about the likelihood of future events.

**Item 9B:** ***Other Information***

None

## PART III

**Item 10:** ***Directors, Executive Officers and Corporate Governance***

Except for the information regarding executive officers required by Item 401 of Regulation S-K (which is included in Part I, Item 1 of this Annual Report on Form 10-K, under "Executive Officers of the Registrant") and code of ethics (which is set forth below), the information required by this item will be provided in accordance with Instruction G(3) to Form 10-K no later than February 25, 2019.

Applied has implemented the Standards of Business Conduct, a code of ethics with which every person who works for Applied and every member of the Board of Directors is expected to comply. If any substantive amendments are made to the Standards of Business Conduct or any waiver is granted, including any implicit waiver, from a provision of the code to Applied's Chief Executive Officer, Chief Financial Officer or Chief Accounting Officer, Applied will disclose the nature of such amendment or waiver on its website or in a report on Form 8-K. The above information, including the Standards of Business Conduct, is available on Applied's website under the Corporate Governance section at *http://www.appliedmaterials.com/company/investor-relations/governance.* This website address is intended to be an inactive, textual reference only. None of the materials on, or accessible through, this website is part of this report or is incorporated by reference herein.

**Item 11:** ***Executive Compensation***

The information required by this Item will be provided in accordance with Instruction G(3) to Form 10-K no later than February 25, 2019.

**Item 12:** ***Security Ownership of Certain Beneficial Owners and Management and Related Stockholder Matters***

Except for the information regarding securities authorized for issuance under equity compensation plans (which is set forth below), the information required by this Item will be provided in accordance with Instruction G(3) to Form 10-K no later than February 25, 2019.

The following table summarizes information with respect to equity awards under Applied's equity compensation plans as of October 28, 2018:

**Equity Compensation Plan Information**

| Plan Category | (a) Number of Securities to be Issued Upon Exercise of Outstanding Options, Warrants and Rights(1) | (b) Weighted Average Exercise Price of Outstanding Options, Warrants and Rights(2) | (c) Number of Securities Available for Future Issuance Under Equity Compensation Plans (Excluding Securities Reflected in Column(a)) |
|---|---|---|---|
| | | (In millions, except prices) | |
| Equity compensation plans approved by security holders | 20 | $ 15.06 | 94 (3) |
| Equity compensation plans not approved by security holders | — (4) | $ 7.27 | 4 (5) |
| Total | 20 | $ 15.02 | 98 |

(1) Includes only options, restricted stock units and performance shares outstanding under Applied's equity compensation plans, as no stock warrants or other rights were outstanding as of October 28, 2018.

(2) The weighted average exercise price calculation does not take into account any restricted stock units or performance shares.

(3) Includes 13 million shares of Applied common stock available for future issuance under the Applied Materials, Inc. Employees' Stock Purchase Plan. Of these 13 million shares, 1 million are subject to purchase during the purchase period in effect as of October 28, 2018.

(4) Includes options to purchase 5 thousand shares of Applied common stock assumed through various mergers and acquisitions, after giving effect to the applicable exchange ratios. The assumed options had a weighted average exercise price of $7.27 per share. No further shares are available for issuance under the plans under which these assumed awards were granted.

(5) Includes 4 million shares of Applied common stock available for future issuance under the Applied Materials, Inc. Stock Purchase Plan for Offshore Employees. Of these 4 million shares, 1 million are subject to purchase during the purchase period in effect as of October 28, 2018.

Applied has the following equity compensation plans that have not been approved by stockholders:

*Stock Purchase Plan for Offshore Employees.* The Stock Purchase Plan for Offshore Employees (the Offshore ESPP) was adopted effective as of October 16, 1995 for the benefit of employees of Applied's participating affiliates. The Offshore ESPP provides for the grant of options to purchase shares of Applied common stock through payroll deductions pursuant to one or more offerings. The administrator of the Offshore ESPP (the Board of Directors of Applied or a committee appointed by the Board) determines the terms and conditions of all options prior to the start of an offering, including the purchase price of shares, the number of shares covered by the option and when the option may be exercised. All options granted as part of an offering must be granted on the same date. As of October 28, 2018, a total of 36 million shares have been authorized for issuance under the Offshore ESPP, and 4 million shares remain available for issuance.

Applied Materials Profit Sharing Scheme. The Applied Materials Profit Sharing Scheme was adopted effective July 3, 1996 to enable employees of Applied Materials Ireland Limited and its participating subsidiaries to purchase Applied common stock at 100% of fair market value on the purchase date. Under this plan, eligible employees may elect to forego a certain portion of their base salary and certain bonuses they have earned and that otherwise would be payable in cash to purchase shares of Applied common stock at full fair market value. Since the eligible employees pay full fair market value for the shares, there is no reserved amount of shares under this plan and, accordingly, the table above does not include any set number of shares available for future issuance under the plan.

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**Item 13:** ***Certain Relationships and Related Transactions, and Director Independence***

The information required by this Item will be provided in accordance with Instruction G(3) to Form 10-K no later than February 25, 2019.

**Item 14:** ***Principal Accounting Fees and Services***

The information required by this Item will be provided in accordance with Instruction G(3) to Form 10-K no later than February 25, 2019.

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## PART IV

**Item 15:** ***Exhibits, Financial Statement Schedules***

(a) The following documents are filed as part of this Annual Report on Form 10-K:


| | Page Number |
|---|---|
| (1) Financial Statements: | |
| Reports of Independent Registered Public Accounting Firm | 54 |
| Consolidated Statements of Operations | 56 |
| Consolidated Statements of Comprehensive Income | 57 |
| Consolidated Balance Sheets | 58 |
| Consolidated Statements of Stockholders' Equity | 59 |
| Consolidated Statements of Cash Flows | 60 |
| Notes to Consolidated Financial Statements | 61 |
| (2) Exhibits: | |
| The exhibits listed in the accompanying Index to Exhibits are filed or incorporated by reference as part of this Annual Report on Form 10-K | 99 |


All other schedules are omitted because they are not applicable or the required information is shown in the Consolidated Financial Statements or Notes thereto.

**Item 16:** ***Form 10-K Summary***

None.

## REPORT OF INDEPENDENT REGISTERED PUBLIC ACCOUNTING FIRM

The Board of Directors and Stockholders
Applied Materials, Inc.:

*Opinion on the Consolidated Financial Statements*

We have audited the accompanying consolidated balance sheets of Applied Materials, Inc. and subsidiaries (the Company) as of October 28, 2018 and October 29, 2017, the related consolidated statements of operations, comprehensive income, stockholders' equity, and cash flows for each of the years in the three-year period ended October 28, 2018, and the related notes (collectively, the consolidated financial statements). In our opinion, the consolidated financial statements present fairly, in all material respects, the financial position of the Company as of October 28, 2018 and October 29, 2017, and the results of its operations and its cash flows for each of the years in the three-year period ended October 28, 2018, in conformity with U.S. generally accepted accounting principles.

We also have audited, in accordance with the standards of the Public Company Accounting Oversight Board (United States) (PCAOB), the Company's internal control over financial reporting as of October 28, 2018, based on criteria established in *Internal Control - Integrated Framework (2013)* issued by the Committee of Sponsoring Organizations of the Treadway Commission, and our report dated December 13, 2018 expressed an unqualified opinion on the effectiveness of the Company's internal control over financial reporting.

*Basis for Opinion*

These consolidated financial statements are the responsibility of the Company's management. Our responsibility is to express an opinion on these consolidated financial statements based on our audits. We are a public accounting firm registered with the PCAOB and are required to be independent with respect to the Company in accordance with the U.S. federal securities laws and the applicable rules and regulations of the Securities and Exchange Commission and the PCAOB.

We conducted our audits in accordance with the standards of the PCAOB. Those standards require that we plan and perform the audit to obtain reasonable assurance about whether the consolidated financial statements are free of material misstatement, whether due to error or fraud. Our audits included performing procedures to assess the risks of material misstatement of the consolidated financial statements, whether due to error or fraud, and performing procedures that respond to those risks. Such procedures included examining, on a test basis, evidence regarding the amounts and disclosures in the consolidated financial statements. Our audits also included evaluating the accounting principles used and significant estimates made by management, as well as evaluating the overall presentation of the consolidated financial statements. We believe that our audits provide a reasonable basis for our opinion.

/s/ KPMG LLP

KPMG LLP

We have served as the Company's auditor since 2004.

Santa Clara, California
December 13, 2018

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## REPORT OF INDEPENDENT REGISTERED PUBLIC ACCOUNTING FIRM

The Board of Directors and Stockholders
Applied Materials, Inc.:

*Opinion on Internal Control Over Financial Reporting*

We have audited Applied Materials, Inc. and subsidiaries (the Company) internal control over financial reporting as of October 28, 2018, based on criteria established in *Internal Control - Integrated Framework (2013)* issued by the Committee of Sponsoring Organizations of the Treadway Commission. In our opinion, the Company maintained, in all material respects, effective internal control over financial reporting as of October 28, 2018, based on criteria established in *Internal Control - Integrated Framework (2013)* issued by the Committee of Sponsoring Organizations of the Treadway Commission.

We also have audited, in accordance with the standards of the Public Company Accounting Oversight Board (United States) (PCAOB), the consolidated balance sheets of the Company as of October 28, 2018 and October 29, 2017, the related consolidated statements of operations, comprehensive income, stockholders' equity, and cash flows for each of the years in the three-year period ended October 28, 2018, and the related notes (collectively, the consolidated financial statements), and our report dated December 13, 2018 expressed an unqualified opinion on those consolidated financial statements.

*Basis for Opinion*

The Company's management is responsible for maintaining effective internal control over financial reporting and for its assessment of the effectiveness of internal control over financial reporting, included in the accompanying Management's Report on Internal Control over Financial Reporting in Item 9A. Our responsibility is to express an opinion on the Company's internal control over financial reporting based on our audit. We are a public accounting firm registered with the PCAOB and are required to be independent with respect to the Company in accordance with the U.S. federal securities laws and the applicable rules and regulations of the Securities and Exchange Commission and the PCAOB.

We conducted our audit in accordance with the standards of the PCAOB. Those standards require that we plan and perform the audit to obtain reasonable assurance about whether effective internal control over financial reporting was maintained in all material respects. Our audit of internal control over financial reporting included obtaining an understanding of internal control over financial reporting, assessing the risk that a material weakness exists, and testing and evaluating the design and operating effectiveness of internal control based on the assessed risk. Our audit also included performing such other procedures as we considered necessary in the circumstances. We believe that our audit provides a reasonable basis for our opinion.

*Definition and Limitations of Internal Control Over Financial Reporting*

A company's internal control over financial reporting is a process designed to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles. A company's internal control over financial reporting includes those policies and procedures that (1) pertain to the maintenance of records that, in reasonable detail, accurately and fairly reflect the transactions and dispositions of the assets of the company; (2) provide reasonable assurance that transactions are recorded as necessary to permit preparation of financial statements in accordance with generally accepted accounting principles, and that receipts and expenditures of the company are being made only in accordance with authorizations of management and directors of the company; and (3) provide reasonable assurance regarding prevention or timely detection of unauthorized acquisition, use, or disposition of the company's assets that could have a material effect on the financial statements.

Because of its inherent limitations, internal control over financial reporting may not prevent or detect misstatements. Also, projections of any evaluation of effectiveness to future periods are subject to the risk that controls may become inadequate because of changes in conditions, or that the degree of compliance with the policies or procedures may deteriorate.

/s/ KPMG LLP

KPMG LLP

Santa Clara, California
December 13, 2018

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## APPLIED MATERIALS, INC.

## CONSOLIDATED STATEMENTS OF OPERATIONS

**(In millions, except per share amounts)**

| Fiscal Year | 2018 | 2017 | 2016 |
|---|---|---|---|
| Net sales | $ 17,253 | $ 14,537 | $ 10,825 |
| Cost of products sold | 9,436 | 8,005 | 6,314 |
| Gross profit | 7,817 | 6,532 | 4,511 |
| Operating expenses: | | | |
| Research, development and engineering | 2,019 | 1,774 | 1,540 |
| Marketing and selling | 521 | 456 | 429 |
| General and administrative | 481 | 434 | 390 |
| Total operating expenses | 3,021 | 2,664 | 2,359 |
| Income from operations | 4,796 | 3,868 | 2,152 |
| Interest expense | 234 | 198 | 155 |
| Interest and other income, net | 132 | 61 | 16 |
| Income before income taxes | 4,694 | 3,731 | 2,013 |
| Provision for income taxes | 1,381 | 297 | 292 |
| Net income | $ 3,313 | $ 3,434 | $ 1,721 |
| Earnings per share: | | | |
| Basic | $ 3.27 | $ 3.20 | $ 1.56 |
| Diluted | $ 3.23 | $ 3.17 | $ 1.54 |
| Weighted average number of shares: | | | |
| Basic | 1,013 | 1,073 | 1,107 |
| Diluted | 1,026 | 1,084 | 1,116 |

See accompanying Notes to Consolidated Financial Statements.

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**APPLIED MATERIALS, INC.**

**CONSOLIDATED STATEMENTS OF COMPREHENSIVE INCOME**

**(In millions)**

| Fiscal Year | 2018 | 2017 | 2016 |
|---|---|---|---|
| Net income | $ 3,313 | $ 3,434 | $ 1,721 |
| Other comprehensive income (loss), net of tax: | | | |
| Change in unrealized net gain on investments | (51) | 23 | 16 |
| Change in unrealized net loss on derivative instruments | 4 | 7 | (3) |
| Change in defined and postretirement benefit plans | (17) | 21 | (36) |
| Other comprehensive income (loss), net of tax | (64) | 51 | (23) |
| Comprehensive income | $ 3,249 | $ 3,485 | $ 1,698 |

See accompanying Notes to Consolidated Financial Statements.

**APPLIED MATERIALS, INC.**

**CONSOLIDATED BALANCE SHEETS**

**(In millions, except per share amounts)**

| | October 28, 2018 | October 29, 2017 |
|---|---|---|
| **ASSETS** | | |
| Current assets: | | |
| Cash and cash equivalents | $ 3,440 | $ 5,010 |
| Short-term investments | 590 | 2,266 |
| Accounts receivable, net | 2,565 | 2,338 |
| Inventories | 3,722 | 2,930 |
| Other current assets | 430 | 374 |
| Total current assets | 10,747 | 12,918 |
| Long-term investments | 1,568 | 1,143 |
| Property, plant and equipment, net | 1,407 | 1,066 |
| Goodwill | 3,368 | 3,368 |
| Purchased technology and other intangible assets, net | 213 | 412 |
| Deferred income taxes and other assets | 470 | 512 |
| Total assets | $ 17,773 | $ 19,419 |
| **LIABILITIES AND STOCKHOLDERS' EQUITY** | | |
| Current liabilities: | | |
| Accounts payable and accrued expenses | $ 2,721 | $ 2,450 |
| Customer deposits and deferred revenue | 1,347 | 1,665 |
| Total current liabilities | 4,068 | 4,115 |
| Long-term debt | 5,309 | 5,304 |
| Income taxes payable | 1,254 | 392 |
| Other liabilities | 303 | 259 |
| Total liabilities | 10,934 | 10,070 |
| Commitments and contingencies (Note 14) | | |
| Stockholders' equity: | | |
| Preferred stock: $.01 par value per share; 1 shares authorized; no shares issued | — | — |
| Common stock: $.01 par value per share; 2,500 shares authorized; 967 and 1,060 shares outstanding at 2018 and 2017, respectively | 10 | 11 |
| Additional paid-in capital | 7,274 | 7,056 |
| Retained earnings | 20,874 | 18,258 |
| Treasury stock: 1,019 and 917 shares at 2018 and 2017, respectively | (21,194) | (15,912) |
| Accumulated other comprehensive loss | (125) | (64) |
| Total stockholders' equity | 6,839 | 9,349 |
| Total liabilities and stockholders' equity | $ 17,773 | $ 19,419 |

See accompanying Notes to Consolidated Financial Statements.

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## APPLIED MATERIALS, INC.

## CONSOLIDATED STATEMENTS OF STOCKHOLDERS' EQUITY

**(In millions)**

| | Common Stock | | Additional Paid-In Capital | Retained Earnings | Treasury Stock | | Accumulated Other Comprehensive Income (Loss) | Total |
|---|---|---|---|---|---|---|---|---|
| | Shares | Amount | | | Shares | Amount | | |
| **Balance at October 25, 2015** | 1,160 | $ 11 | $ 6,575 | $ 13,967 | 793 | $ (12,848) | $ (92) | $ 7,613 |
| Net income | — | — | — | 1,721 | — | — | — | 1,721 |
| Other comprehensive loss, net of tax | — | — | — | — | — | — | (23) | (23) |
| Dividends | — | — | — | (436) | — | — | — | (436) |
| Share-based compensation | — | — | 201 | — | — | — | — | 201 |
| Issuance under stock plans, net of a tax benefit of $23 and other | 14 | — | 33 | — | — | — | — | 33 |
| Common stock repurchases | (96) | — | — | — | 96 | (1,892) | — | (1,892) |
| **Balance at October 30, 2016** | 1,078 | $ 11 | $ 6,809 | $ 15,252 | 889 | $ (14,740) | $ (115) | $ 7,217 |
| Net income | — | — | — | 3,434 | — | — | — | 3,434 |
| Other comprehensive income, net of tax | — | — | — | — | — | — | 51 | 51 |
| Dividends | — | — | — | (428) | — | — | — | (428) |
| Share-based compensation | — | — | 220 | — | — | — | — | 220 |
| Issuance under stock plans, net of a tax benefit of $55 and other | 10 | — | 27 | — | — | — | — | 27 |
| Common stock repurchases | (28) | — | — | — | 28 | (1,172) | — | (1,172) |
| **Balance at October 29, 2017** | 1,060 | $ 11 | $ 7,056 | $ 18,258 | 917 | $ (15,912) | $ (64) | $ 9,349 |
| Adoption of new accounting standard (a) | — | — | — | (3) | — | — | 3 | — |
| Net income | — | — | — | 3,313 | — | — | — | 3,313 |
| Other comprehensive loss, net of tax | — | — | — | — | — | — | (64) | (64) |
| Dividends | — | — | — | (694) | — | — | — | (694) |
| Share-based compensation | — | — | 258 | — | — | — | — | 258 |
| Issuance under stock plans | 9 | — | (40) | — | — | — | — | (40) |
| Common stock repurchases | (102) | (1) | — | — | 102 | (5,282) | — | (5,283) |
| **Balance at October 28, 2018** | 967 | $ 10 | $ 7,274 | $ 20,874 | 1,019 | $ (21,194) | $ (125) | $ 6,839 |

(a) - Represents the reclassification adjustment related to the early adoption of Accounting Standards Update (ASU) 2018-02 *Reclassification of Certain Tax Effects from Accumulated Other Comprehensive Income.* See Note 1.

See accompanying Notes to Consolidated Financial Statements.

**APPLIED MATERIALS, INC.**

**CONSOLIDATED STATEMENTS OF CASH FLOWS**

**(In millions)**

| Fiscal Year | 2018 | 2017 | 2016 |
|---|---|---|---|
| Cash flows from operating activities: | | | |
| Net income | $ 3,313 | $ 3,434 | $ 1,721 |
| Adjustments required to reconcile net income to cash provided by operating activities: | | | |
| Depreciation and amortization | 457 | 407 | 389 |
| Deferred income taxes | 94 | (11) | 21 |
| Other | 4 | (9) | 38 |
| Share-based compensation | 258 | 220 | 201 |
| Changes in operating assets and liabilities, net of amounts acquired: | | | |
| Accounts receivable | (226) | (37) | (542) |
| Inventories | (792) | (879) | (216) |
| Other current and non-current assets | (93) | (157) | 30 |
| Accounts payable and accrued expenses | 179 | 370 | 184 |
| Customer deposits and deferred revenue | (318) | 289 | 611 |
| Income taxes payable | 886 | 121 | 173 |
| Other liabilities | 25 | 41 | (44) |
| Cash provided by operating activities | 3,787 | 3,789 | 2,566 |
| Cash flows from investing activities: | | | |
| Capital expenditures | (622) | (345) | (253) |
| Cash paid for acquisitions, net of cash acquired | (6) | (68) | (16) |
| Proceeds from sales and maturities of investments | 3,276 | 2,743 | 1,234 |
| Purchases of investments | (2,077) | (4,856) | (1,390) |
| Cash provided by (used in) investing activities | 571 | (2,526) | (425) |
| Cash flows from financing activities: | | | |
| Debt borrowings, net of issuance costs | — | 2,176 | — |
| Debt repayments | — | (205) | (1,207) |
| Proceeds from common stock issuances | 124 | 97 | 88 |
| Common stock repurchases | (5,283) | (1,172) | (1,892) |
| Tax withholding payments for vested equity awards | (164) | (125) | (77) |
| Payments of dividends to stockholders | (605) | (430) | (444) |
| Cash provided by (used in) financing activities | (5,928) | 341 | (3,532) |
| Increase (decrease) in cash and cash equivalents | (1,570) | 1,604 | (1,391) |
| Cash and cash equivalents — beginning of year | 5,010 | 3,406 | 4,797 |
| Cash and cash equivalents — end of year | $ 3,440 | $ 5,010 | $ 3,406 |
| Supplemental cash flow information: | | | |
| Cash payments for income taxes | $ 300 | $ 194 | $ 157 |
| Cash refunds from income taxes | $ 63 | $ 61 | $ 113 |
| Cash payments for interest | $ 219 | $ 186 | $ 151 |

See accompanying Notes to Consolidated Financial Statements.

**APPLIED MATERIALS, INC.**

**NOTES TO CONSOLIDATED FINANCIAL STATEMENTS**

## Note 1 Summary of Significant Accounting Policies

*Principles of Consolidation and Basis of Presentation*

The consolidated financial statements include the accounts of Applied Materials, Inc. and its subsidiaries (Applied or the Company) after elimination of intercompany balances and transactions. All references to a fiscal year apply to Applied's fiscal year which ends on the last Sunday in October. Fiscal 2018, 2017 and 2016 contained 52, 52, and 53 weeks, respectively. Each fiscal quarter of 2018 and 2017 contained 13 weeks. The first fiscal quarter of 2016 contained 14 weeks, while the second, third and fourth quarters of fiscal 2016 contained 13 weeks.

Certain prior year amounts have been reclassified to conform to current year presentation.

*Use of Estimates*

The preparation of financial statements in conformity with accounting principles generally accepted in the United States of America requires management to make judgments, estimates and assumptions that affect the amounts reported in the financial statements and accompanying notes. Actual results could differ materially from those estimates. On an ongoing basis, Applied evaluates its estimates, including those related to accounts receivable and sales allowances, fair values of financial instruments, inventories, intangible assets and goodwill, useful lives of intangible assets and property and equipment, fair values of share-based awards, and income taxes, among others. Applied bases its estimates on historical experience and on various other assumptions that are believed to be reasonable, the results of which form the basis for making judgments about the carrying values of assets and liabilities.

*Cash Equivalents*

All highly-liquid investments with a remaining maturity of three months or less at the time of purchase are considered to be cash equivalents. Cash equivalents consist primarily of investments in institutional money market funds.

*Investments*

All of Applied's investments, except equity investments held in privately-held companies, are classified as available-for-sale at the respective balance sheet dates. Investments classified as available-for-sale are recorded at fair value based upon quoted market prices, and any temporary difference between the cost and fair value of an investment is presented as a separate component of accumulated other comprehensive income (loss). The specific identification method is used to determine the gains and losses on investments. Interest earned on cash and investments, as well as realized gains and losses on sale of securities, are included in interest and other income, net in the accompanying Consolidated Statements of Operations.

Equity investments in privately-held companies are generally accounted for under the cost method of accounting and are periodically assessed for other-than-temporary impairment when an event or circumstance indicates that an other-than-temporary decline in value may have occurred.

*Allowance for Doubtful Accounts*

Applied maintains an allowance for doubtful accounts for estimated losses resulting from the inability of its customers to make required payments. This allowance is based on historical experience, credit evaluations, specific customer collection history and any customer-specific issues Applied has identified. Changes in circumstances, such as an unexpected material adverse change in a major customer's ability to meet its financial obligation to Applied or its payment trends, may require Applied to further adjust its estimates of the recoverability of amounts due to Applied. Bad debt expense and any reversals are recorded in marketing and selling expenses in the Consolidated Statement of Operations.

*Inventories*

Inventories are stated at the lower of cost or net realizable value, with cost determined on a first-in, first-out (FIFO) basis. Applied adjusts inventory carrying value for estimated obsolescence equal to the difference between the cost of inventory and the estimated net realizable value based upon assumptions about future demand and market conditions. Applied fully writes down inventories and noncancelable purchase orders for inventory deemed obsolete. Applied performs periodic reviews of inventory items to identify excess inventories on hand by comparing on-hand balances to anticipated usage using recent historical activity as well as anticipated or forecasted demand. If estimates of customer demand diminish further or market conditions become less favorable than those projected by Applied, additional inventory adjustments may be required.

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**APPLIED MATERIALS, INC.**

**NOTES TO CONSOLIDATED FINANCIAL STATEMENTS — (Continued)**

*Property, Plant and Equipment*

Property, plant and equipment is stated at cost. Depreciation is provided over the estimated useful lives of the assets using the straight-line method. Estimated useful lives for financial reporting purposes are as follows: buildings and improvements, 3 to 30 years; demonstration and manufacturing equipment, 3 to 5 years; software, 3 to 5 years; and furniture, fixtures and other equipment, 3 to 5 years. Land improvements are amortized over the shorter of 15 years or the estimated useful life. Leasehold improvements are amortized over the shorter of five years or the lease term.

*Intangible Assets*

Goodwill and indefinite-lived assets are not amortized, but are reviewed for impairment annually during the fourth quarter of each fiscal year and whenever events or changes in circumstances indicate that the carrying value of an asset may not be recoverable. Purchased technology and other intangible assets are presented at cost, net of accumulated amortization, and are amortized over their estimated useful lives of 1 to 15 years using the straight-line method.

*Long-Lived Assets*

Applied reviews long-lived assets for impairment whenever events or changes in circumstances indicate that the carrying amount of these assets or asset group may not be recoverable. Applied assesses these assets for impairment based on estimated future cash flows from these assets.

*Research, Development and Engineering Costs*

Research, development and engineering costs are expensed as incurred.

*Sales and Value Added Taxes*

Taxes collected from customers and remitted to governmental authorities are presented on a net basis in the accompanying Consolidated Statements of Operations.

*Warranty*

Applied provides for the estimated cost of warranty when revenue is recognized. Estimated warranty costs are determined by analyzing specific product, current and historical configuration statistics and regional warranty support costs. Applied's warranty obligation is affected by product and component failure rates, material usage and labor costs incurred in correcting product failures during the warranty period. If actual warranty costs differ substantially from Applied's estimates, revisions to the estimated warranty liability would be required.

*Income Taxes*

Applied recognizes a current tax liability for the estimated amount of income tax payable on tax returns for the current fiscal year. Deferred tax assets and liabilities are recognized for the estimated future tax effects of temporary differences between the book and tax bases of assets and liabilities. Deferred tax assets are also recognized for net operating loss and tax credit carryovers. Deferred tax assets are offset by a valuation allowance to the extent it is more likely than not that they are not expected to be realized.

Applied recognizes tax benefits from uncertain tax positions only if it is more likely than not that the tax position will be sustained upon examination by the taxing authorities, based on the technical merits of the position. The tax benefits recognized from such positions are estimated based on the largest benefit that has a greater than 50% likelihood of being realized upon ultimate settlement. Any changes in judgment related to uncertain tax positions are recognized in Applied's provision for income taxes in the quarter in which such change occurs. Interest and penalties related to uncertain tax positions are recognized in Applied's provision for income taxes.

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**APPLIED MATERIALS, INC.**

**NOTES TO CONSOLIDATED FINANCIAL STATEMENTS — (Continued)**

*Revenue Recognition*

Applied recognizes revenue when all four revenue recognition criteria have been met: persuasive evidence of an arrangement exists; delivery has occurred or services have been rendered; seller's price to buyer is fixed or determinable; and collectability is probable. Applied's shipping terms are customarily FOB Applied shipping point or equivalent terms. Applied's revenue recognition policy generally results in revenue recognition at the following points: (1) for all transactions where legal title passes to the customer upon shipment or delivery, Applied recognizes revenue upon passage of title for all products that have been demonstrated to meet product specifications prior to shipment; the portion of revenue associated with certain installation-related tasks is deferred, and that revenue is recognized upon completion of the installation-related tasks; (2) for products that have not been demonstrated to meet product specifications prior to shipment, revenue is recognized at customer technical acceptance; (3) for transactions where legal title does not pass at shipment or delivery, revenue is recognized when legal title passes to the customer, which is generally at customer technical acceptance; and (4) for arrangements containing multiple elements, the revenue relating to the undelivered elements is deferred using the relative selling price method utilizing estimated sales prices until delivery of the deferred elements. Applied limits the amount of revenue recognition for delivered elements to the amount that is not contingent on the future delivery of products or services, future performance obligations or subject to customer-specified return or adjustment. In cases where Applied has sold products that have been demonstrated to meet product specifications prior to shipment, Applied believes that at the time of delivery, it has an enforceable claim to amounts recognized as revenue. Spare parts revenue is generally recognized upon shipment, and services revenue is generally recognized over the period that the services are provided.

When a sales arrangement contains multiple elements, such as hardware and services and/or software products, Applied allocates revenue to each element based on a selling price hierarchy. The selling price for a deliverable is based on its vendor specific objective evidence (VSOE) if available, third party evidence (TPE) if VSOE is not available, or estimated selling price (ESP) if neither VSOE nor TPE is available. Applied generally utilizes the ESP due to the nature of its products. In multiple element arrangements where more-than-incidental software deliverables are included, revenue is allocated to each separate unit of accounting for each of the non-software deliverables and to the software deliverables as a group using the relative selling prices of each of the deliverables in the arrangement based on the aforementioned selling price hierarchy. If the arrangement contains more than one software deliverable, the arrangement consideration allocated to the software deliverables as a group is then allocated to each software deliverable using the guidance for recognizing software revenue.

*Derivative Financial Instruments*

Applied uses financial instruments, such as forward exchange and currency option contracts, to hedge a portion of, but not all, existing and anticipated foreign currency denominated transactions typically expected to occur within 24 months. The purpose of Applied's foreign currency management is to mitigate the effect of exchange rate fluctuations on certain foreign currency denominated revenues, costs and eventual cash flows. In certain cases, Applied also uses interest rate swap or lock agreements to hedge against the variability of cash flows due to changes in the benchmark interest rate of fixed rate debt. The terms of derivative financial instruments used for hedging purposes are generally consistent with the timing of the transactions being hedged. All of Applied's derivative financial instruments are recorded at fair value based upon quoted market prices for comparable instruments. For derivative instruments designated and qualifying as cash flow hedges, the effective portion of the gain or loss on these hedges is reported as a component of accumulated other comprehensive income (loss) in stockholders' equity, and is reclassified into earnings when the hedged transaction affects earnings. If the transaction being hedged fails to occur, or if a portion of any derivative is ineffective, the gain or loss on the associated financial instrument is recorded promptly in earnings. For derivative instruments used to hedge existing foreign currency denominated assets or liabilities, the gain or loss on these hedges is recorded promptly in earnings to offset the changes in the fair value of the assets or liabilities being hedged. Applied does not use derivative financial instruments for trading or speculative purposes.

*Foreign Currencies*

As of October 28, 2018, all of Applied's subsidiaries use the United States dollar as their functional currency. Accordingly, assets and liabilities of these subsidiaries are remeasured using exchange rates in effect at the end of the period, except for non-monetary assets, such as inventories and property, plant and equipment, which are remeasured using historical exchange rates. Foreign currency-denominated revenues and costs are remeasured using average exchange rates for the period, except for costs related to those balance sheet items that are remeasured using historical exchange rates. The resulting remeasurement gains and losses are included in general and administrative expenses in the Consolidated Statements of Operations as incurred.

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**APPLIED MATERIALS, INC.**

**NOTES TO CONSOLIDATED FINANCIAL STATEMENTS — (Continued)**

*Concentrations of Credit Risk*

Financial instruments that potentially subject Applied to significant concentrations of credit risk consist principally of cash equivalents, investments, trade accounts receivable and derivative financial instruments used in hedging activities. Applied invests in a variety of financial instruments, such as, but not limited to, commercial paper, corporate and municipal bonds, United States Treasury and agency securities, and asset-backed and mortgage-backed securities, and, by policy, limits the amount of credit exposure with any one financial institution or commercial issuer. Applied performs ongoing credit evaluations of its customers' financial condition and generally requires no collateral to secure accounts receivable. Applied maintains an allowance reserve for potentially uncollectible accounts receivable based on its assessment of the collectability of accounts receivable. Applied regularly reviews the allowance by considering factors such as historical experience, credit quality, age of the accounts receivable balances, and current economic conditions that may affect a customer's ability to pay. In addition, Applied utilizes letters of credit to mitigate credit risk when considered appropriate. Applied is exposed to credit-related losses in the event of nonperformance by counterparties to derivative financial instruments, but does not expect any counterparties to fail to meet their obligations. In some instances, Applied has entered into security arrangements which require the counterparties to post collateral to further mitigate credit exposure.

*Recent Accounting Pronouncements*

*Accounting Standards Adopted*

*Inventory Measurement.* In July 2015, the Financial Accounting Standard Board (FASB) issued authoritative guidance that requires inventory to be measured at the lower of cost and net realizable value instead of at lower of cost or market. This guidance does not apply to inventory that is measured using last-in, first out (LIFO) or the retail inventory method but applies to all other inventory including those measured using first-in, first-out (FIFO) or the average cost method. Applied adopted this authoritative guidance in the first quarter of fiscal 2018 prospectively to the measurement of inventory after the effective date. The adoption of this guidance did not have a material impact on Applied's consolidated financial statements.

*Share-Based Compensation.* In March 2016, the FASB issued authoritative guidance that simplifies several aspects of the accounting for share-based payment transactions, including forfeitures, income tax, and classification on the statement of cash flows. Applied adopted this guidance in the first quarter of fiscal 2018. Upon adoption, Applied elected to continue to estimate forfeitures expected to occur to determine the amount of compensation costs to be recognized in each period. The new standard also required recognition of excess tax benefits in the provision for income taxes rather than additional paid-in capital prospectively. In fiscal 2018, Applied recognized an excess tax benefit of $51 million in the Consolidated Condensed Statements of Operations. Additionally, Applied elected to apply the presentation requirements for cash flows related to excess tax benefits and employee taxes paid for withheld shares retrospectively. Adopting this guidance increased cash provided by operating activities by $180 million and $100 million with corresponding net decreases in cash provided by financing activities for fiscal 2017 and 2016, respectively.

*Reclassification of Certain Tax Effects from Accumulated Other Comprehensive Income.* In February 2018, the FASB issued authoritative guidance that allows stranded tax effects of the Tax Cuts and Jobs Act (the "Tax Act") to be reclassified from accumulated other comprehensive income to retained earnings. Applied adopted this guidance at the beginning of the third quarter of fiscal 2018. The stranded income tax effects related to the Tax Act were reclassified from accumulated other comprehensive income to retained earnings in the consolidated statements of stockholders' equity as of October 28, 2018.

*Share-Based Compensation: Modification Accounting.* In May 2017, the FASB issued an update to clarify when to account for a change to the terms or conditions of a share-based payment award as a modification. Under the new guidance, modification accounting is required only if the fair value, the vesting conditions, or the classification of the award changes as a result of the change in terms or conditions. Applied adopted this guidance in the third quarter of fiscal 2018 on a prospective basis. The impact of the adoption of this guidance will depend on whether the Company makes any future modifications of share-based payment awards.

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**APPLIED MATERIALS, INC.**

**NOTES TO CONSOLIDATED FINANCIAL STATEMENTS — (Continued)**

*Accounting Standards Not Yet Adopted*

*Retirement Benefits: Changes to the Disclosure Requirements for Defined Benefit and other Postretirement Plans.* In August 2018, the FASB issued authoritative guidance that adds, removes, and clarifies disclosure requirements for defined benefit and other postretirement plans. This authoritative guidance will be effective for Applied in fiscal 2021 on a retrospective basis, with early adoption permitted. Applied is currently evaluating the effect of this new guidance on Applied's consolidated financial statements.

*Fair Value Measurement: Changes to the Disclosure Requirements for Fair Value Measurement.* In August 2018, the FASB issued authoritative guidance that eliminates, amends, and adds disclosure requirements for fair value measurements. While the amended and new disclosure requirements primarily relate to Level 3 fair value measurements, the authoritative guidance also eliminates disclosure requirements related to the amount and reasons for transfer between Level 1 and Level 2 of fair value hierarchy, policy for timing of transfer between levels, and the valuation processes for Level 3 fair value measurements. The authoritative guidance will be effective for Applied in the first quarter of fiscal 2020. Early adoption is permitted only for the removal and amendment of certain disclosures, while the new disclosures requirements are to be applied prospectively. Applied is currently evaluating the effect of this new guidance on Applied's consolidated financial statements.

*Derivatives and Hedging.* In August 2017, the FASB issued authoritative guidance that modifies the recognition and presentation of hedge accounting to better align an entity's risk management strategies and financial reporting for hedging relationships. The authoritative guidance expands the application of hedge accounting for non-financial and financial risk components and eases certain hedge effectiveness assessment requirements. The authoritative guidance will be effective for Applied in the first quarter of fiscal 2020, with early adoption permitted. Applied is currently evaluating the effect of this new guidance on Applied's consolidated financial statements.

*Receivables: Nonrefundable Fees and Other Costs.* In March 2017, the FASB issued authoritative guidance that will shorten the amortization period for certain callable debt securities held at a premium to the earliest call date to more closely align with expectations incorporated in market pricing. This authoritative guidance will be effective for Applied in the first quarter of fiscal 2020 on a modified retrospective basis, with early adoption permitted. Applied is currently evaluating the impact of adopting this new accounting guidance on Applied's consolidated financial statements.

*Retirement Benefits.* In March 2017, the FASB issued authoritative guidance which requires companies to present the service cost component of net benefit cost in the same line items in which they report compensation cost. The other components of net benefit cost are required to be presented in the income statement separately from the service cost component and outside a subtotal of income from operations, if one is presented. The authoritative guidance will be effective for Applied in the first quarter of fiscal 2019 on a retrospective basis. The adoption of this guidance is only expected to result in reclassification of other components of net benefit costs outside of income from operations and is not expected to have a significant impact on Applied's consolidated financial statements.

*Goodwill Impairment.* In January 2017, the FASB issued authoritative guidance that simplifies the process required to test goodwill for impairment. The authoritative guidance will be effective for Applied in the first quarter of fiscal 2021. Early adoption is permitted. The adoption of this guidance is not expected to have a significant impact on Applied's consolidated financial statements.

*Business Combinations.* In January 2017, the FASB issued authoritative guidance that clarifies the definition of a business to help companies evaluate whether acquisition or disposal transactions should be accounted for as asset groups or as businesses. The authoritative guidance will be effective for Applied in the first quarter of fiscal 2019 on a prospective basis. The impact of the adoption depends on the facts and circumstances of future acquisition or disposal transactions.

*Income Taxes: Intra-Entity Asset Transfers.* In October 2016, the FASB issued authoritative guidance that requires entities to recognize at the transaction date the income tax consequences of intercompany asset transfers other than inventory. Applied will adopt this standard in the first quarter of fiscal 2019 using a modified-retrospective approach resulting in a cumulative-effect adjustment to retained earnings. Applied estimates an increase in deferred tax assets of approximately $1.6 billion with a corresponding increase in retained earnings related to the recognition of income tax effects of intra-entity asset transfers that occurred prior to the adoption date.

*Classification of Certain Cash Receipts and Cash Payments.* In August 2016, the FASB issued authoritative guidance which addresses classification of certain cash receipts and cash payments related to the statement of cash flows. The authoritative guidance will be effective for Applied in the first quarter of fiscal 2019. The adoption of this guidance is not expected to have a significant impact on Applied's consolidated financial statements.

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**APPLIED MATERIALS, INC.**

**NOTES TO CONSOLIDATED FINANCIAL STATEMENTS — (Continued)**

*Financial Instruments: Credit Losses.* In June 2016, the FASB issued authoritative guidance that modifies the impairment model for certain financial assets by requiring use of an expected loss methodology, which will result in more timely recognition of credit losses. The authoritative guidance will be effective for Applied in the first quarter of fiscal 2021. Early adoption is permitted beginning in the first quarter of fiscal 2020. Applied is currently evaluating the effect of this new guidance on Applied's consolidated financial statements.

*Leases.* In February 2016, the FASB issued authoritative guidance for lease accounting, which requires lessees to recognize lease assets and liabilities on the balance sheet for certain lease arrangements that are classified as operating leases under the previous standard, and to provide for enhanced disclosures. The authoritative guidance will be effective for Applied in the first quarter of fiscal 2020 and should be applied using a modified retrospective approach. Early adoption is permitted. Applied is currently evaluating the effect of this new guidance on Applied's consolidated financial statements.

*Financial Instruments: Classification and Measurement.* In January 2016, the FASB issued authoritative guidance that requires equity investments that do not result in consolidation, and are not accounted for under the equity method, to be measured at fair value, and requires recognition of any changes in fair value in net income unless the investments qualify for a new practicability exception. For financial liabilities measured at fair value, the change in fair value caused by a change in instrument-specific credit risk will be required to be presented separately in other comprehensive income. Applied will adopt this standard in the first quarter of fiscal year 2019. Applied will make elections on whether to apply fair value or the measurement alternative for equity investments without readily determinable fair value on an investment-by-investment basis. For the equity securities without readily determinable fair values, fair value will be applied prospectively to all equity investments that exist as of adoption. Upon adoption, Applied will record a cumulative-effect adjustment to retained earnings for the unrealized gains and losses associated with equity investments with readily determinable fair values. Applied currently estimates a net increase to retained earnings of approximately $21 million with the corresponding net decrease in accumulated other comprehensive income.

*Revenue Recognition.* In May 2014, the FASB issued authoritative guidance that requires revenue recognition to depict the transfer of promised goods or services to a customer in an amount that reflects the consideration to which the entity expects to be entitled in exchange for those goods or services and requires certain additional disclosures. This new standard will supersede most current revenue recognition guidance, including industry-specific guidance. Entities will have the option of using either a full retrospective or modified retrospective approach to adopting the guidance. In August 2015, the FASB issued an amendment to defer the effective date by one year and allow entities to early adopt no earlier than the original effective date. With this amendment, the guidance will be effective for Applied in the first quarter of fiscal 2019, which is the Company's planned adoption date. Applied currently anticipates adopting this new guidance using the full retrospective approach.

In fiscal 2016, Applied established a project steering committee and cross-functional implementation team that has identified potential differences that would result from applying the requirements of the new standard to Applied's revenue contracts. In addition, the implementation team has identified and is now implementing changes to business processes, systems and controls to support recognition and disclosure under the new standard.

While the Company's evaluation of the impact of this new guidance is not complete, Applied currently expects that the new standard will have the following impacts:

- Revenue related to the sale of equipment and spares will generally continue to be recognized at a point in time upon the transfer of control. Under this new guidance, the point of time at which revenue is recognized for certain of these products is expected to be earlier than under current revenue recognition guidance as there will be fewer constraints related to customer acceptance.
- Revenue related to the sale of services will generally continue to be recognized over time as the services are performed.
- Disclosures related to revenue recognition, including contract balances and revenue disaggregation, will be expanded.
- Applied continues to evaluate the effect that this new guidance will have on the financial position and results of operations for fiscal 2018. However, for fiscal 2017, Applied expects an increase of our net sales between $140 million and $180 million, or approximately 1%.

Applied is nearing completion of retrospectively adjusting financial information and will continue to evaluate the effect that the standard will have on its consolidated financial statements, including disclosures, and its preliminary assessments are subject to change.

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**APPLIED MATERIALS, INC.**

**NOTES TO CONSOLIDATED FINANCIAL STATEMENTS — (Continued)**

## Note 2 Earnings Per Share

Basic earnings per share is determined using the weighted average number of common shares outstanding during the period. Diluted earnings per share is determined using the weighted average number of common shares and potential common shares (representing the dilutive effect of stock options, restricted stock units, and employee stock purchase plan shares) outstanding during the period. Applied's net income has not been adjusted for any period presented for purposes of computing basic or diluted earnings per share due to the Company's non-complex capital structure.

| Fiscal Year | 2018 | 2017 | 2016 |
|---|---|---|---|
| | (In millions, except per share amounts) | | |
| Numerator: | | | |
| Net income | $ 3,313 | $ 3,434 | $ 1,721 |
| Denominator: | | | |
| Weighted average common shares outstanding | 1,013 | 1,073 | 1,107 |
| Effect of dilutive stock options, restricted stock units and employee stock purchase plan shares | 13 | 11 | 9 |
| Denominator for diluted earnings per share | 1,026 | 1,084 | 1,116 |
| Basic earnings per share | $ 3.27 | $ 3.20 | $ 1.56 |
| Diluted earnings per share | $ 3.23 | $ 3.17 | $ 1.54 |
| Potentially dilutive securities | — | — | — |

Potentially dilutive securities attributable to outstanding stock options and restricted stock units are excluded from the calculation of diluted earnings per share where the combined exercise price and average unamortized fair value are greater than the average market price of Applied common stock, and therefore their inclusion would be anti-dilutive. Prior to the adoption of Accounting Standards Update (ASU) 2016-09 *Compensation—Stock Compensation (Topic 718): Improvements to Employee Share-Based Payment Accounting* in the first quarter of fiscal 2018, the assumed tax benefits upon the exercise of options and vesting of restricted stock units were also included in this calculation.

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**APPLIED MATERIALS, INC.**

**NOTES TO CONSOLIDATED FINANCIAL STATEMENTS — (Continued)**

## Note 3 Cash, Cash Equivalents and Investments

### *Summary of Cash, Cash Equivalents and Investments*

The following tables summarize Applied's cash, cash equivalents and investments by security type:

| October 28, 2018 | Cost | Gross Unrealized Gains | Gross Unrealized Losses | Estimated Fair Value |
|---|---|---|---|---|
| | | (In millions) | | |
| Cash | $ 1,489 | $ — | $ — | $ 1,489 |
| Cash equivalents: | | | | |
| Money market funds | 1,599 | — | — | 1,599 |
| Commercial paper, corporate bonds and medium-term notes | 352 | — | — | 352 |
| Total Cash equivalents | 1,951 | — | — | 1,951 |
| Total Cash and Cash equivalents | $ 3,440 | $ — | $ — | $ 3,440 |
| Short-term and long-term investments: | | | | |
| U.S. Treasury and agency securities | $ 335 | $ — | $ 2 | $ 333 |
| Non-U.S. government securities* | 10 | — | — | 10 |
| Municipal securities | 399 | — | 4 | 395 |
| Commercial paper, corporate bonds and medium-term notes | 705 | — | 3 | 702 |
| Asset-backed and mortgage-backed securities | 595 | — | 4 | 591 |
| Total fixed income securities | 2,044 | — | 13 | 2,031 |
| Publicly traded equity securities | 17 | 25 | 4 | 38 |
| Equity investments in privately-held companies | 89 | — | — | 89 |
| Total short-term and long-term investments | $ 2,150 | $ 25 | $ 17 | $ 2,158 |
| Total Cash, Cash equivalents and Investments | $ 5,590 | $ 25 | $ 17 | $ 5,598 |

* Includes agency debt securities guaranteed by Canada.

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**APPLIED MATERIALS, INC.**

**NOTES TO CONSOLIDATED FINANCIAL STATEMENTS — (Continued)**

| October 29, 2017 | Cost | Gross Unrealized Gains | Gross Unrealized Losses | Estimated Fair Value |
|---|---|---|---|---|
| | | (In millions) | | |
| Cash | $ 1,346 | $ — | $ — | $ 1,346 |
| Cash equivalents: | | | | |
| Money market funds | 2,658 | — | — | 2,658 |
| U.S. Treasury and agency securities | 15 | — | — | 15 |
| Non-U.S. government securities | 55 | — | — | 55 |
| Municipal securities | 341 | — | — | 341 |
| Commercial paper, corporate bonds and medium-term notes | 595 | — | — | 595 |
| Total Cash equivalents | 3,664 | — | — | 3,664 |
| Total Cash and Cash equivalents | $ 5,010 | $ — | $ — | $ 5,010 |
| Short-term and long-term investments: | | | | |
| U.S. Treasury and agency securities | $ 667 | $ — | $ 1 | $ 666 |
| Non-U.S. government securities* | 161 | — | — | 161 |
| Municipal securities | 1,007 | — | — | 1,007 |
| Commercial paper, corporate bonds and medium-term notes | 1,024 | 1 | 1 | 1,024 |
| Asset-backed and mortgage-backed securities | 379 | — | 1 | 378 |
| Total fixed income securities | 3,238 | 1 | 3 | 3,236 |
| Publicly traded equity securities | 22 | 78 | 1 | 99 |
| Equity investments in privately-held companies | 74 | — | — | 74 |
| Total short-term and long-term investments | $ 3,334 | $ 79 | $ 4 | $ 3,409 |
| Total Cash, Cash equivalents and Investments | $ 8,344 | $ 79 | $ 4 | $ 8,419 |

* Includes agency debt securities guaranteed by non-U.S. governments, which consist of Canada and Germany.

*Maturities of Investments*

The following table summarizes the contractual maturities of Applied's investments at October 28, 2018:

| | Cost | Estimated Fair Value |
|---|---|---|
| | (In millions) | |
| Due in one year or less | $ 503 | $ 502 |
| Due after one through five years | 946 | 938 |
| No single maturity date** | 701 | 718 |
| Total | $ 2,150 | $ 2,158 |

** Securities with no single maturity date include publicly-traded and privately-held equity securities, and asset-backed and mortgage-backed securities.

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**APPLIED MATERIALS, INC.**

**NOTES TO CONSOLIDATED FINANCIAL STATEMENTS — (Continued)**

*Gains and Losses on Investments*

Gross realized gains and losses on sales of investments for each fiscal year were as follows:

| | 2018 | 2017 | 2016 |
|---|---|---|---|
| | | (In millions) | |
| Gross realized gains | $ 29 | $ 14 | $ 10 |
| Gross realized losses | $ 3 | $ 1 | $ 2 |

At October 28, 2018, gross unrealized losses related to Applied's investment portfolio were not material. Applied regularly reviews its investment portfolio to identify and evaluate investments that have indications of possible impairment. Factors considered in determining whether an unrealized loss is considered to be temporary, or other-than-temporary and therefore impaired, include: the length of time and extent to which fair value has been lower than the cost basis; the financial condition, credit quality and near-term prospects of the investee; and whether it is more likely than not that Applied will be required to sell the security prior to recovery. Generally, the contractual terms of investments in marketable securities do not permit settlement at prices less than the amortized cost of the investments. Applied determined that the gross unrealized losses on its marketable fixed income securities at October 28, 2018, October 29, 2017 and October 30, 2016 were temporary in nature and therefore it did not recognize any impairment of its marketable fixed income securities for fiscal 2018, 2017 or 2016. During fiscal 2018, 2017 and 2016, Applied determined that certain of its equity investments were other-than-temporarily impaired and, accordingly, recognized impairment charges of $5 million, $10 million and $8 million, respectively. These impairment charges are included in interest and other income, net in the Consolidated Statement of Operations.

Unrealized gains and temporary losses on investments classified as available-for-sale are included within accumulated other comprehensive income (loss), net of any related tax effect. Upon realization, those amounts are reclassified from accumulated other comprehensive income (loss) to results of operations.

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**APPLIED MATERIALS, INC.**

**NOTES TO CONSOLIDATED FINANCIAL STATEMENTS — (Continued)**

## Note 4 Fair Value Measurements

Applied's financial assets are measured and recorded at fair value, except for equity investments in privately-held companies. These equity investments are generally accounted for under the cost method of accounting and are periodically assessed for other-than-temporary impairment when events or circumstances indicate that an other-than-temporary decline in value may have occurred. Applied's nonfinancial assets, such as goodwill, intangible assets, and property, plant and equipment, are recorded at cost and are assessed for impairment whenever events or changes in circumstances indicate that the carrying value of an asset may not be recoverable.

*Fair Value Hierarchy*

Applied uses the following fair value hierarchy, which prioritizes the inputs to valuation techniques used to measure fair value into three levels and bases the categorization within the hierarchy upon the lowest level of input that is available and significant to the fair value measurement:

- Level 1 — Quoted prices in active markets for identical assets or liabilities;
- Level 2 — Inputs other than Level 1 that are observable, either directly or indirectly, such as quoted prices for similar assets or liabilities, quoted prices in markets that are not active, or other inputs that are observable or can be corroborated by observable market data for substantially the full term of the assets or liabilities; and
- Level 3 — Unobservable inputs that are supported by little or no market activity and that are significant to the fair value of the assets or liabilities.

Applied's investments consist primarily of debt securities that are classified as available-for-sale and recorded at their fair values. In determining the fair value of investments, Applied uses pricing information from pricing services that value securities based on quoted market prices and models that utilize observable market inputs. In the event a fair value estimate is unavailable from a pricing service, Applied generally obtains non-binding price quotes from brokers. Applied then reviews the information provided by the pricing services or brokers to determine the fair value of its short-term and long-term investments. In addition, to validate pricing information obtained from pricing services, Applied periodically performs supplemental analysis on a sample of securities. Applied reviews any significant unanticipated differences identified through this analysis to determine the appropriate fair value.

Investments with remaining effective maturities of 12 months or less from the balance sheet date are classified as short-term investments. Investments with remaining effective maturities of more than 12 months from the balance sheet date are classified as long-term investments. As of October 28, 2018, substantially all of Applied's available-for-sale, short-term and long-term investments were recognized at fair value that was determined based upon observable inputs.

*Assets Measured at Fair Value on a Recurring Basis*

Financial assets (excluding cash balances) measured at fair value on a recurring basis are summarized below:

| | October 28, 2018 | | | October 29, 2017 | | |
|---|---|---|---|---|---|---|
| | Level 1 | Level 2 | Total | Level 1 | Level 2 | Total |
| | (In millions) | | | | | |
| Assets: | | | | | | |
| Money market funds | $ 1,599 | $ — | $ 1,599 | $ 2,658 | $ — | $ 2,658 |
| U.S. Treasury and agency securities | 297 | 36 | 333 | 192 | 489 | 681 |
| Non-U.S. government securities | — | 10 | 10 | — | 216 | 216 |
| Municipal securities | — | 395 | 395 | — | 1,348 | 1,348 |
| Commercial paper, corporate bonds and medium-term notes | — | 1,054 | 1,054 | — | 1,619 | 1,619 |
| Asset-backed and mortgage-backed securities | — | 591 | 591 | — | 378 | 378 |
| Publicly traded equity securities | 38 | — | 38 | 99 | — | 99 |
| Total | $ 1,934 | $ 2,086 | $ 4,020 | $ 2,949 | $ 4,050 | $ 6,999 |

There were no transfers between Level 1 and Level 2 fair value measurements during fiscal 2018 and 2017, and Applied did not have any financial assets measured at fair value on a recurring basis within Level 3 fair value measurements as of October 28, 2018 or October 29, 2017.

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**APPLIED MATERIALS, INC.**

**NOTES TO CONSOLIDATED FINANCIAL STATEMENTS — (Continued)**

*Assets and Liabilities Measured at Fair Value on a Non-recurring Basis*

Equity investments in privately-held companies are generally accounted for under the cost method of accounting and are periodically assessed for other-than-temporary impairment when an event or circumstance indicates that an other-than-temporary decline in value may have occurred. If Applied determines that an other-than-temporary impairment has occurred, the investment will be written down to its estimated fair value based on available information, such as pricing in recent rounds of financing, current cash positions, earnings and cash flow forecasts, recent operational performance and any other readily available market data. Equity investments in privately-held companies totaled $89 million at October 28, 2018, of which $72 million of investments were accounted for under the cost method of accounting and $17 million of investments had been measured at fair value on a non-recurring basis within Level 3 fair value measurements due to an other-than-temporary decline in value. Equity investments in privately-held companies totaled $74 million at October 29, 2017, of which $65 million of investments were accounted for under the cost method of accounting and $9 million of investments had been measured at fair value on a non-recurring basis within Level 3 fair value measurements due to an other-than-temporary decline in value.

During fiscal 2018, 2017 and 2016, Applied determined that certain of its equity investments were other-than-temporarily impaired and, accordingly, recognized impairment charges of $5 million, $10 million and $8 million, respectively.

*Other*

The carrying amounts of Applied's financial instruments, including cash and cash equivalents, accounts receivable, notes payable - short term, and accounts payable and accrued expenses, approximate fair value due to their short maturities. At October 28, 2018, the aggregate principal and estimated fair value amounts of long-term debt were both $5.4 billion. At October 29, 2017, the aggregate principal amount of long-term debt was $5.4 billion, and the estimated fair value was $5.8 billion. The estimated fair value of long-term debt is determined by Level 2 inputs and is based primarily on quoted market prices for the same or similar issues. See Note 10 of the Notes to the Consolidated Financial Statements for further detail of existing debt.

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## Note 5 Derivative Instruments and Hedging Activities

*Derivative Financial Instruments*

Applied conducts business in a number of foreign countries, with certain transactions denominated in local currencies, such as the Japanese yen, euro, Israeli shekel and Taiwanese dollar. Applied uses derivative financial instruments, such as forward exchange contracts and currency option contracts, to hedge certain forecasted foreign currency denominated transactions expected to occur typically within the next 24 months. The purpose of Applied's foreign currency management is to mitigate the effect of exchange rate fluctuations on certain foreign currency denominated revenues, costs and eventual cash flows. The terms of currency instruments used for hedging purposes are generally consistent with the timing of the transactions being hedged.

During fiscal 2017, Applied entered into and settled interest rate lock agreements, with a total notional amount of $700 million to hedge against the variability of benchmark interest rates prior to the issuance of the debt. These instruments were designated as cash flow hedges at inception and settled in conjunction with the issuance of debt in March 2017. The $14 million loss from the settlement of the interest rate lock agreement, which was included in AOCI in stockholders' equity, is being amortized to interest expense over the term of the senior unsecured 10-year notes issued in March 2017.

Applied does not use derivative financial instruments for trading or speculative purposes. Derivative instruments and hedging activities, including foreign currency exchange and interest rate contracts, are recognized on the balance sheet at fair value. Changes in the fair value of derivatives that do not qualify for hedge treatment, as well as the ineffective portion of any hedges, are recognized currently in earnings. All of Applied's derivative financial instruments are recorded at their fair value in other current assets or in accounts payable and accrued expenses.

Hedges related to anticipated transactions are designated and documented at the inception of the hedge as cash flow hedges and foreign exchange derivatives are typically entered into once per month. Cash flow hedges are evaluated for effectiveness quarterly. The effective portion of the gain or loss on these hedges is reported as a component of AOCI in stockholders' equity and is reclassified into earnings when the hedged transaction affects earnings. The majority of the after-tax net income or loss related to foreign exchange derivative instruments included in AOCI at October 28, 2018 is expected to be reclassified into earnings within 12 months. Changes in the fair value of currency forward exchange and option contracts due to changes in time value are excluded from the assessment of effectiveness. Both ineffective hedge amounts and hedge components excluded from the assessment of effectiveness are recognized in earnings. If the transaction being hedged is no longer probable to occur, or if a portion of any derivative is deemed to be ineffective, Applied promptly recognizes the gain or loss on the associated financial instrument in earnings. The amount recognized due to discontinuance of cash flow hedges that were probable not to occur by the end of the originally specified time period were not significant for fiscal years 2018 and 2017 and was a loss of $8 million for fiscal 2016.

Additionally, forward exchange contracts are generally used to hedge certain foreign currency denominated assets or liabilities. These derivatives are typically entered into once per month and are not designated for hedge accounting treatment. Accordingly, changes in the fair value of these hedges are recorded in earnings to offset the changes in the fair value of the assets or liabilities being hedged.

The fair values of foreign exchange derivative instruments at October 28, 2018 and October 29, 2017 were not material.

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**APPLIED MATERIALS, INC.**

**NOTES TO CONSOLIDATED FINANCIAL STATEMENTS — (Continued)**

The effects of derivative instruments and hedging activities on the Consolidated Statements of Operations were as follows:

| Derivatives in Cash Flow Hedging Relationships | Location of Gain or (Loss) | Effective Portion: Gain or (Loss) | Effective Portion: Gain or (Loss) Reclassified from AOCI into Income | Ineffective Portion and Amount Excluded from Effectiveness Testing: Gain or (Loss) Recognized in Income |
|---|---|---|---|---|
| | | | (In millions) | |
| **2018** | | | | |
| Foreign exchange contracts | AOCI | $ 3 | $ — | $ — |
| Foreign exchange contracts | Cost of products sold | — | 4 | 19 |
| Foreign exchange contracts | General and administrative | — | (3) | (7) |
| Interest rate contracts | Interest expense | — | (3) | — |
| Total | | $ 3 | $ (2) | $ 12 |
| **2017** | | | | |
| Foreign exchange contracts | AOCI | $ 35 | $ — | $ — |
| Foreign exchange contracts | Cost of products sold | — | 7 | (3) |
| Foreign exchange contracts | General and administrative | — | 7 | (2) |
| Interest rate contracts | AOCI | (14) | — | — |
| Interest rate contracts | Interest expense | — | (3) | — |
| Total | | $ 21 | $ 11 | $ (5) |
| **2016** | | | | |
| Foreign exchange contracts | AOCI | $ (53) | $ — | $ — |
| Foreign exchange contracts | Cost of products sold | — | (46) | 2 |
| Foreign exchange contracts | General and administrative | — | — | (11) |
| Interest rate contracts | Interest expense | — | (2) | — |
| Total | | $ (53) | $ (48) | $ (9) |

| Derivatives Not Designated as Hedging Instruments | Location of Gain or (Loss) Recognized in Income | Amount of Gain or (Loss) Recognized in Income: 2018 | 2017 | 2016 |
|---|---|---|---|---|
| | | | (In millions) | |
| Foreign exchange contracts | General and administrative | $ (5) | $ 39 | $ (75) |
| Total | | $ (5) | $ 39 | $ (75) |

*Credit Risk Contingent Features*

If Applied's credit rating were to fall below investment grade, it would be in violation of credit risk contingent provisions of the derivative instruments discussed above, and certain counterparties to the derivative instruments could request immediate payment on derivative instruments in net liability positions. The aggregate fair value of all derivative instruments with credit-risk related contingent features that were in a net liability position was immaterial as of October 28, 2018 and October 29, 2017.

Entering into derivative contracts with banks exposes Applied to credit-related losses in the event of the banks' nonperformance. However, Applied's exposure is not considered significant.

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**APPLIED MATERIALS, INC.**

**NOTES TO CONSOLIDATED FINANCIAL STATEMENTS — (Continued)**

## Note 6 Accounts Receivable, Net

Applied has agreements with various financial institutions to sell accounts receivable and discount promissory notes from selected customers. Applied sells its accounts receivable without recourse. Applied, from time to time, also discounts letters of credit issued by customers through various financial institutions. The discounting of letters of credit depends on many factors, including the willingness of financial institutions to discount the letters of credit and the cost of such arrangements.

Applied sold $1.6 billion, $746 million and $75 million of accounts receivable during fiscal 2018, 2017 and 2016, respectively. Applied discounted letters of credit issued by customers of $37 million in fiscal 2018. There was no discounting of promissory notes in fiscal 2018. Applied did not discount letters of credit issued by customers or discount promissory notes during fiscal 2017 or 2016. Financing charges on the sale of receivables and discounting of letters of credit are included in interest expense in the accompanying Consolidated Statements of Operations and were not material for all years presented.

Accounts receivable are presented net of allowance for doubtful accounts of $33 million and $34 million at October 28, 2018 and October 29, 2017, respectively. Changes in allowance for doubtful accounts in each fiscal year were as follows:

| | 2018 | 2017 | 2016 |
|---|---|---|---|
| | | (In millions) | |
| Beginning balance | $ 34 | $ 51 | $ 49 |
| Provision | — | — | 3 |
| Deductions[1] | (1) | (17) | (1) |
| Ending balance | $ 33 | $ 34 | $ 51 |

[1] Fiscal 2018, 2017 and 2016 deductions primarily represent releases of allowance for doubtful accounts credited to expense as a result of an overall lower risk profile of Applied's customers and cash collections.

Applied sells its products principally to manufacturers within the semiconductor and display industries. While Applied believes that its allowance for doubtful accounts is adequate and represents its best estimate as of October 28, 2018, it continues to closely monitor customer liquidity and industry and economic conditions, which may result in changes to Applied's estimates.

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APPLIED MATERIALS, INC.

NOTES TO CONSOLIDATED FINANCIAL STATEMENTS — (Continued)

**Note 7 Balance Sheet Detail**

| | October 28, 2018 | October 29, 2017 |
|---|---|---|
| | (In millions) | |
| **Inventories** | | |
| Customer service spares | $ 989 | $ 595 |
| Raw materials | 1,020 | 603 |
| Work-in-process | 505 | 468 |
| Finished goods | 1,208 | 1,264 |
| | $ 3,722 | $ 2,930 |

Included in finished goods inventory is $19 million at October 28, 2018 and $331 million at October 29, 2017, of newly-introduced systems at customer locations where the sales transaction did not meet Applied's revenue recognition criteria as set forth in Note 1. Finished goods inventory includes $350 million and $281 million of evaluation inventory at October 28, 2018 and October 29, 2017, respectively.

| | October 28, 2018 | October 29, 2017 |
|---|---|---|
| | (In millions) | |
| **Other Current Assets** | | |
| Prepaid income taxes and income taxes receivable | $ 40 | $ 57 |
| Prepaid expenses and other | 390 | 317 |
| | $ 430 | $ 374 |

| | Useful Life | October 28, 2018 | October 29, 2017 |
|---|---|---|---|
| | (In years) | (In millions) | |
| **Property, Plant and Equipment, Net** | | | |
| Land and improvements | | $ 245 | $ 160 |
| Buildings and improvements | 3-30 | 1,448 | 1,315 |
| Demonstration and manufacturing equipment | 3-5 | 1,282 | 1,129 |
| Furniture, fixtures and other equipment | 3-5 | 634 | 572 |
| Construction in progress | | 203 | 135 |
| Gross property, plant and equipment | | 3,812 | 3,311 |
| Accumulated depreciation | | (2,405) | (2,245) |
| | | $ 1,407 | $ 1,066 |

Depreciation expense was $258 million, $214 million and $200 million for fiscal 2018, 2017 and 2016 respectively.

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**APPLIED MATERIALS, INC.**

**NOTES TO CONSOLIDATED FINANCIAL STATEMENTS — (Continued)**

| | October 28, 2018 | October 29, 2017 |
|---|---|---|
| | (In millions) | |
| **Accounts Payable and Accrued Expenses** | | |
| Accounts payable | $ 996 | $ 945 |
| Compensation and employee benefits | 639 | 666 |
| Warranty | 209 | 199 |
| Dividends payable | 193 | 106 |
| Income taxes payable | 136 | 112 |
| Other accrued taxes | 112 | 70 |
| Interest payable | 38 | 38 |
| Other | 398 | 314 |
| | $ 2,721 | $ 2,450 |

| | October 28, 2018 | October 29, 2017 |
|---|---|---|
| | (In millions) | |
| **Customer Deposits and Deferred Revenue** | | |
| Customer deposits | $ 444 | $ 381 |
| Deferred revenue | 903 | 1,284 |
| | $ 1,347 | $ 1,665 |

Applied typically receives deposits on future deliverables from customers in the Display and Adjacent Markets segment and, in certain instances, may also receive deposits from customers in the Applied Global Services segment.

| | October 28, 2018 | October 29, 2017 |
|---|---|---|
| | (In millions) | |
| **Other Liabilities** | | |
| Defined and postretirement benefit plans | $ 177 | $ 160 |
| Other | 126 | 99 |
| | $ 303 | $ 259 |

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## Note 8 Business Combinations

During fiscal 2017, Applied completed three acquisitions to complement Applied's existing product offerings and to provide opportunities for future growth within Applied's Display and Adjacent Markets and Applied Global Services segments.

Pro forma results of operations for these acquisitions have not been presented because they are not material to Applied's consolidated results of operations. The acquired businesses are included in the results for the Display and Adjacent Markets and Applied Global Services segments.

The following table represents the final aggregated purchase price allocation for acquisitions completed in fiscal year 2017:

| | Estimated Fair Values (In millions) |
|---|---|
| Fair value of net assets acquired | $ 23 |
| Goodwill | 55 |
| Purchased technology | 31 |
| Purchase price allocated | $ 109 |

Intangible assets are being amortized on a straight-line basis over an estimated weighted-average useful life of 3.5 years. Total transaction costs related to these acquisitions were not material and were expensed as incurred in general and administrative expenses in the Consolidated Statement of Operations.

## Note 9 Goodwill, Purchased Technology and Other Intangible Assets

### *Goodwill and Purchased Intangible Assets*

Applied's methodology for allocating the purchase price relating to purchase acquisitions is determined through established and generally accepted valuation techniques. Goodwill is measured as the excess of the purchase price over the sum of the amounts assigned to tangible and identifiable intangible assets acquired less liabilities assumed. Applied assigns assets acquired (including goodwill) and liabilities assumed to one or more reporting units as of the date of acquisition. Typically, acquisitions relate to a single reporting unit and thus do not require the allocation of goodwill to multiple reporting units. If the products obtained in an acquisition are assigned to multiple reporting units, the goodwill is distributed to the respective reporting units as part of the purchase price allocation process.

Goodwill and purchased intangible assets with indefinite useful lives are not amortized, but are reviewed for impairment annually during the fourth quarter of each fiscal year and whenever events or changes in circumstances indicate that the carrying value of an asset may not be recoverable. The process of evaluating the potential impairment of goodwill and intangible assets requires significant judgment, especially in emerging markets. Applied regularly monitors current business conditions and considers other factors including, but not limited to, adverse industry or economic trends, restructuring actions and lower projections of profitability that may impact future operating results.

To test goodwill for impairment, Applied first performs a qualitative assessment to determine whether it is more likely than not that the fair value of a reporting unit is less than its carrying value. If it is concluded that this is the case, Applied then performs the two-step goodwill impairment test. Otherwise, the two-step goodwill impairment test is not required. Under the two-step goodwill impairment test, Applied would in the first step compare the estimated fair value of each reporting unit to its carrying value. Applied determines the fair value of each of its reporting units based on a weighting of income and market approaches. If the carrying value of a reporting unit exceeds its fair value, Applied would then perform the second step of the impairment test in order to determine the implied fair value of the reporting unit's goodwill. If Applied determines that the carrying value of a reporting unit's goodwill exceeds its implied fair value, Applied would record an impairment charge equal to the difference.

As of October 28, 2018, Applied's reporting units include Semiconductor Product Group and Imaging and Process Control Group, which combine to form the Semiconductor Systems reporting segment, Applied Global Services, and Display and Adjacent Markets.

In the fourth quarter of fiscal 2018, Applied performed a qualitative assessment to test goodwill for all of its reporting units for impairment. Applied determined that it was more likely than not that each of its reporting units' fair values exceeded their respective carrying values and that it was not necessary to perform the two-step goodwill impairment test for any of its reporting units.

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**APPLIED MATERIALS, INC.**

**NOTES TO CONSOLIDATED FINANCIAL STATEMENTS — (Continued)**

The evaluation of goodwill and intangible assets for impairment requires the exercise of significant judgment. In the event of future changes in business conditions, Applied will be required to reassess and update its forecasts and estimates used in future impairment analyses. If the results of these future analyses are lower than current estimates, a material impairment charge may result at that time.

Details of goodwill were as follows:

| | October 28, 2018 | October 29, 2017 |
|---|---|---|
| | (In millions) | |
| Semiconductor Systems | $ 2,151 | $ 2,151 |
| Applied Global Services | 1,018 | 1,018 |
| Display and Adjacent Markets | 199 | 199 |
| Carrying amount | $ 3,368 | $ 3,368 |

A summary of Applied's purchased technology and intangible assets is set forth below:

| | October 28, 2018 | October 29, 2017 |
|---|---|---|
| | (In millions) | |
| Purchased technology, net | $ 109 | $ 288 |
| Intangible assets - finite-lived, net | 104 | 124 |
| | $ 213 | $ 412 |

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**APPLIED MATERIALS, INC.**

**NOTES TO CONSOLIDATED FINANCIAL STATEMENTS — (Continued)**

***Finite-Lived Purchased Intangible Assets***

Applied amortizes purchased intangible assets with finite lives using the straight-line method over the estimated economic lives of the assets, ranging from 1 to 15 years.

Applied evaluates long-lived assets for impairment whenever events or changes in circumstances indicate the carrying value of an asset group may not be recoverable. Applied assesses the fair value of the assets based on the amount of the undiscounted future cash flow that the assets are expected to generate and recognizes an impairment loss when estimated undiscounted future cash flow expected to result from the use of the asset, plus net proceeds expected from disposition of the asset, if any, are less than the carrying value of the asset. When Applied identifies an impairment, Applied reduces the carrying value of the group of assets to comparable market values, when available and appropriate, or to its estimated fair value based on a discounted cash flow approach.

Intangible assets, such as purchased technology, are generally recorded in connection with a business acquisition. The value assigned to intangible assets is usually based on estimates and judgments regarding expectations for the success and life cycle of products and technology acquired. Applied evaluates the useful lives of its intangible assets each reporting period to determine whether events and circumstances require revising the remaining period of amortization. In addition, Applied reviews intangible assets for impairment when events or changes in circumstances indicate their carrying value may not be recoverable. Management considers such indicators as significant differences in actual product acceptance from the estimates, changes in the competitive and economic environments, technological advances, and changes in cost structure.

Details of finite-lived intangible assets were as follows:

| | October 28, 2018 | | | October 29, 2017 | | |
|---|---|---|---|---|---|---|
| | Purchased Technology | Other Intangible Assets | Total | Purchased Technology | Other Intangible Assets | Total |
| | (In millions) | | | | | |
| **Gross carrying amount:** | | | | | | |
| Semiconductor Systems | $ 1,449 | $ 252 | $ 1,701 | $ 1,449 | $ 252 | $ 1,701 |
| Applied Global Services | 33 | 44 | 77 | 33 | 44 | 77 |
| Display and Adjacent Markets | 163 | 38 | 201 | 163 | 38 | 201 |
| Corporate and Other | — | 9 | 9 | — | 9 | 9 |
| Gross carrying amount | $ 1,645 | $ 343 | $ 1,988 | $ 1,645 | $ 343 | $ 1,988 |
| **Accumulated amortization:** | | | | | | |
| Semiconductor Systems | $ (1,375) | $ (150) | $ (1,525) | $ (1,210) | $ (131) | $ (1,341) |
| Applied Global Services | (29) | (44) | (73) | (28) | (44) | (72) |
| Display and Adjacent Markets | (132) | (36) | (168) | (119) | (35) | (154) |
| Corporate and Other | — | (9) | (9) | — | (9) | (9) |
| Accumulated amortization | $ (1,536) | $ (239) | $ (1,775) | $ (1,357) | $ (219) | $ (1,576) |
| Carrying amount | $ 109 | $ 104 | $ 213 | $ 288 | $ 124 | $ 412 |

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**APPLIED MATERIALS, INC.**

**NOTES TO CONSOLIDATED FINANCIAL STATEMENTS — (Continued)**

Details of amortization expense for each fiscal year by segment were as follows:

| | 2018 | 2017 | 2016 |
|---|---|---|---|
| | | (In millions) | |
| Semiconductor Systems | $ 184 | $ 185 | $ 185 |
| Applied Global Services | 1 | 1 | 1 |
| Display and Adjacent Markets | 14 | 7 | — |
| Corporate and Other | — | — | 3 |
| Total | $ 199 | $ 193 | $ 189 |

Amortization expense for each fiscal year was charged to the following categories:

| | 2018 | 2017 | 2016 |
|---|---|---|---|
| | | (In millions) | |
| Cost of products sold | $ 180 | $ 173 | $ 167 |
| Research, development and engineering | 1 | 1 | 2 |
| Marketing and selling | 18 | 19 | 20 |
| Total | $ 199 | $ 193 | $ 189 |

As of October 28, 2018, future estimated amortization expense is expected to be as follows:

| | Amortization Expense |
|---|---|
| | (In millions) |
| 2019 | $ 57 |
| 2020 | 52 |
| 2021 | 39 |
| 2022 | 65 |
| Total | $ 213 |

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**APPLIED MATERIALS, INC.**

**NOTES TO CONSOLIDATED FINANCIAL STATEMENTS — (Continued)**

## Note 10 Borrowing Facilities and Debt

Applied has credit facilities for unsecured borrowings in various currencies of up to $1.6 billion, of which $1.5 billion is comprised of a committed revolving credit agreement with a group of banks that is scheduled to expire in September 2021. This agreement provides for borrowings in United States dollars at interest rates keyed to one of various benchmark rates selected by Applied for each advance, plus a margin based on Applied's public debt rating and includes financial and other covenants. Remaining credit facilities in the amount of approximately $71 million are with Japanese banks. Applied's ability to borrow under these facilities is subject to bank approval at the time of the borrowing request, and any advances will be at rates indexed to the banks' prime reference rate denominated in Japanese yen. No amounts were outstanding under any of these facilities at both October 28, 2018 and October 29, 2017, and Applied has not utilized these credit facilities. In fiscal 2011, Applied established a short-term commercial paper program of up to $1.5 billion. At October 28, 2018 and October 29, 2017, Applied did not have any commercial paper outstanding.

In March 2017, Applied issued senior unsecured notes in the aggregate principal amount of $2.2 billion and in May 2017, used a portion of the net proceeds to redeem the outstanding $200 million in principal amount of its 7.125% senior notes due in October 2017.

Debt outstanding as of October 28, 2018 and October 29, 2017 was as follows:

| | Principal Amount | | Effective Interest Rate | Interest Pay Dates |
|---|---|---|---|---|
| | October 28, 2018 | October 29, 2017 | | |
| | (In millions) | | | |
| Long-term debt: | | | | |
| 2.625% Senior Notes Due 2020 | $ 600 | $ 600 | 2.640% | April 1, October 1 |
| 4.300% Senior Notes Due 2021 | 750 | 750 | 4.326% | June 15, December 15 |
| 3.900% Senior Notes Due 2025 | 700 | 700 | 3.944% | April 1, October 1 |
| 3.300% Senior Notes Due 2027 | 1,200 | 1,200 | 3.342% | April 1, October 1 |
| 5.100% Senior Notes Due 2035 | 500 | 500 | 5.127% | April 1, October 1 |
| 5.850% Senior Notes Due 2041 | 600 | 600 | 5.879% | June 15, December 15 |
| 4.350% Senior Notes Due 2047 | 1,000 | 1,000 | 4.361% | April 1, October 1 |
| | 5,350 | 5,350 | | |
| Total unamortized discount | (11) | (12) | | |
| Total unamortized debt issuance costs | (30) | (34) | | |
| Total long-term debt | $ 5,309 | $ 5,304 | | |

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**APPLIED MATERIALS, INC.**

**NOTES TO CONSOLIDATED FINANCIAL STATEMENTS — (Continued)**

## Note 11 Stockholders' Equity, Comprehensive Income and Share-Based Compensation

*Accumulated Other Comprehensive Income (Loss)*

Changes in the components of accumulated other comprehensive income (AOCI), net of tax, were as follows:

| | Unrealized Gain (Loss) on Investments, Net | Unrealized Gain (Loss) on Derivative Instruments Qualifying as Cash Flow Hedges | Defined and Postretirement Benefit Plans | Cumulative Translation Adjustments | Total |
|---|---|---|---|---|---|
| | | | (In millions) | | |
| Balance at October 25, 2015 | $ 14 | $ (15) | $ (105) | $ 14 | (92) |
| Other comprehensive income (loss) before reclassifications | 14 | (33) | (42) | — | (61) |
| Amounts reclassified out of AOCI | 2 | 30 | 6 | — | 38 |
| Other comprehensive income (loss), net of tax | 16 | (3) | (36) | — | (23) |
| Balance at October 30, 2016 | $ 30 | $ (18) | $ (141) | $ 14 | $ (115) |
| Other comprehensive income (loss) before reclassifications | 24 | 13 | 29 | — | 66 |
| Amounts reclassified out of AOCI | (1) | (6) | (8) | — | (15) |
| Other comprehensive income, net of tax | 23 | 7 | 21 | — | 51 |
| Balance at October 29, 2017 | $ 53 | $ (11) | $ (120) | $ 14 | $ (64) |
| Adoption of new accounting standards (a) | 5 | (2) | — | — | 3 |
| Other comprehensive income (loss) before reclassifications | (66) | 5 | (23) | — | (84) |
| Amounts reclassified out of AOCI | 15 | (1) | 6 | — | 20 |
| Other comprehensive income (loss), net of tax | (51) | 4 | (17) | — | (64) |
| Balance at October 28, 2018 | $ 7 | $ (9) | $ (137) | $ 14 | $ (125) |

(a) - Represents the reclassification adjustment related to the early adoption of Accounting Standards Update (ASU) 2018-02 *Reclassification of Certain Tax Effects from Accumulated Other Comprehensive Income.* See Note 1.

The tax effects on net income of amounts reclassified from AOCI for fiscal 2016 was $22 million. The tax effects on net income of amounts reclassified from AOCI for the fiscal years 2018 and 2017, were not material.

*Stock Repurchase Programs*

In September 2017, Applied's Board of Directors approved a common stock repurchase program authorizing up to $3.0 billion in repurchases. In February 2018, the Board of Directors approved a new common stock repurchase program authorizing up to an additional $6.0 billion in repurchases. At October 28, 2018, $4.3 billion remained available for future stock repurchases under this repurchase program.

The following table summarizes Applied's stock repurchases for each fiscal year:

| | 2018 | 2017 | 2016 |
|---|---|---|---|
| | (In millions, except per share amounts) | | |
| Shares of common stock repurchased | 102 | 28 | 96 |
| Cost of stock repurchased | $ 5,283 | $ 1,172 | $ 1,892 |
| Average price paid per share | $ 51.55 | $ 42.08 | $ 19.82 |

Applied records treasury stock purchases under the cost method using the first-in, first-out (FIFO) method. Upon reissuance of treasury stock, amounts in excess of the acquisition cost are credited to additional paid in capital. If Applied reissues treasury stock at an amount below its acquisition cost and additional paid in capital associated with prior treasury stock transactions is insufficient to cover the difference between the acquisition cost and the reissue price, this difference is recorded against retained earnings.

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*Dividends*

During fiscal 2018, Applied's Board of Directors declared three quarterly cash dividends of $0.20 per share and one quarterly cash dividend of $0.10 per share. During each of fiscal 2017 and 2016, Applied's Board of Directors declared four quarterly cash dividends in the amount of $0.10 per share. Dividends paid during fiscal 2018, 2017 and 2016 amounted to $605 million, $430 million and $444 million, respectively. Applied currently anticipates that cash dividends will continue to be paid on a quarterly basis, although the declaration of any future cash dividend is at the discretion of the Board of Directors and will depend on Applied's financial condition, results of operations, capital requirements, business conditions and other factors, as well as a determination by the Board of Directors that cash dividends are in the best interests of Applied's stockholders.

*Share-Based Compensation*

Applied has a stockholder-approved equity plan, the Employee Stock Incentive Plan, which permits grants to employees of share-based awards, including stock options, restricted stock, restricted stock units, performance shares and performance units. In addition, the plan provides for the automatic grant of restricted stock units to non-employee directors and permits the grant of share-based awards to non-employee directors and consultants. Share-based awards made under the plan may be subject to accelerated vesting under certain circumstances in the event of a change in control of Applied. Applied also has two Employee Stock Purchase Plans, one generally for United States employees and a second for employees of international subsidiaries (collectively, ESPP), which enable eligible employees to purchase Applied common stock.

Applied recognized share-based compensation expense related to stock options, ESPP shares, restricted stock, restricted stock units, performance shares and performance units, and related tax benefits for each fiscal year as follows:

| | 2018 | 2017 | 2016 |
|---|---|---|---|
| | | (In millions) | |
| Share-based compensation | $ 258 | $ 220 | $ 201 |
| Tax benefit recognized | $ 45 | $ 60 | $ 63 |

The effect of share-based compensation on the results of operations for each fiscal year was as follows:

| | 2018 | 2017 | 2016 |
|---|---|---|---|
| | | (In millions) | |
| Cost of products sold | $ 87 | $ 69 | $ 62 |
| Research, development, and engineering | 96 | 83 | 76 |
| Marketing and selling | 31 | 28 | 26 |
| General and administrative | 44 | 40 | 37 |
| Total share-based compensation | $ 258 | $ 220 | $ 201 |

The cost associated with share-based awards that are subject solely to time-based vesting requirements, less expected forfeitures, is recognized over the awards' service period for the entire award on a straight-line basis. The cost associated with performance-based equity awards is recognized for each tranche over the service period, based on an assessment of the likelihood that the applicable performance goals will be achieved.

At October 28, 2018, Applied had $368 million in total unrecognized compensation expense, net of estimated forfeitures, related to grants of share-based awards and shares issued under Applied's ESPP, which will be recognized over a weighted average period of 2.4 years. At October 28, 2018, there were 81 million shares available for grants of share-based awards under the Employee Stock Incentive Plan, and an additional 17 million shares available for issuance under the ESPP.

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**APPLIED MATERIALS, INC.**

**NOTES TO CONSOLIDATED FINANCIAL STATEMENTS — (Continued)**

*Stock Options*

Stock options are rights to purchase, at future dates, shares of Applied common stock. The exercise price of each stock option equals the fair market value of Applied common stock on the date of grant. Options typically vest over three to four years, subject to the grantee's continued service with Applied through the scheduled vesting date, and expire no later than seven years from the grant date. There were no stock options granted during fiscal 2018, 2017 and 2016. Outstanding stock options at the end of fiscal 2018 were not material to the consolidated financial statements.

*Restricted Stock Units, Restricted Stock, Performance Shares and Performance Units*

Restricted stock units are converted into shares of Applied common stock upon vesting on a one-for-one basis. Restricted stock has the same rights as other issued and outstanding shares of Applied common stock except these shares generally have no right to dividends and are held in escrow until the award vests. Performance shares and performance units are awards that result in a payment to a grantee, generally in shares of Applied common stock on a one-for-one basis, if performance goals and/or other vesting criteria established by the Human Resources and Compensation Committee of Applied's Board of Directors are achieved or the awards otherwise vest. Restricted stock units, restricted stock, performance shares and performance units typically vest over four years and vesting is usually subject to the grantee's continued service with Applied and, in some cases, achievement of specified performance goals. The compensation expense related to the service-based awards is determined using the fair market value of Applied common stock on the date of the grant, and the compensation expense is recognized over the vesting period.

Certain executive officers were granted awards that are subject to the achievement of specified performance goals (performance-based awards). These awards become eligible to vest only if performance goals are achieved and will vest only if the grantee remains employed by Applied through each applicable vesting date. The fair value of these awards is estimated on the date of grant. If the goals are achieved, the awards will vest, provided that the grantee remains employed by Applied through each scheduled vesting date. If the performance goals are not met as of the end of the performance period, no compensation expense is recognized and any previously recognized compensation expense is reversed. The expected cost is based on the awards that are probable to vest and is reflected over the service period and reduced for estimated forfeitures.

For performance-based awards granted in fiscal 2018 and 2017, certain awards require the achievement of positive adjusted operating profit and vest ratably over three years. Other awards require the achievement of targeted levels of adjusted operating profit margin and wafer fabrication equipment market share, and the number of shares that may vest in full after three years ranges from 0% to 200% of the target amount. Performance-based awards granted in fiscal 2016 require the achievement of targeted levels of adjusted annual operating profit margin, and additional shares become eligible for time-based vesting if Applied achieves certain levels of total shareholder return relative to a peer group, comprised of companies in the Standard & Poor's 500 Information Technology Index, measured at the end of a two-year period.

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**APPLIED MATERIALS, INC.**

**NOTES TO CONSOLIDATED FINANCIAL STATEMENTS — (Continued)**

A summary of the changes in restricted stock units, restricted stock, performance shares and performance units outstanding under Applied's equity compensation plans is presented below:

| | Shares | Weighted Average Grant Date Fair Value | Weighted Average Remaining Contractual Term | Aggregate Intrinsic Value |
|---|---|---|---|---|
| | (In millions, except per share amounts) | | | |
| Non-vested restricted stock units, restricted stock, performance shares and performance units at October 25, 2015 | 27 | $ 16.41 | 2.2 years | $ 440 |
| Granted | 11 | $ 18.54 | | |
| Vested | (11) | $ 14.25 | | |
| Canceled | (2) | $ 17.57 | | |
| Non-vested restricted stock units, restricted stock, performance shares and performance units at October 30, 2016 | 25 | $ 18.28 | 2.3 years | $ 718 |
| Granted | 8 | $ 31.79 | | |
| Vested | (10) | $ 16.50 | | |
| Canceled | (1) | $ 21.25 | | |
| Non-vested restricted stock units, restricted stock, performance shares and performance units at October 29, 2017 | 22 | $ 23.96 | 2.2 years | $ 1,239 |
| Granted | 6 | $ 50.62 | | |
| Vested | (9) | $ 22.15 | | |
| Canceled | (1) | $ 30.19 | | |
| Non-vested restricted stock units, restricted stock, performance shares and performance units at October 28, 2018 | 18 | $ 32.64 | 2.0 years | $ 600 |
| Non-vested restricted stock units, restricted stock, performance shares and performance units expected to vest | 16 | $ 31.11 | 1.9 years | $ 523 |

At October 28, 2018, 1 million additional performance-based awards could be earned based upon achievement of certain levels of specified performance goals.

*Employee Stock Purchase Plans*

Under the ESPP, substantially all employees may purchase Applied common stock through payroll deductions at a price equal to 85 percent of the lower of the fair market value of Applied common stock at the beginning or end of each 6-month purchase period, subject to certain limits. Applied issued 3 million shares in each of fiscal 2018 and 2017, and 6 million shares during fiscal 2016, under the ESPP. Compensation expense is calculated using the fair value of the employees' purchase rights under the Black-Scholes model. Underlying assumptions used in the model are outlined in the following table:

| | 2018 | 2017 | 2016 |
|---|---|---|---|
| **ESPP:** | | | |
| Dividend yield | 1.68% | 0.99% | 1.76% |
| Expected volatility | 34.4% | 26.3% | 29.3% |
| Risk-free interest rate | 2.09% | 0.92% | 0.47% |
| Expected life (in years) | 0.5 | 0.5 | 0.5 |
| Weighted average estimated fair value | $12.02 | $9.14 | $5.48 |

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## Note 12 Employee Benefit Plans

*Employee Bonus Plans*

Applied has various employee bonus plans. A discretionary bonus plan provides for the distribution of a percentage of pre-tax income to Applied employees who are not participants in other performance-based incentive plans, up to a maximum percentage of eligible compensation. Other plans provide for bonuses to Applied's executives and other key contributors based on the achievement of profitability and/or other specified performance criteria. Charges under these plans for fiscal 2018, 2017 and 2016 were $382 million, $449 million and $312 million, respectively.

*Employee Savings and Retirement Plan*

Applied's Employee Savings and Retirement Plan (the 401(k) Plan) is qualified under Sections 401(a) and (k) of the Internal Revenue Code (the Code). Eligible employees may make salary deferral and catch-up contributions under the 401(k) Plan on a pre-tax basis and on a Roth basis, subject to an annual dollar limit established by the Code. Applied matches 100% of participant salary and/or Roth deferral contributions up to the first 3% of eligible contribution and then 50% of every dollar between 4% and 6% of eligible contribution. Applied does not make matching contributions on any catch-up contributions made by participants. Plan participants who were employed by Applied or any of its affiliates became 100% vested in their Applied matching contribution account balances. Applied's matching contributions under the 401(k) Plan were approximately $45 million for fiscal 2018 and $38 million for each of fiscal 2017 and 2016.

*Defined Benefit Pension Plans of Foreign Subsidiaries and Other Post-Retirement Benefits*

Several of Applied's foreign subsidiaries have defined benefit pension plans covering substantially all of their eligible employees. Benefits under these plans are typically based on years of service and final average compensation levels. The plans are managed in accordance with applicable local statutes and practices. Applied deposits funds for certain of these plans with insurance companies, pension trustees, government-managed accounts, and/or accrues the expense for the unfunded portion of the benefit obligation on its Consolidated Financial Statements. Applied's practice is to fund the various pension plans in amounts sufficient to meet the minimum requirements as established by applicable local governmental oversight and taxing authorities. Depending on the design of the plan, local custom and market circumstances, the liabilities of a plan may exceed the qualified plan assets. The differences between the aggregate projected benefit obligations and aggregate plan assets of these plans have been recorded as liabilities by Applied and are included in other liabilities and accrued expenses in the Consolidated Balance Sheets.

Through December 31, 2017, Applied also sponsored a U.S. post-retirement plan that provided covered medical and vision benefits to certain eligible retirees. An eligible retiree also could elect coverage for an eligible spouse or domestic partner who was not eligible for Medicare. Coverage ended entirely for all participants when the plan terminated on December 31, 2017. In addition, Applied also has a post-retirement benefit plan as a result of the acquisition of Varian. Applied's liability under these post-retirement plans, which was included in other liabilities in the Consolidated Balance Sheets, were immaterial at each of October 28, 2018 and October 29, 2017.

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**APPLIED MATERIALS, INC.**

**NOTES TO CONSOLIDATED FINANCIAL STATEMENTS — (Continued)**

A summary of the changes in benefit obligations and plan assets, which includes post-retirement benefits, for each fiscal year is presented below:

| | 2018 | 2017 | 2016 |
|---|---|---|---|
| | (In millions, except percentages) | | |
| **Change in projected benefit obligation** | | | |
| Beginning projected benefit obligation | $ 506 | $ 495 | $ 471 |
| Service cost | 12 | 13 | 13 |
| Interest cost | 11 | 10 | 13 |
| Plan participants' contributions | 1 | 2 | 1 |
| Actuarial (gain) loss | 24 | (35) | 77 |
| Curtailments, settlements and special termination benefits | (1) | (1) | (6) |
| Foreign currency exchange rate changes | (16) | 34 | (42) |
| Benefits paid | (12) | (12) | (10) |
| Plan amendments and business combinations | (1) | — | (22) |
| Ending projected benefit obligation | $ 524 | $ 506 | $ 495 |
| **Ending accumulated benefit obligation** | $ 490 | $ 472 | $ 460 |
| **Range of assumptions to determine benefit obligations** | | | |
| Discount rate | 0.6% - 3.1% | 0.5% - 3.4% | 0.5% - 3.1% |
| Rate of compensation increase | 2.4% - 3.5% | 2.2% - 3.5% | 1.6% - 3.6% |
| **Change in plan assets** | | | |
| Beginning fair value of plan assets | $ 361 | $ 310 | $ 281 |
| Return on plan assets | 17 | 18 | 37 |
| Employer contributions | 11 | 16 | 50 |
| Plan participants' contributions | 1 | 2 | 1 |
| Foreign currency exchange rate changes | (12) | 28 | (45) |
| Divestitures, settlements and business combinations | (1) | (1) | (4) |
| Benefits paid | (12) | (12) | (10) |
| Ending fair value of plan assets | $ 365 | $ 361 | $ 310 |
| Funded status | $ (159) | $ (145) | $ (185) |
| **Amounts recognized in the consolidated balance sheets** | | | |
| Noncurrent asset | $ 19 | $ 17 | $ 11 |
| Current liability | (1) | (1) | (2) |
| Noncurrent liability | (177) | (161) | (194) |
| Total | $ (159) | $ (145) | $ (185) |
| **Estimated amortization from accumulated other comprehensive loss into net periodic benefit cost over the next fiscal period** | | | |
| Actuarial loss | $ 8 | $ 6 | $ 6 |
| Prior service credit | (1) | (4) | (16) |
| Total | $ 7 | $ 2 | $ (10) |
| **Amounts recognized in accumulated other comprehensive loss** | | | |
| Net actuarial loss | $ 161 | $ 141 | $ 186 |
| Prior service credit | (2) | (4) | (21) |
| Total | $ 159 | $ 137 | $ 165 |
| **Plans with projected benefit obligations in excess of plan assets** | | | |
| Projected benefit obligation | $ 365 | $ 326 | $ 341 |
| Fair value of plan assets | $ 186 | $ 142 | $ 145 |

| | | | |
|---|---|---|---|
| **Plans with accumulated benefit obligations in excess of plan assets** | | | |
| Accumulated benefit obligation | $ 331 | $ 293 | $ 307 |
| Fair value of plan assets | $ 186 | $ 142 | $ 145 |

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**APPLIED MATERIALS, INC.**

**NOTES TO CONSOLIDATED FINANCIAL STATEMENTS — (Continued)**

| | 2018 | 2017 |
|---|---|---|
| **Plan assets — allocation** | | |
| Equity securities | 47% | 47% |
| Debt securities | 32% | 39% |
| Insurance contracts | 10% | 11% |
| Other investments | 10% | 3% |
| Cash | 1% | —% |

The following table presents a summary of the ending fair value of the plan assets:

| | October 28, 2018 | | | | October 29, 2017 | | | |
|---|---|---|---|---|---|---|---|---|
| | Level 1 | Level 2 | Level 3 | Total | Level 1 | Level 2 | Level 3 | Total |
| | (In millions) | | | | | | | |
| Equity securities | $ 86 | $ — | $ — | $ 86 | $ 83 | $ — | $ — | $ 83 |
| Debt securities | 19 | — | — | 19 | 16 | — | — | 16 |
| Insurance contracts | — | — | 36 | 36 | — | — | 38 | 38 |
| Other investments | — | 14 | — | 14 | — | 13 | — | 13 |
| Cash | 2 | — | — | 2 | 2 | — | — | 2 |
| Total assets at fair value | $ 107 | $ 14 | $ 36 | 157 | $ 101 | $ 13 | $ 38 | 152 |
| Assets measured at net asset value | | | | 208 | | | | 209 |
| Total | | | | $ 365 | | | | $ 361 |

The following table presents the activity in Level 3 instruments for each fiscal year:

| | 2018 | 2017 |
|---|---|---|
| | (In millions) | |
| Balance, beginning of year | $ 38 | $ 38 |
| Actual return on plan assets: | | |
| Relating to assets still held at reporting date | (1) | (3) |
| Purchases, sales, settlements, net | — | 1 |
| Currency impact | (1) | 2 |
| Balance, end of year | $ 36 | $ 38 |

Applied's investment strategy for its defined benefit plans is to invest plan assets in a prudent manner, maintaining well-diversified portfolios with the long-term objective of meeting the obligations of the plans as they come due. Asset allocation decisions are typically made by plan fiduciaries with input from Applied's international pension committee. Applied's asset allocation strategy incorporates a sufficient equity exposure in order for the plans to benefit from the expected better long-term performance of equities relative to the plans' liabilities. Applied retains investment managers, where appropriate, to manage the assets of the plans. Performance of investment managers is monitored by plan fiduciaries with the assistance of local investment consultants. The investment managers make investment decisions within the guidelines set forth by plan fiduciaries. Risk management practices include diversification across asset classes and investment styles, and periodic rebalancing toward target asset allocation ranges. Investment managers may use derivative instruments for efficient portfolio management purposes. Plan assets do not include any of Applied's own equity or debt securities.

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**APPLIED MATERIALS, INC.**

**NOTES TO CONSOLIDATED FINANCIAL STATEMENTS — (Continued)**

A summary of the components of net periodic benefit costs and the weighted average assumptions used for net periodic benefit cost calculations for each fiscal year is presented below:

| | 2018 | 2017 | 2016 |
|---|---|---|---|
| | (In millions, except percentages) | | |
| **Components of net periodic benefit cost** | | | |
| Service cost | $ 12 | $ 13 | $ 13 |
| Interest cost | 11 | 10 | 13 |
| Expected return on plan assets | (20) | (18) | (14) |
| Amortization of actuarial loss and prior service credit | 3 | (10) | 3 |
| Settlement and curtailment loss | — | — | (5) |
| Net periodic benefit cost (income) | $ 6 | $ (5) | $ 10 |
| **Weighted average assumptions** | | | |
| Discount rate | 2.16% | 1.88% | 2.82% |
| Expected long-term return on assets | 5.41% | 5.38% | 5.38% |
| Rate of compensation increase | 2.66% | 2.69% | 2.71% |

Asset return assumptions are derived based on actuarial and statistical methodologies, from analysis of long-term historical data relevant to the country in which each plan is in effect and the investments applicable to the corresponding plan. The discount rate for each plan was derived by reference to appropriate benchmark yields on high quality corporate bonds, allowing for the approximate duration of both plan obligations and the relevant benchmark yields.

Future expected benefit payments for the pension plans and the post-retirement plan over the next ten fiscal years are as follows:

| | Benefit Payments |
|---|---|
| | (In millions) |
| 2019 | $ 12 |
| 2020 | 12 |
| 2021 | 13 |
| 2022 | 13 |
| 2023 | 12 |
| 2024-2028 | 77 |
| | $ 139 |

Company contributions to these plans for fiscal 2019 are expected to be approximately $15 million.

*Executive Deferred Compensation Plans*

Applied sponsors two unfunded deferred compensation plans, the Executive Deferred Compensation Plan (Predecessor EDCP) and the 2016 Deferred Compensation Plan (2016 DCP) (formerly known as the 2005 Executive Deferred Compensation Plan), under which certain employees may elect to defer a portion of their following year's eligible earnings. The Predecessor EDCP was frozen as of December 31, 2004 such that no new deferrals could be made under the plan after that date and the plan would qualify for "grandfather" relief under Section 409A of the Code. The Predecessor EDCP participant accounts continue to be maintained under the plan and credited with deemed interest. The 2016 DCP was originally implemented by Applied effective as of January 1, 2005, and amended and restated as of October 12, 2015, and is intended to comply with the requirements of Section 409A of the Code. In addition, Applied also sponsors a non-qualified deferred compensation plan as a result of the acquisition of Varian. Amounts payable, including accrued deemed interest, totaled $92 million and $63 million at October 28, 2018 and October 29, 2017, respectively, which were included in other liabilities in the Consolidated Balance Sheets.

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**APPLIED MATERIALS, INC.**

**NOTES TO CONSOLIDATED FINANCIAL STATEMENTS — (Continued)**

## Note 13 Income Taxes

The components of income before income taxes for each fiscal year were as follows:

| | 2018 | 2017 | 2016 |
|---|---|---|---|
| | | (In millions) | |
| U.S. | $ 439 | $ 514 | $ 199 |
| Foreign | 4,255 | 3,217 | 1,814 |
| | $ 4,694 | $ 3,731 | $ 2,013 |

The components of the provision for income taxes for each fiscal year were as follows:

| | 2018 | 2017 | 2016 |
|---|---|---|---|
| | | (In millions) | |
| **Current:** | | | |
| U.S. | $ 1,035 | $ 67 | $ (36) |
| Foreign | 126 | 233 | 351 |
| State | 22 | 9 | (2) |
| | 1,183 | 309 | 313 |
| **Deferred:** | | | |
| U.S. | 151 | (11) | 55 |
| Foreign | 57 | (7) | (89) |
| State | (10) | 6 | 13 |
| | 198 | (12) | (21) |
| | $ 1,381 | $ 297 | $ 292 |

A reconciliation between the statutory U.S. federal income tax rate and Applied's actual effective income tax rate for each fiscal year is presented below:

| | 2018 | 2017 | 2016 |
|---|---|---|---|
| Tax provision at U.S. statutory rate | 23.4 % | 35.0 % | 35.0 % |
| Changes in U.S. tax law | 23.7 | — | — |
| Resolutions of prior years' income tax filings | (0.7) | (1.9) | 3.9 |
| Effect of foreign operations taxed at various rates | (15.6) | (24.9) | (24.1) |
| State income taxes, net of federal benefit | 0.2 | 0.3 | 0.6 |
| Research and other tax credits | (0.7) | (0.7) | (1.3) |
| U.S. domestic production deduction | (0.1) | (0.2) | (0.2) |
| Share-based compensation | (0.7) | 0.4 | 0.4 |
| Other | (0.1) | — | 0.2 |
| | 29.4 % | 8.0 % | 14.5 % |

On December 22, 2017, the U.S. government enacted the Tax Cuts and Jobs Act (the "Tax Act"). The Tax Act includes a reduction to the U.S. federal corporate tax rate from 35.0 percent to 21.0 percent and requires a one-time transition tax on certain unrepatriated earnings of foreign subsidiaries payable over eight years. U.S. deferred tax assets and liabilities were subject to remeasurement due to the reduction of the U.S. federal corporate tax rate. Applied has a blended U.S. federal corporate tax rate of 23.4 percent for fiscal 2018 based on the number of days before and after the effective date of the Tax Act.

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**APPLIED MATERIALS, INC.**

**NOTES TO CONSOLIDATED FINANCIAL STATEMENTS — (Continued)**

The U.S. Securities and Exchange Commission issued Staff Accounting Bulletin No. 118 (SAB 118), which provides guidance on accounting for the income tax effects of the Tax Act. SAB 118 provides a measurement period for companies to complete this accounting. Pursuant to SAB 118, provisional adjustments were recorded when reasonable estimates could be determined. These provisional estimates will be revised as information becomes available and as guidance is issued by the Internal Revenue Service. The accounting for the income tax effects of the Tax Act will be completed during the measurement period, which will not extend beyond one year from the Tax Act enactment date. Applied continues to evaluate certain unrepatriated earnings of foreign subsidiaries used to calculate the transition tax. The remeasurement of U.S. deferred tax assets and liabilities is complete.

The Tax Act also includes provisions that do not affect Applied in fiscal 2018, including a provision designed to tax global intangible low-taxed income ("GILTI"). Due to the complexity of the GILTI tax rules, this provision and related tax accounting will continue to be evaluated. An accounting policy choice is allowed to either treat taxes due on future U.S. inclusions related to GILTI in taxable income as a current-period expense when incurred (the "period cost method") or factor such amounts into the measurement of deferred taxes (the "deferred method"). The calculation of the deferred balance with respect to the new GILTI tax provisions will depend, in part, on analyzing global income to determine whether future U.S. inclusions in taxable income are expected related to GILTI and, if so, what the impact is expected to be. An accounting policy choice has not yet been made.

Before the Tax Act, U.S. income tax had not been provided for certain unrepatriated earnings that were considered indefinitely reinvested. Income tax is now provided for all unrepatriated earnings.

The effective tax rate for fiscal 2018 was higher than fiscal 2017 primarily due to tax expense of $1.1 billion for the transition tax and remeasurement of deferred tax assets as a result of the Tax Act, partially offset by changes in the geographical composition of income, tax benefits from the lower U.S. federal corporate tax rate, adoption of authoritative guidance for share-based compensation, and the resolution of tax liabilities for uncertain tax positions. In addition, fiscal 2017 included tax benefits from the recognition of previously unrecognized foreign tax credits.

The effective tax rate for fiscal 2017 was lower than fiscal 2016 primarily due to the recognition of previously unrecognized foreign tax credits and changes in the geographical composition of income. In addition, fiscal 2016 included unfavorable resolutions and changes related to income tax liabilities for uncertain tax positions as well as the reinstatement of the U.S. federal R&D tax credit retroactive to its expiration in December of 2015 which did not reoccur in fiscal 2017.

In the reconciliation between the statutory U.S. federal income tax rate and the effective income tax rate, the effect of foreign operations taxed at various rates represents the difference between an income tax provision at the U.S. federal statutory income tax rate and the recorded income tax provision, with the difference expressed as a percentage of worldwide income before income taxes. This effect is substantially related to the tax effect of pre-tax income in jurisdictions with lower statutory tax rates. The foreign operations with the most significant effective tax rate impact are in Singapore. The statutory tax rate for fiscal 2018 for Singapore is 17%. Applied has been granted conditional reduced tax rates that expire in fiscal 2025, excluding potential renewal and subject to certain conditions with which Applied expects to comply. The tax benefit arising from these tax rates was $272 million for fiscal 2018 or $0.28 per diluted share.

Deferred tax assets and liabilities are recognized for the estimated future tax effects of temporary differences between the book and tax bases of assets and liabilities. Deferred tax assets are also recognized for net operating loss and tax credit carryovers. Deferred tax assets are offset by a valuation allowance to the extent it is more likely than not that they are not expected to be realized. The components of deferred income tax assets and liabilities were as follows:

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**APPLIED MATERIALS, INC.**

**NOTES TO CONSOLIDATED FINANCIAL STATEMENTS — (Continued)**

| | October 28, 2018 | October 29, 2017 |
|---|---|---|
| | (In millions) | |
| **Deferred tax assets:** | | |
| Allowance for doubtful accounts | $ 8 | $ 13 |
| Inventory reserves and basis difference | 117 | 156 |
| Installation and warranty reserves | 7 | 1 |
| Accrued liabilities | 20 | 31 |
| Deferred revenue | 9 | 15 |
| Tax credits | 236 | 317 |
| Deferred compensation | 79 | 81 |
| Share-based compensation | 37 | 53 |
| Other | 48 | 67 |
| Gross deferred tax assets | 561 | 734 |
| Valuation allowance | (230) | (227) |
| Total deferred tax assets | 331 | 507 |
| **Deferred tax liabilities:** | | |
| Fixed assets | (48) | (36) |
| Intangible assets | (38) | (76) |
| Undistributed foreign earnings | (32) | (11) |
| Foreign exchange | (3) | (4) |
| Total gross deferred tax liabilities | (121) | (127) |
| Net deferred tax assets | $ 210 | $ 380 |

The following table presents a summary of non-current deferred tax assets and liabilities:

| | October 28, 2018 | October 29, 2017 |
|---|---|---|
| | (In millions) | |
| Non-current deferred tax asset | $ 222 | $ 385 |
| Non-current deferred tax liability | (12) | (5) |
| | $ 210 | $ 380 |

A valuation allowance is recorded to reflect the estimated amount of net deferred tax assets that may not be realized. Changes in the valuation allowance in each fiscal year were as follows:

| | 2018 | 2017 | 2016 |
|---|---|---|---|
| | | (In millions) | |
| Beginning balance | $ 227 | $ 207 | $ 207 |
| Increases | 8 | 20 | 27 |
| Decreases | (5) | — | (27) |
| Ending balance | $ 230 | $ 227 | $ 207 |

At October 28, 2018, Applied has state research and development tax credit carryforwards of $236 million, including $227 million of credits that are carried over until exhausted and $9 million that are carried over for 15 years and begin to expire in fiscal 2028. Management believes it is more likely than not that all tax credit carryforwards, net of valuation allowance, will be utilized.

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**APPLIED MATERIALS, INC.**

**NOTES TO CONSOLIDATED FINANCIAL STATEMENTS — (Continued)**

Applied maintains liabilities for uncertain tax positions. These liabilities involve considerable judgment and estimation and are continuously monitored by management based on the best information available. Gross unrecognized tax benefits are classified as non-current income taxes payable. A reconciliation of the beginning and ending balances of gross unrecognized tax benefits in each fiscal year is as follows:

| | 2018 | 2017 | 2016 |
|---|---|---|---|
| | | (In millions) | |
| Beginning balance of gross unrecognized tax benefits | $ 391 | $ 320 | $ 177 |
| Settlements with tax authorities | (152) | (42) | (25) |
| Lapses of statutes of limitation | (37) | (15) | (2) |
| Increases in tax positions for current year | 91 | 95 | 62 |
| Increases in tax positions for prior years | 83 | 33 | 109 |
| Decreases in tax positions for prior years | (2) | — | (1) |
| Ending balance of gross unrecognized tax benefits | $ 374 | $ 391 | $ 320 |

Tax expense for interest and penalties on unrecognized tax benefits for fiscal 2018, 2017 and 2016 was $12 million, $17 million and $24 million, respectively. The income tax liability for interest and penalties for fiscal 2018, 2017 and 2016 was $26 million, $46 million and $33 million, respectively, and was classified as non-current income taxes payable.

Included in the balance of unrecognized tax benefits for fiscal 2018, 2017 and 2016 are $294 million, $284 million, and $302 million, respectively, of tax benefits that, if recognized, would affect the effective tax rate.

In fiscal 2018, Applied paid $158 million, including interest and penalties, as a result of a settlement in Israel for fiscal 2011 through fiscal 2015 resulting in the recognition of a tax benefit of $6 million. In fiscal 2017, Applied paid $29 million, including interest and penalties, as a result of a settlement in Italy for fiscal 2011 resulting in the recognition of a tax expense of $6 million. In fiscal 2016, Applied paid $25 million, including interest and penalties, as a result of a settlement in Switzerland for fiscal 2011 through fiscal 2015 resulting in the recognition of a tax expense of $19 million.

Applied's tax returns remain subject to examination by taxing authorities. These include U.S. returns for fiscal 2010 and later years, and foreign tax returns for fiscal 2009 and later years.

The timing of the resolution of income tax examinations, as well as the amounts and timing of various tax payments that may be part of the settlement process, is highly uncertain. This could cause fluctuations in Applied's financial condition and results of operations. Applied continues to have ongoing negotiations with various taxing authorities throughout the year.

## Note 14 Warranty, Guarantees, Commitments and Contingencies

### *Leases*

Applied leases some of its facilities and equipment under non-cancelable operating leases and has options to renew most leases, with rentals to be negotiated. Total rent expense for fiscal 2018, 2017 and 2016, was $50 million, $34 million and $38 million, respectively.

As of October 28, 2018, future minimum lease payments are expected to be as follows:

| Fiscal | Lease Payments (In millions) |
|---|---|
| 2019 | $ 50 |
| 2020 | 40 |
| 2021 | 28 |
| 2022 | 20 |
| 2023 | 13 |
| Thereafter | 22 |
| | $ 173 |

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**APPLIED MATERIALS, INC.**

**NOTES TO CONSOLIDATED FINANCIAL STATEMENTS — (Continued)**

***Warranty***

Changes in the warranty reserves during each fiscal year were as follows:

| | 2018 | 2017 | 2016 |
|---|---|---|---|
| | | (In millions) | |
| Beginning balance | $ 199 | $ 153 | $ 126 |
| Provisions for warranty | 183 | 166 | 135 |
| Changes in reserves related to preexisting warranty | 3 | 1 | (12) |
| Consumption of reserves | (176) | (121) | (96) |
| Ending balance | $ 209 | $ 199 | $ 153 |

Applied products are generally sold with a warranty for a 12-month period following installation. The provision for the estimated cost of warranty is recorded when revenue is recognized. Parts and labor are covered under the terms of the warranty agreement. The warranty provision is based on historical experience by product, configuration and geographic region. Quarterly warranty consumption is generally associated with sales that occurred during the preceding four quarters, and quarterly warranty provisions are generally related to the current quarter's sales.

***Guarantees***

In the ordinary course of business, Applied provides standby letters of credit or other guarantee instruments to third parties as required for certain transactions initiated by either Applied or its subsidiaries. As of October 28, 2018, the maximum potential amount of future payments that Applied could be required to make under these guarantee agreements was approximately $58 million. Applied has not recorded any liability in connection with these guarantee agreements beyond that required to appropriately account for the underlying transaction being guaranteed. Applied does not believe, based on historical experience and information currently available, that it is probable that any amounts will be required to be paid under these guarantee agreements.

Applied also has agreements with various banks to facilitate subsidiary banking operations worldwide, including overdraft arrangements, issuance of bank guarantees, and letters of credit. As of October 28, 2018, Applied has provided parent guarantees to banks for approximately $149 million to cover these arrangements.

***Legal Matters***

From time to time, Applied receives notification from third parties, including customers and suppliers, seeking indemnification, litigation support, payment of money or other actions by Applied in connection with claims made against them. In addition, from time to time, Applied receives notification from third parties claiming that Applied may be or is infringing or misusing their intellectual property or other rights. Applied also is subject to various other legal proceedings and claims, both asserted and unasserted, that arise in the ordinary course of business.

Although the outcome of the above-described matters, claims and proceedings cannot be predicted with certainty, Applied does not believe that any will have a material effect on its consolidated financial condition or results of operations.

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**Note 15 Industry Segment Operations**

Applied's three reportable segments are: Semiconductor Systems, Applied Global Services, and Display and Adjacent Markets. As defined under the accounting literature, Applied's chief operating decision-maker has been identified as the President and Chief Executive Officer, who reviews operating results to make decisions about allocating resources and assessing performance for the entire Company. Segment information is presented based upon Applied's management organization structure as of October 28, 2018 and the distinctive nature of each segment. Future changes to this internal financial structure may result in changes to Applied's reportable segments.

The Semiconductor Systems reportable segment includes semiconductor capital equipment for etch, rapid thermal processing, deposition, chemical mechanical planarization, metrology and inspection, wafer packaging, and ion implantation.

The Applied Global Services segment provides integrated solutions to optimize equipment and fab performance and productivity, including spares, upgrades, services, certain remanufactured earlier generation equipment and factory automation software for semiconductor, display and other products.

The Display and Adjacent Markets segment includes products for manufacturing liquid crystal displays (LCDs), organic light-emitting diodes (OLEDs), equipment upgrades and flexible coating systems and other display technologies for TVs, monitor, laptops, personal computers, smart phones, and other consumer-oriented devices.

Each operating segment is separately managed and has separate financial results that are reviewed by Applied's chief operating decision-maker. Each reportable segment contains closely related products that are unique to the particular segment. Segment operating income is determined based upon internal performance measures used by Applied's chief operating decision-maker. The chief operating decision-maker does not evaluate operating segments using total asset information.

Applied derives the segment results directly from its internal management reporting system. The accounting policies Applied uses to derive reportable segment results are substantially the same as those used for external reporting purposes. Management measures the performance of each reportable segment based upon several metrics including orders, net sales and operating income. Management uses these results to evaluate the performance of, and to assign resources to, each of the reportable segments.

The Corporate and Other category includes revenues from products, as well as costs of products sold, for fabricating solar photovoltaic cells and modules, and certain operating expenses that are not allocated to its reportable segments and are managed separately at the corporate level. These operating expenses include costs related to share-based compensation; certain management, finance, legal, human resources, and research, development and engineering functions provided at the corporate level; and unabsorbed information technology and occupancy. In addition, Applied does not allocate to its reportable segments restructuring and asset impairment charges and any associated adjustments related to restructuring actions, unless these actions pertain to a specific reportable segment. Segment operating income also excludes interest income/expense and other financial charges and income taxes. Management does not consider the unallocated costs in measuring the performance of the reportable segments.

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**APPLIED MATERIALS, INC.**

**NOTES TO CONSOLIDATED FINANCIAL STATEMENTS — (Continued)**

Information for each reportable segment for and as of the end of each fiscal year were as follows:

| | Net Sales | Operating Income (Loss) | Depreciation/ Amortization | Capital Expenditures | Accounts Receivable | Inventories |
|---|---|---|---|---|---|---|
| | | | (In millions) | | | |
| 2018: | | | | | | |
| Semiconductor Systems | $ 10,903 | $ 3,634 | $ 303 | $ 168 | $ 1,755 | $ 2,213 |
| Applied Global Services | 3,754 | 1,102 | 21 | 33 | 610 | 1,243 |
| Display and Adjacent Markets | 2,498 | 679 | 20 | 39 | 248 | 249 |
| Corporate and Other | 98 | (619) | 113 | 382 | (48) | 17 |
| Total | $ 17,253 | $ 4,796 | $ 457 | $ 622 | $ 2,565 | $ 3,722 |
| 2017: | | | | | | |
| Semiconductor Systems | $ 9,517 | $ 3,173 | $ 286 | $ 150 | $ 1,626 | $ 1,760 |
| Applied Global Services | 3,017 | 817 | 15 | 21 | 564 | 762 |
| Display and Adjacent Markets | 1,900 | 502 | 12 | 17 | 190 | 367 |
| Corporate and Other | 103 | (624) | 94 | 157 | (42) | 41 |
| Total | $ 14,537 | $ 3,868 | $ 407 | $ 345 | $ 2,338 | $ 2,930 |
| 2016: | | | | | | |
| Semiconductor Systems | $ 6,873 | $ 1,807 | $ 277 | $ 114 | $ 1,524 | $ 1,188 |
| Applied Global Services | 2,589 | 682 | 12 | 14 | 559 | 594 |
| Display and Adjacent Markets | 1,206 | 245 | 5 | 6 | 238 | 215 |
| Corporate and Other | 157 | (582) | 95 | 119 | (42) | 53 |
| Total | $ 10,825 | $ 2,152 | $ 389 | $ 253 | $ 2,279 | $ 2,050 |

The reconciling items included in Corporate and Other were as follows:

| | 2018 | 2017 | 2016 |
|---|---|---|---|
| | | (In millions) | |
| Unallocated net sales | $ 98 | $ 103 | $ 157 |
| Unallocated cost of products sold and expenses | (459) | (507) | (538) |
| Share-based compensation | (258) | (220) | (201) |
| Total | $ (619) | $ (624) | $ (582) |

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**APPLIED MATERIALS, INC.**

**NOTES TO CONSOLIDATED FINANCIAL STATEMENTS — (Continued)**

For geographical reporting, revenue by geographic location is determined by the location of customers' facilities to which products were shipped. Long-lived assets consist primarily of property, plant and equipment and are attributed to the geographic location in which they are located. Net sales and long-lived assets by geographic region for and as of each fiscal year were as follows:

| | 2018 | 2017 | 2016 |
|---|---|---|---|
| | | (In millions) | |
| Net sales: | | | |
| United States | $ 1,532 | $ 1,474 | $ 1,143 |
| China | 5,113 | 2,746 | 2,259 |
| Korea | 3,603 | 4,052 | 1,883 |
| Taiwan | 2,732 | 3,291 | 2,843 |
| Japan | 2,405 | 1,518 | 1,279 |
| Europe | 1,066 | 816 | 615 |
| Southeast Asia | 802 | 640 | 803 |
| Total outside United States | 15,721 | 13,063 | 9,682 |
| Consolidated total | $ 17,253 | $ 14,537 | $ 10,825 |

| | October 28, 2018 | October 29, 2017 |
|---|---|---|
| | (In millions) | |
| Long-lived assets: | | |
| United States | $ 1,414 | $ 915 |
| China | 13 | 47 |
| Korea | 21 | 21 |
| Taiwan | 29 | 50 |
| Japan | 9 | 8 |
| Europe | 50 | 47 |
| Southeast Asia | 25 | 98 |
| Total outside United States | 147 | 271 |
| Consolidated total | $ 1,561 | $ 1,186 |

The following customers accounted for at least 10 percent of Applied's net sales in each fiscal year, which were for products and services in multiple reportable segments:

| | 2018 | 2017 | 2016 |
|---|---|---|---|
| Samsung Electronics Co., Ltd. | 13% | 23% | 13% |
| Taiwan Semiconductor Manufacturing Company Limited | 11% | 15% | 16% |
| Intel Corporation | 11% | * | 11% |
| Micron Technology, Inc. | * | * | 11% |

* Less than 10%

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**APPLIED MATERIALS, INC.**

**NOTES TO CONSOLIDATED FINANCIAL STATEMENTS — (Continued)**

## Note 16 Unaudited Quarterly Consolidated Financial Data

| | Fiscal Quarter | | | | |
|---|---|---|---|---|---|
| | First | Second | Third | Fourth | Fiscal Year |
| | (In millions, except per share amounts) | | | | |
| **2018:** | | | | | |
| Net sales | $ 4,204 | $ 4,567 | $ 4,468 | $ 4,014 | $ 17,253 |
| Gross profit | $ 1,920 | $ 2,090 | $ 2,027 | $ 1,780 | $ 7,817 |
| Net income | $ 135 | $ 1,129 | $ 1,173 | $ 876 | $ 3,313 |
| Earnings per diluted share | $ 0.13 | $ 1.09 | $ 1.17 | $ 0.89 | $ 3.23 |
| **2017:** | | | | | |
| Net sales | $ 3,278 | $ 3,546 | $ 3,744 | $ 3,969 | $ 14,537 |
| Gross profit | $ 1,445 | $ 1,600 | $ 1,700 | $ 1,787 | $ 6,532 |
| Net income | $ 703 | $ 824 | $ 925 | $ 982 | $ 3,434 |
| Earnings per diluted share | $ 0.65 | $ 0.76 | $ 0.85 | $ 0.91 | $ 3.17 |

## INDEX TO EXHIBITS

These Exhibits are numbered in accordance with the Exhibit Table of Item 601 of Regulation S-K:

| | | Incorporated by Reference | | | |
|---|---|---|---|---|---|
| Exhibit No. | Description | Form | File No. | Exhibit No. | Filing Date |
| 3.1 | Certificate of Incorporation of Applied Materials, Inc., as amended and restated through March 10, 2009 | 10-Q | 000-06920 | 3.1 | 6/3/2009 |
| 3.2 | Certificate of Designation, Preferences and Rights of the Terms of the Series A Junior Participating Preferred Stock dated as of July 9, 1999 | 10-Q | 000-06920 | 3(i)(a) | 9/14/1999 |
| 3.3 | Amended and Restated Bylaws of Applied Materials, Inc., amended as of December 8, 2015 | 8-K | 000-06920 | 3.1 | 12/11/2015 |
| 4.1 | Indenture, dated June 8, 2011, by and between Applied Materials, Inc. and U.S. Bank National Association, as Trustee | 8-K | 000-06920 | 4.1 | 6/10/2011 |
| 4.2 | First Supplemental Indenture, dated June 8, 2011, by and between Applied Materials, Inc. and U.S. Bank National Association, as Trustee | 8-K | 000-06920 | 4.2 | 6/10/2011 |
| 4.3 | Second Supplemental Indenture, dated September 24, 2015, by and between Applied Materials, Inc. and U.S. Bank National Association, as Trustee | 8-K | 000-06920 | 4.1 | 9/24/2015 |
| 4.4 | Third Supplemental Indenture, dated March 31, 2017, by and between Applied Materials, Inc. and U.S. Bank National Association, as Trustee | 8-K | 000-06920 | 4.1 | 3/31/2017 |
| 10.1 | Form of Indemnification Agreement between Applied Materials, Inc. and Non-Employee Directors | 10-K | 000-06920 | 10.44 | 1/31/2000 |
| 10.2 | Form of Indemnification Agreement between Applied Materials, Inc. and certain of its officers | 10-K | 000-06920 | 10.46 | 1/31/2000 |
| 10.3 | Applied Materials, Inc. Profit Sharing Scheme (Ireland) | S-8 | 333-45011 | 4.1 | 1/27/1998 |
| 10.4* | Applied Materials, Inc. amended and restated Relocation Policy | 8-K | 000-06920 | 10.46 | 10/31/2005 |
| 10.5* | Applied Materials Inc. Employee Financial Assistance Plan, amended and restated as of December 18, 2008 | 10-Q | 000-06920 | 10.58 | 3/3/2009 |

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| Exhibit No. | Description | Incorporated by Reference | | | |
|---|---|---|---|---|---|
| | | Form | File No. | Exhibit No. | Filing Date |
| 10.6 | Deed of Amendment to Applied Materials Profit Sharing Scheme, dated February 7, 2006, to amend Clause 20 of the Trust Deed thereunder | 10-K | 000-06920 | 10.48 | 12/12/2008 |
| 10.7 | Deed of Amendment to Applied Materials Profit Sharing Scheme, dated February 7, 2006, to amend the definition of Eligible Employee in the First Schedule to the Trust Deed thereunder. | 10-K | 000-06920 | 10.49 | 12/12/2008 |
| 10.8* | Form of Restricted Stock Unit Agreement for use under the amended and restated Applied Materials, Inc. Employee Stock Incentive Plan | 10-Q | 000-06920 | 10.1 | 8/23/2018 |
| 10.9* | Form of Restricted Stock Unit Agreement for Nonemployee Directors for use under the amended and restated Applied Materials, Inc. Employee Stock Incentive Plan | 10-Q | 000-06920 | 10.4 | 5/24/2012 |
| 10.10* | Form of Performance Shares Agreement for use under the amended and restated Applied Materials, Inc. Employee Stock Incentive Plan | 10-Q | 000-06920 | 10.5 | 5/24/2012 |
| 10.11* | Form of Restricted Stock Agreement for use under the amended and restated Applied Materials, Inc. Employee Stock Incentive Plan | 10-Q | 000-06920 | 10.3 | 8/23/2012 |
| 10.12* | Applied Materials, Inc. Employees' Stock Purchase Plan, amended and restated effective October 28, 2012 | 10-K | 000-06920 | 10.54 | 12/5/2012 |
| 10.13* | Offer Letter, dated August 14, 2013, between Applied Materials, Inc. and Gary E. Dickerson | 10-Q | 000-06920 | 10.2 | 8/22/2013 |
| 10.14* | Form of Non-Qualified Stock Option Agreement for Employees for use under the Applied Materials, Inc. Employee Stock Incentive Plan, as amended | 10-Q | 000-06920 | 10.4 | 8/22/2013 |
| 10.15* | Form of Performance Unit Agreement for use under the Applied Materials, Inc. Employee Stock Incentive Plan, as amended | 10-Q | 000-06920 | 10.2 | 2/20/2014 |
| 10.16* | Form of Letter of Understanding for Long-Term Assignment | 10-K | 000-06920 | 10.49 | 12/17/2014 |
| 10.17* | Applied Materials, Inc. Applied Incentive Plan, amended and restated effective October 27, 2014 | 10-Q | 000-06920 | 10.4 | 2/19/2015 |
| 10.18* | Form of Performance Shares Agreement for fiscal 2015 performance-based equity awards for certain executive officers under the amended and restated Applied Materials, Inc. Employee Stock Incentive Plan | 10-Q | 000-06920 | 10.5 | 2/19/2015 |
| 10.19 | Credit Agreement, dated as of September 3, 2015, among Applied Materials, Inc., JPMorgan Chase Bank, N.A., as administrative agent, and other lenders named therein | 8-K | 000-06920 | 10.1 | 9/9/2015 |
| 10.20 | Extension Agreement, dated as of September 3, 2017, to Credit Agreement, dated as of September 3, 2015, among Applied Materials, Inc., JPMorgan Chase Bank, N.A., as administrative agent, and the lenders party thereto. | 8-K | 000-06920 | 10.1 | 9/5/2017 |
| 10.21* | Applied Materials, Inc. Stock Purchase Plan for Offshore Employees, amended and restated effective July 15, 2015 | 10-K | 000-06920 | 10.32 | 12/9/2015 |
| 10.22* | Applied Materials, Inc. 2016 Deferred Compensation Plan | 10-K | 000-06920 | 10.33 | 12/9/2015 |
| 10.23* | Applied Materials, Inc. Employee Stock Incentive Plan, as amended and restated effective March 9, 2017 | 10-Q | 000-06920 | 10.1 | 5/25/2017 |
| 10.24* | Applied Materials, Inc. Senior Executive Bonus Plan, as amended and restated effective March 9, 2017 | 10-Q | 000-06920 | 10.2 | 5/25/2017 |
| 21 | Subsidiaries of Applied Materials, Inc.† | | | | |
| 23 | Consent of Independent Registered Public Accounting Firm, KPMG LLP† | | | | |
| 24 | Power of Attorney (included on the signature page of this Annual Report on Form 10-K)† | | | | |

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| Exhibit No. | Description | Incorporated by Reference | | | |
|---|---|---|---|---|---|
| | | Form | File No. | Exhibit No. | Filing Date |
| 31.1 | Certification of Chief Executive Officer Pursuant to Section 302 of the Sarbanes-Oxley Act of 2002† | | | | |
| 31.2 | Certification of Chief Financial Officer Pursuant to Section 302 of the Sarbanes-Oxley Act of 2002† | | | | |
| 32.1 | Certification Pursuant to 18 U.S.C. Section 1350, as adopted pursuant to Section 906 of the Sarbanes-Oxley Act of 2002‡ | | | | |
| 32.2 | Certification Pursuant to 18 U.S.C. Section 1350, as adopted pursuant to Section 906 of the Sarbanes-Oxley Act of 2002‡ | | | | |
| 101.INS | XBRL Instance Document‡ | | | | |
| 101.SCH | XBRL Taxonomy Extension Schema Document‡ | | | | |
| 101.CAL | XBRL Taxonomy Extension Calculation Linkbase Document‡ | | | | |
| 101.DEF | XBRL Taxonomy Extension Definition Linkbase Document‡ | | | | |
| 101.LAB | XBRL Taxonomy Extension Label Linkbase Document‡ | | | | |
| 101.PRE | XBRL Taxonomy Extension Presentation Linkbase Document‡ | | | | |

* Indicates a management contract or compensatory plan or arrangement, as required by Item 15(a)(3).

† Filed herewith.

‡ Furnished herewith.

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## SIGNATURES

Pursuant to the requirements of Section 13 or 15(d) of the Securities Exchange Act of 1934, the registrant has duly caused this report to be signed on its behalf by the undersigned, thereunto duly authorized.

APPLIED MATERIALS, INC.

By: /s/ GARY E. DICKERSON
Gary E. Dickerson
*President, Chief Executive Officer*

Dated: December 13, 2018

## POWER OF ATTORNEY

KNOW ALL PERSONS BY THESE PRESENTS, that each person whose signature appears below constitutes and appoints Gary E. Dickerson, Daniel J. Durn and Thomas F. Larkins, jointly and severally, his or her attorneys-in-fact, each with the power of substitution, for him or her in any and all capacities, to sign any amendments to this Annual Report on Form 10-K, and to file the same, with exhibits thereto and other documents in connection therewith, with the Securities and Exchange Commission, hereby ratifying and confirming all that each of said attorneys-in-fact, or his substitute or substitutes, may do or cause to be done by virtue hereof.

******

Pursuant to the requirements of the Securities Exchange Act of 1934, this report has been signed below by the following persons on behalf of the registrant and in the capacities and on the dates indicated.

| | Title | Date |
|---|---|---|
| /s/ GARY E. DICKERSON<br>Gary E. Dickerson | President, Chief Executive Officer and Director (Principal Executive Officer) | December 13, 2018 |
| /s/ DANIEL J. DURN<br>Daniel J. Durn | Senior Vice President, Chief Financial Officer (Principal Financial Officer) | December 13, 2018 |
| /s/ CHARLES W. READ<br>Charles W. Read | Corporate Vice President, Corporate Controller and Chief Accounting Officer (Principal Accounting Officer) | December 13, 2018 |
| /s/ THOMAS J. IANNOTTI<br>Thomas J. Iannotti | Chairman of the Board | December 13, 2018 |
| /s/ JUDY BRUNER<br>Judy Bruner | Director | December 13, 2018 |
| /s/ XUN CHEN<br>Xun Chen | Director | December 13, 2018 |
| /s/ AART J. DE GEUS<br>Aart J. de Geus | Director | December 13, 2018 |
| /s/ STEPHEN R. FORREST<br>Stephen R. Forrest | Director | December 13, 2018 |
| /s/ ALEXANDER A. KARSNER<br>Alexander A. Karsner | Director | December 13, 2018 |
| /s/ ADRIANNA C. MA<br>Adrianna C. Ma | Director | December 13, 2018 |
| /s/ SCOTT A. MCGREGOR<br>Scott A. McGregor | Director | December 13, 2018 |
| /s/ DENNIS D. POWELL<br>Dennis D. Powell | Director | December 13, 2018 |